

Particulate Matter

Week in Review

Week ending 26 September 2026 | 20 September 2026 to 26 September 2026 | 162 records pooled from 7 daily issues

162

RECORDS POOLED
(162 UNIQUE DOIs)

33%

MEASUREMENT SIDE
(54 RECORDS)

10

SUBTOPICS
ACTIVE

31

POOLED EFFECT
ESTIMATES (31 WITH A
CI)

34

ABSTRACT-FREE
(21% OF WEEK)

The week in one paragraph

The biggest week in the series by volume, and the worst since August for missing abstracts: 162 records, 34 of them (21 %) carried on metadata alone — double last week's rate. The measurement side fell to a series-low third of the week while *Occupational & indoor* nearly doubled; the indoor turn named a trend last week is now the week's dominant movement.

Scale. 162 records over seven issues on daily totals of 12, 18, 23, 27, 35, 24 and 23 — a $2.9\times$ range, wider than last week's $1.9\times$ because Sunday 20 September returned 12. Against recent rollups: 162 after 148, 126, 150, 109 and 112 — the highest weekly total recorded. All ten subtopics are populated.

The measurement side fell to 33 %. *Exposure assessment & modelling* eased 38 → 34 and *Sensing* fell 27 → 20, so the measurement-side share reads 47, 54, 40, 44, 33 % across five weeks — the lowest in the series. It is not a contraction of the instrumentation literature: 21 of the 54 measurement-side records are metadata-only (12 in *Exposure*, 9 in *Sensing*), and the same Elsevier and ACS deposit pattern that hollowed out those rows also drove the total up. The share will be re-read when the retry list clears.

***Occupational & indoor* 12 → 22.** Last week this watch said the indoor turn had qualified as a trend and asked whether any of it would move from chambers and models into occupied buildings with health outcomes. **Some did.** The 25 September issue carried a 948-home year-round rural heating study (annual mean $109 \pm 207 \mu\text{g m}^{-3}$) and a Xuanwei stove intervention; 24 September a purifier crossover with mitochondrial DNA methylation outcomes; 26 September a cystic-fibrosis home-sensor pilot in which time above $35 \mu\text{g m}^{-3}$, not the mean, carried the cytokine signal. Occupied-building work with a biological outcome now exists; none of it is yet randomised with a clinical endpoint except the AIRVAC protocol entering the registry.

Abstract-free doubled, and one reason was an outage. 34 of 162 (21 %) against 15 of 148 (10 %). Europe PMC returned HTTP 503 for the whole 25 September run, removing both a harvest leg and the second abstract-recovery route, and Elsevier's same-day deposits without abstracts continued. The deferred-retry pass named as the only remaining fix in the last two rollups is **still not built**, and the held list has grown past forty.

Fewer estimates, all with intervals. 31 pooled estimates, one per 5.2 records (series 2.9, 4.4, 2.7, 3.1, 5.2 — the sparsest week), but 31 of 31 carry a CI, the first complete week. Only the 25 September issue reported none.

Subtopic trend and inflection

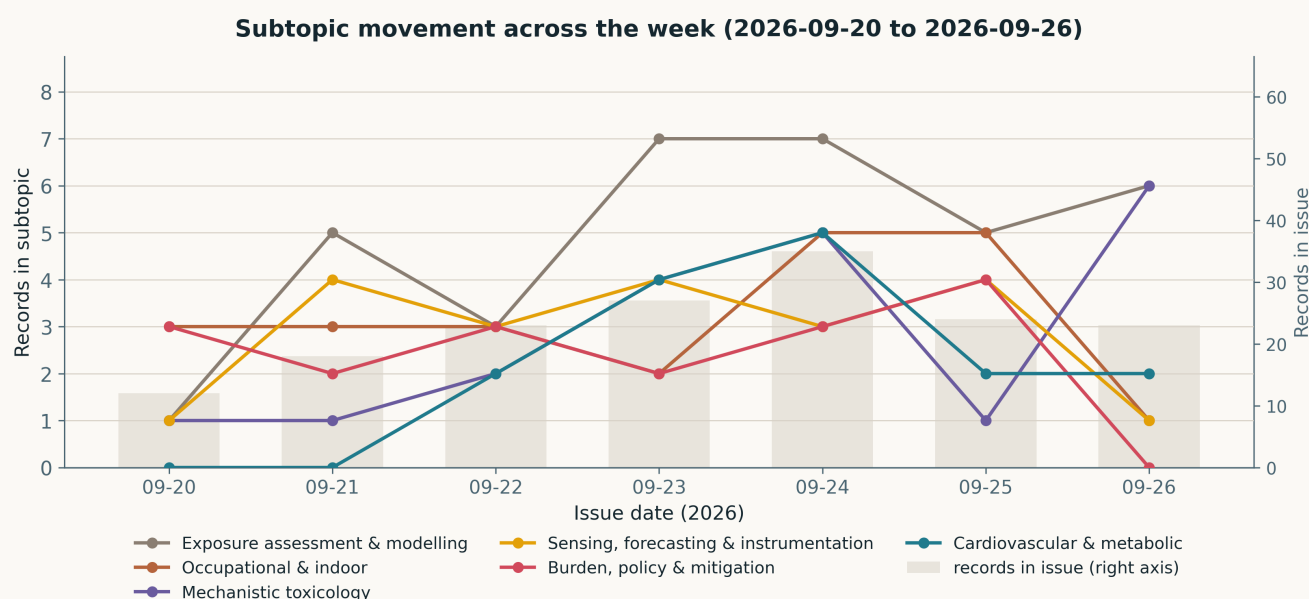


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals 12, 18, 23, 27, 35, 24 and 23 — a $2.9\times$ range. The Sunday low (20 September) returns after a week's absence; the 35-record Thursday (24 September) carried the CHARLS constituent and purifier-methylation records and is a complete day, not a double build. Four of the week's seven issues were built late (the 22:00 rule deferred each by up to a day), none left a gap.

Expanding. *Occupational & indoor* 12 → 22 is the week's largest movement (above). *Mechanistic toxicology* 14 → 20 continues a third week of recovery, led by nitrated-PM inflammation, size-resolved constituent toxicity and the iron-superoxide results. *Cardiovascular & metabolic* 11 → 15 and *Burden, policy & mitigation* 13 → 17 both rise; *Reproductive & developmental* 5 → 8 leaves the bottom of the table for the first time in this series, carried by the Butajira fetal-biometry cohort, the Chengdu monitor-versus-satellite preterm analysis, the Chiang Rai placental microscopy and the Shenzhen Birth Cohort profile.

Contracting. *Sensing* 27 → 20 and *Exposure* 38 → 34, both inflated in metadata-only share; *Respiratory & allergic* 13 → 10. *Neuro* (7 → 8) and *Other clinical* (8 → 8) hold.

The inflection worth naming. Black carbon led the constituent weighting in both cardiovascular constituent analyses of the week — CHARLS incident stroke (per IQR BC HR 1.77, 1.54–2.03, above $PM_{2.5}$ mass at 1.55) on 24 September, and the Kailuan heart-failure cohort (BC 40.2–48.7% of the mixture weight) on 25 September. Two independent Chinese cohorts, two cardiovascular outcomes, one constituent. For a sensor watch this is the most actionable signal of the month: BC is the one health-leading constituent that a low-cost node can measure (filter-based absorption), and mass-only networks cannot see it. The test for next week is whether a non-Chinese cohort shows the same ordering.

Study architecture

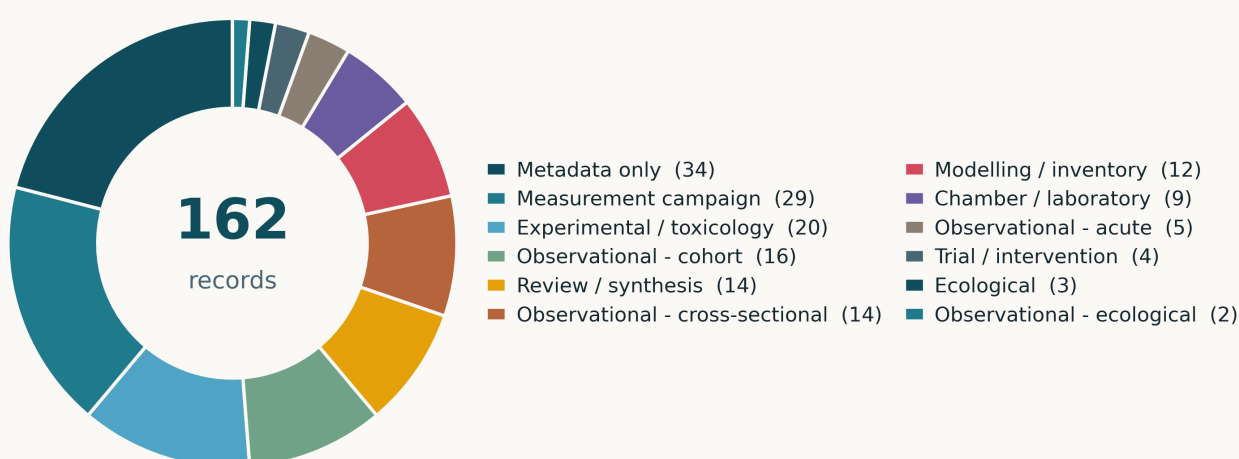


Fig. Study architecture over the pooled week. *Metadata only* is the largest single group at 34, ahead of *Measurement campaign* (29) and *Experimental / toxicology* (20) — the first week since August in which the missing-abstract bucket leads. Primary observational epidemiology (cohort 16, cross-sectional 14, acute 5, ecological 5) totals 40. Four interventional records, against one last week: the AIRVAC and COATED-AIR trial designs, the Xuanwei stove intervention and a synbiotic RCT.

Pooled analytics

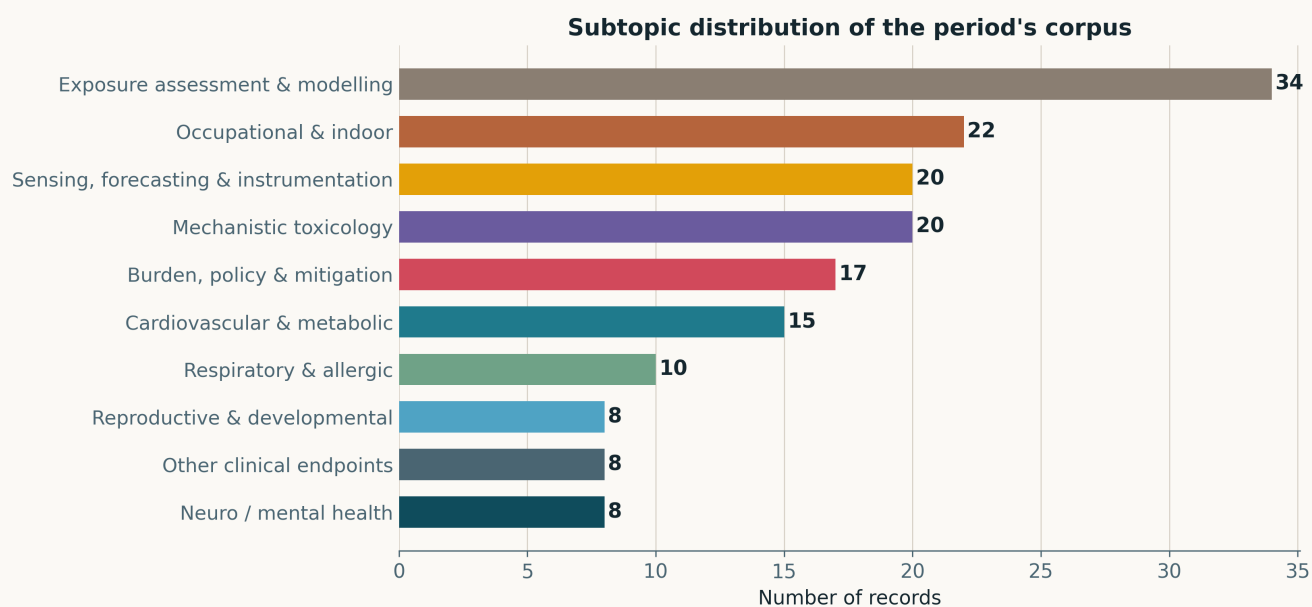


Fig. Pooled subtopic distribution, **all ten bands populated**. *Exposure* 34 and *Sensing* 20 put the measurement side at **54 of 162 (33%)**. *Occupational & indoor* at 22 is now the largest health-side band, ahead of *Mechanistic toxicology* and *Sensing* at 20 and *Burden* at 17.

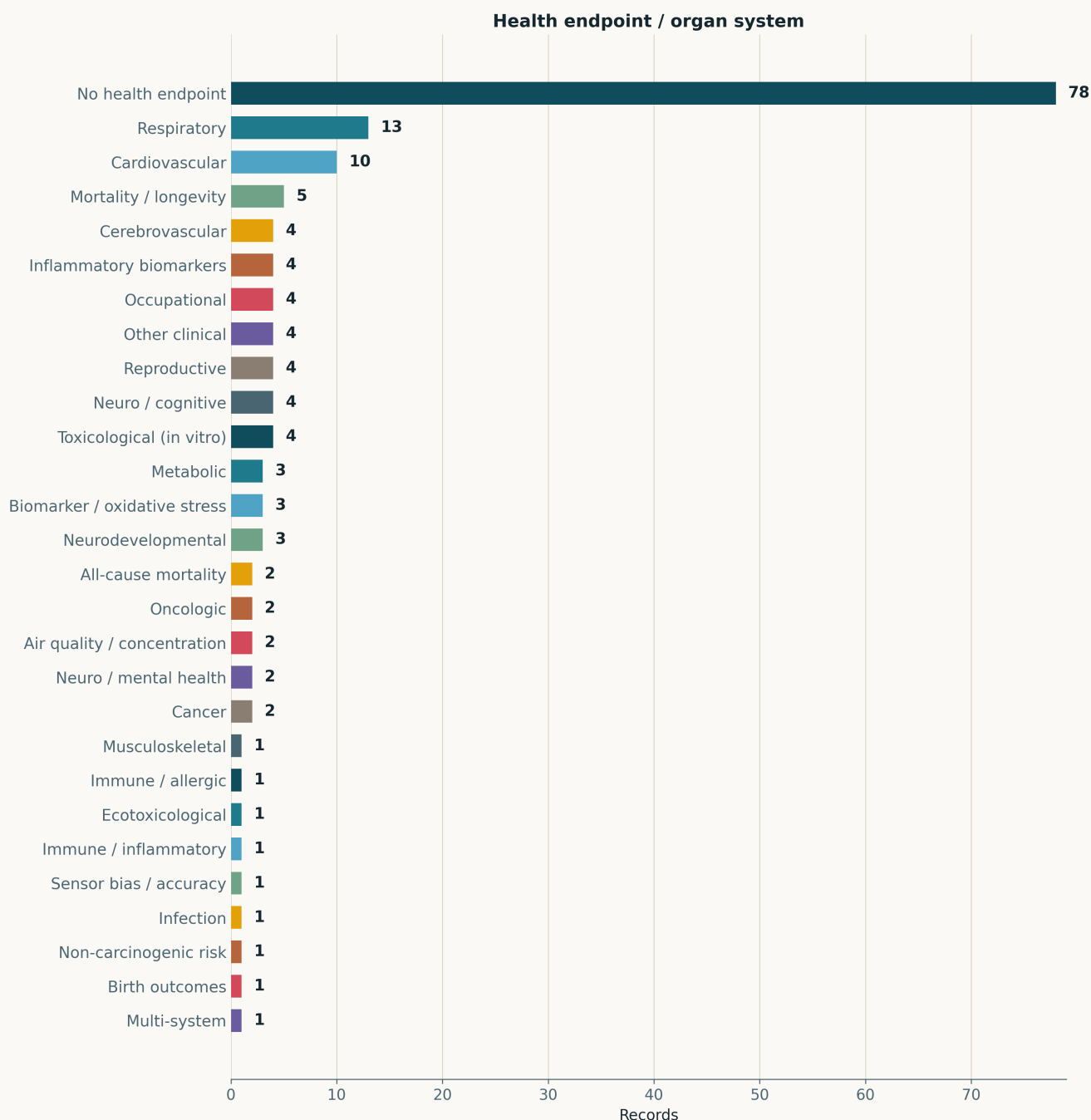


Fig. Health endpoint over the pooled week, canonicalised. “No health endpoint” pools 78 of 162 (48 %) — missing abstracts (24) plus exposure, instrument, chemistry and emission records — up from 41 % last week, the abstract gap showing through. Among human outcomes, **respiratory leads at 13**, cardiovascular 10, mortality/longevity 5, then a run of labels at 4 (cerebrovascular, inflammatory biomarkers, occupational, reproductive, neuro/cognitive, other clinical).

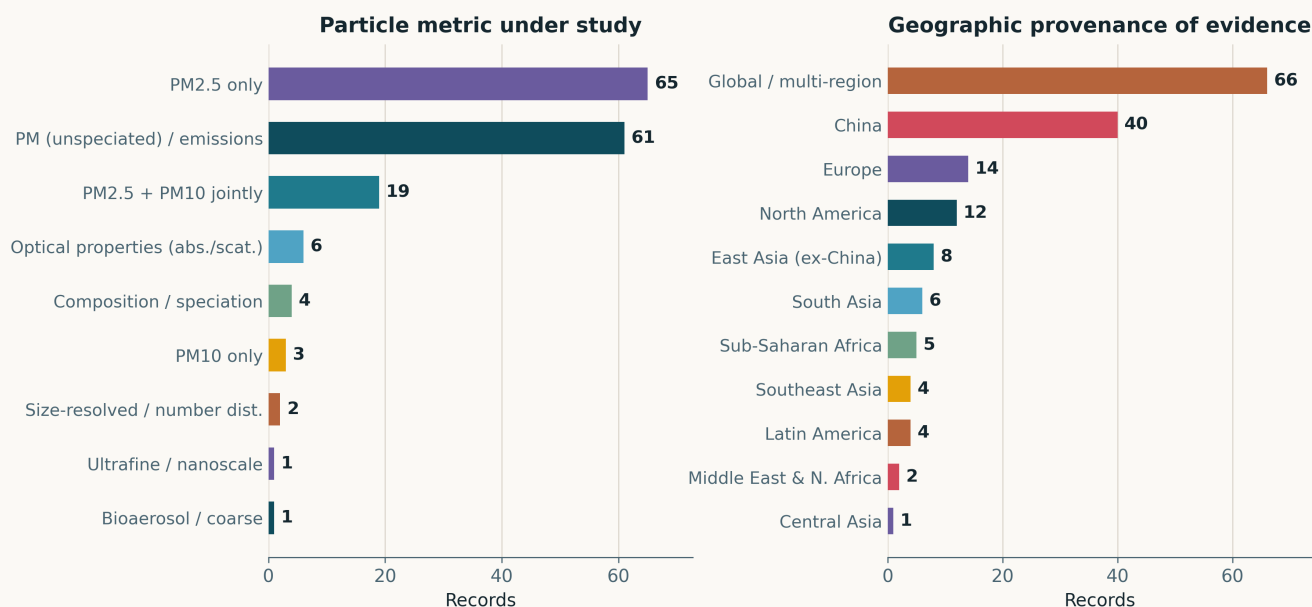


Fig. Left — particle metric. PM_{2.5} alone 65 of 162, unspciated PM or emissions 61, joint PM_{2.5} + PM₁₀ 19. The tail: optical properties 6, composition 4, PM₁₀ alone 3, size-resolved 2 and ultrafine 1, bioaerosol 1. Right — geography. Global / multi-region 66 overstates reach, as every week, pooling reviews, laboratory studies and unrecoverable metadata-only records. China 40 is the largest single-country bar, then Europe 14, North America 12 and East Asia (ex-China) 8; Sub-Saharan Africa reached 5 (Luanda, Durban, Butajira among them).

Life-course windows addressed this period (records may span several stages)

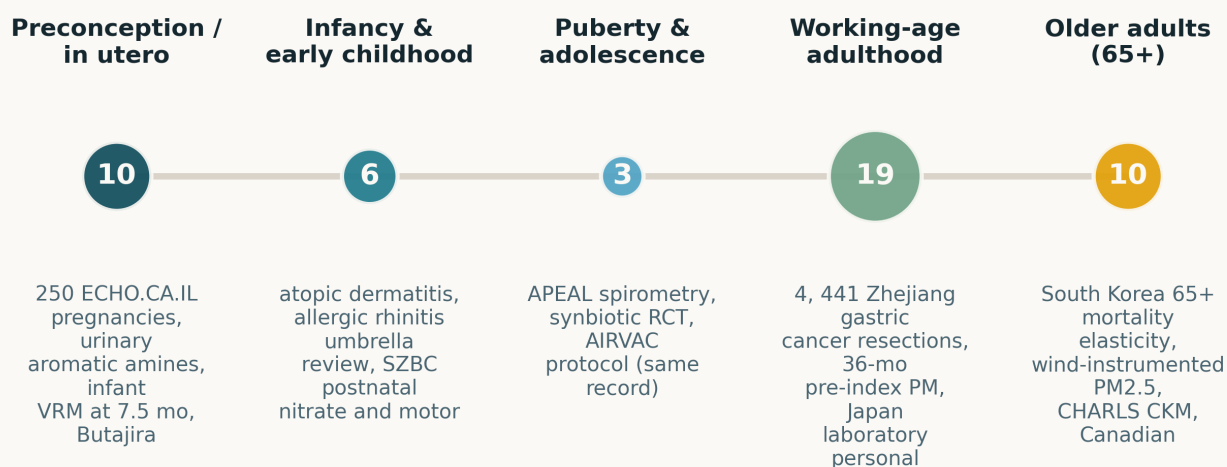


Fig. Life-course windows across the pooled week; counts non-exclusive. Working age 19, preconception / in utero 10 and older adults 10, infancy 6. Adolescence 3 — still the weakest window, carried by APEAL spirometry, a synbiotic RCT and the AIRVAC protocol (also counted in childhood).

Quantitative associations reported in the period's corpus (1 of 2)



Quantitative associations reported in the period's corpus (2 of 2)

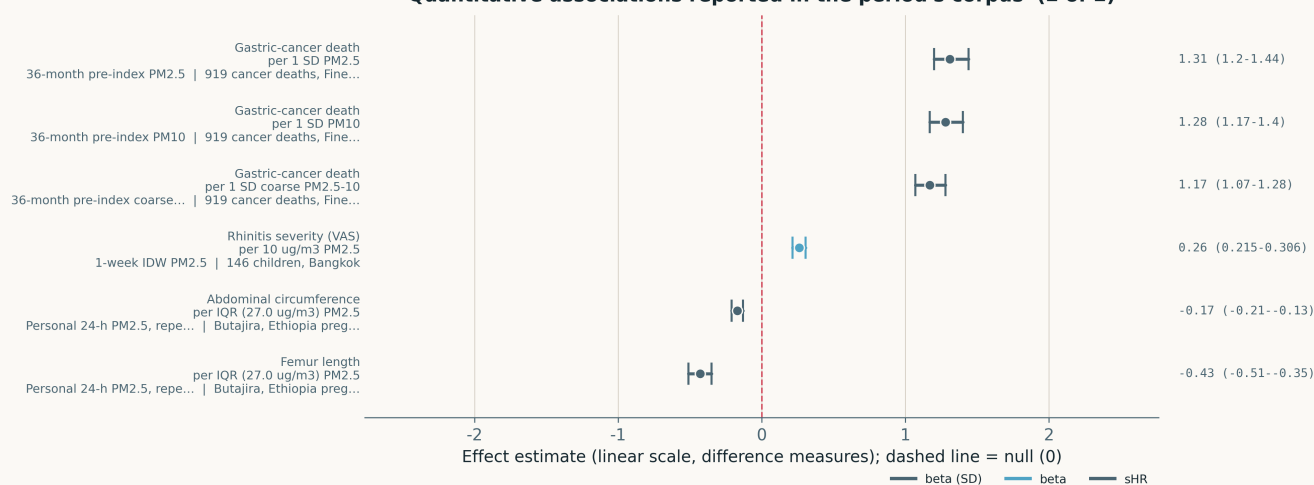


Fig. All 31 pooled estimates, ratio and difference scales in separate panels. **Not mutually comparable and not meta-analysed:** contrasts include per-IQR, per-SD, per-10 and per-1 $\mu\text{g m}^{-3}$, per-doubling, dichotomised and unstated increments, on HR, sHR, OR, RR and RR-equivalent scales; heterogeneity is structural, so no I^2 is reported. Readable points: **the largest ratio is the least precise — OR 4.78 (1.61–14.29)** per doubling of same-day PM_{2.5} for type A aortic dissection, from 40 cases; **the tightest interval is the smallest effect — 0.9977 (0.9972–0.9983)** scholar publications per 1 $\mu\text{g m}^{-3}$ annual PM_{2.5}; three CHARLS analyses (stroke $\times 2$, CVD) share participants and are not independent; and the Taiwan COPD case-control shows the exposure-window dependence directly (1-month OR 1.252,

12-month 1.642, same cases). The difference panel carries the Butajira fetal-biometry deficits (femur length -0.43 SD per IQR) and the Bangkok rhinitis VAS slope.

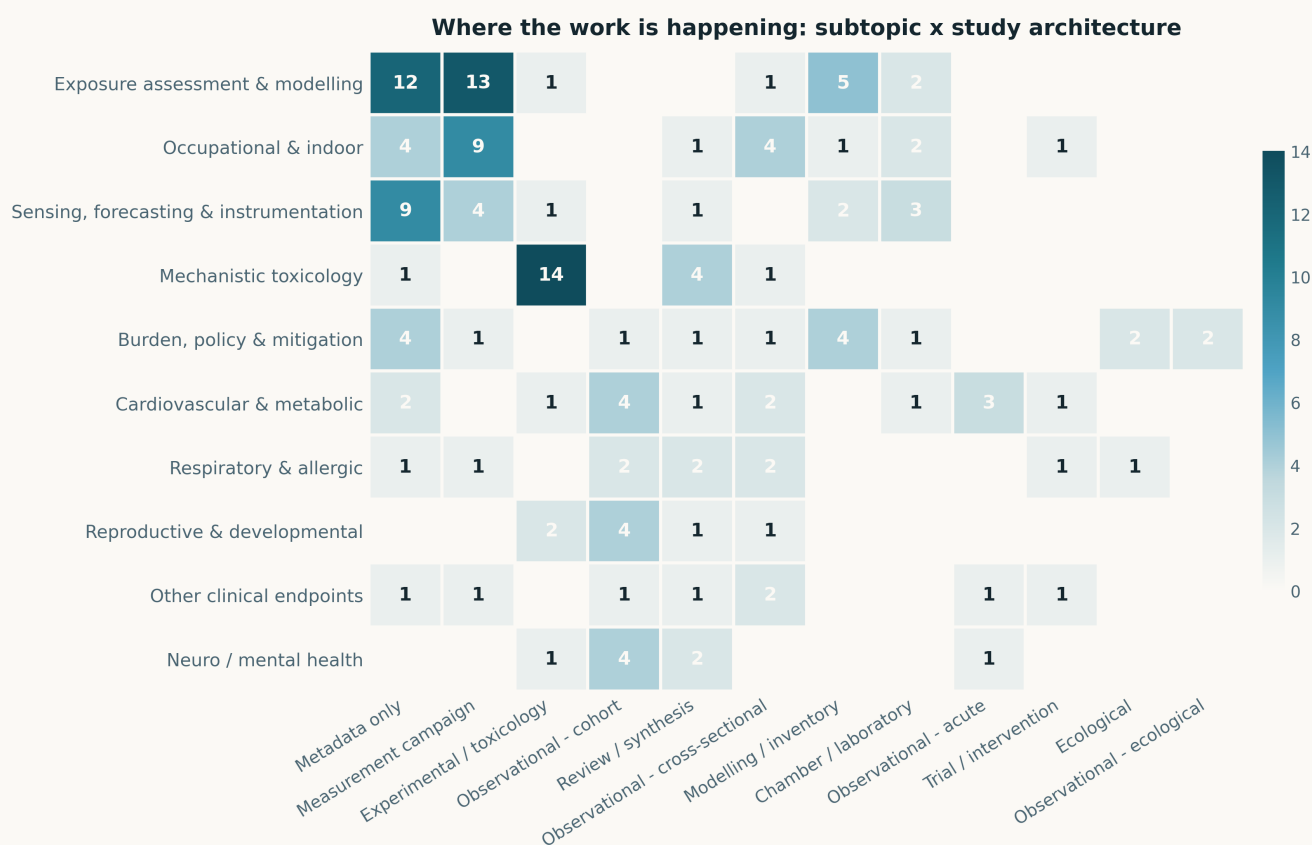


Fig. Subtopic $\times$ architecture, pooled. The densest cell is *Mechanistic toxicology* $\times$ *Experimental* (14), then *Exposure* $\times$ *Measurement campaign* (13), *Exposure* $\times$ *Metadata only* (12) and *Sensing* $\times$ *Metadata only* (9) — the last two are the abstract gap concentrated on the measurement side. The standing finding survives a ninth week: of the 54 measurement-side records, 4 carry any outcome label and 1 observes a human outcome (the PMRI ecological mortality preprint); two are modelled risk indices and one is an ecotoxicity assay.

Venues, trial watch, and what the pipeline did

Venues. A three-way tie at 9 — *Environ Sci Technol*, *Environ Pollut*, *Atmos Environ* — then *Atmos Pollut Res* and *Toxics* at 7, *Atmos Chem Phys*, *Environ Health (Wash)* and *Front Public Health* at 5. The leading venue holds 6%. *Atmos Environ* and *Atmos Pollut Res* are the main sources of metadata-only records (Elsevier same-day deposits). **Preprints supplied 6**, up from 2, all labelled.

Recurring authors. Unchanged: surname frequency is a naming artefact, and **no repeat-group cluster is claimed**. The one real repeat is a *dataset*, not a group: CHARLS supplied three separate PM-cardiometabolic papers in the week.

Trial watch — one new trial, one resolved, one update. NCT07196436 (AIRVAC) entered on 25 September via its *PLoS One* protocol: four arms, active or sham energy recovery ventilator $\times$ active or sham portable air cleaner, >80 Dallas-Fort Worth asthma households; `analyze_endpoints` shows **spirometry and ACT as co-primary** in the registry against ACT alone in the protocol. The **COATED-AIR** design paper (22 September) was resolved to NCT06541691 ($n=3,000$, Tehran): the air-pollution randomisation's primary is a composite of non-fatal ischaemic stroke, type I MI, acute limb events or CV death over 30 months, with **no registered exposure or adherence endpoint**, so whether the mask-and-alert package changes anyone's PM dose will not be known; the record still reads *recruiting* six months after its listed primary completion. NCT02944682 (HAPIN), the completed multi-country LPG trial, posted a registry update. NCT05423665 reappeared twice and was already held. `trials.json` now holds 16 trials.

What the harvesting legs did. The PubMed connector was load-bearing every day, adding PMIDs the harvester's two PubMed queries missed and supplying admissions on each issue. The harvester's PubMed sensing query returned 0 on 20–23 September and 1, 5 and 2 on the last three days. Crossref-by-ISSN remained the highest-yield leg on weekdays. Europe PMC was down for the 25 September run (HTTP 503) and back with 45 records on 26 September. OpenAlex contributed nothing all week (429 / paywalled date filter). arXiv returned HTTP 406 through 25 September and was answering again by the 29 September run (no entry dated inside the window). Consensus returned the right literature and 0 new records every day, for the fifth consecutive week. Scholar Gateway failed with an identity error on its first attempt this week. Proofing caught caption and prose miscounts, one un sourced number and orphaned headings on several days; all were fixed before shipping.

Open gaps

- **The abstract-retry pass, third week running.** 21% of the week shipped without an abstract, and the most watch-relevant titles are disproportionately among them (a year-long personal PM/CO₂ monitor pilot, Korean station-relocation discontinuities, an AS&T real-time

EC monitor comparison, a DMA geometry paper, the NAKO mental-health cohort). This is now the single largest loss of signal in the pipeline.

- ▶ **Black carbon is leading cardiovascular constituent analyses; the watch has no BC sensing record this week with a health link.** The two constituent cohorts use modelled BC; none of the week's absorption-measurement records (Beijing lensing apportionment, Italian 10-year eBC trends) touches an outcome.
- ▶ **One of 54 measurement-side records observes a human outcome,** ninth consecutive week.
- ▶ **Ultrafine and size-resolved number: 3 of 162,** the same share as last week, despite a real-time ultrafine size-distribution paper on 22 September.
- ▶ **Pipeline defects carried forward:** no first-author affiliation field; the `Cancer/Oncologic` split in `ENDPOINT_CANON`; no `\DIGESTDATE-vs--date` build guard; the 2026-07-27 corpus missing `LIFECOURSE`; `.git` debris on the delete-restricted mount. `metrics.csv` continues clean (csv.writer, 500 rows, 0 malformed).

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation

20 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (20 OF 20 RECORDS IN THIS SUBTOPIC)

AIRWISE Delhi 2026 Atmos Chem Phys

Improving aerosol-radiation interactions in the operational forecasting system - AIRWISE.

Replacing globally prescribed refractive indices with Delhi-specific values inside the operational AIRWISE system cut surface downwelling shortwave by up to ~80 W/m² instantaneously (~43 W/m² diurnally) during a severe post-monsoon episode, and monthly mean SWDOWN by 25-37 W/m², a quarter to a third of total clear-sky aerosol-radiation interaction. The meteorological response is coherent and the right sign: surface temperature -0.2 C, near-surface wind -0.4 m/s, boundary layer height -200 m, daytime humidity +3-4%, and consequently higher PM_{2.5} - the shallower, calmer boundary layer traps the aerosol that caused the dimming. Validated against WiFEX surface radiation at Delhi airport with 62% (December) and 35% (January) bias reduction. Limits: one city, one season, refractive indices tuned where they are evaluated, and radiation bias reduction does not demonstrate PM_{2.5} forecast skill improvement, which is the operational claim a reader will assume. Relevant because the feedback runs the wrong way for forecasting: getting optics right makes the pollution episode worse, so a network assimilating into a model with generic refractive indices inherits a systematic low bias during exactly the episodes that matter.

doi: 10.5194/acp-26-13189-2026

Anwer et al. 2026 Atmos Environ

Artificial Intelligence in Sand and Dust Storm Research: A Systematic and Bibliometric Review (1990-2025)

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 26 Sep with no abstract; Crossref, PubMed efetch, Europe PMC and the publisher landing page returned nothing. The title indicates a systematic and bibliometric review of AI methods in sand and dust storm research over 1990-2025. Of interest only if it tabulates forecast skill against PM₁₀ observations; bibliometric reviews usually do not. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122387

Barth et al. 2026 Environ Sci Atmos

Online characterisation of oxidative potential and reactive oxygen species of fresh and aged aircraft turbine and ship engine particulate emissions.

Only a one-sentence abstract was deposited: fresh and aged transport-sector particles from shipping and aircraft contribute significantly to particle oxidative potential. The title establishes the design - online (not filter-offline) OP and ROS measurement of fresh and aged turbine and marine-engine emissions - but no OP values, ageing conditions, fuels or particle metrics can be stated. Carried at tier B for the method: online OP is the toxicity-relevant metric a sensor network cannot yet measure, and fresh-versus-aged contrasts determine whether near-source mass is a fair proxy for harm. Retry list for the full abstract.

doi: 10.1039/d5ea00174a

Broda et al. 2026 Atmos Meas Tech

A new balloon-borne system for measuring the vertical variability of aerosol optical properties using low-cost sensors.

A balloon-borne AeroSonde carrying calibrated SPS30 optical PM sensors was referenced against an Aurora 4000 nephelometer and an LAS 3340 spectrometer to retrieve aerosol scattering coefficient, scattering Angstrom exponent and effective radius, with Pearson r above 0.88. Agreement was best for the scattering coefficient; SAE and effective radius carried substantially larger uncertainty and are honestly restricted to qualitative fine-versus-coarse discrimination. Soundings in June and August 2025 reached 10-27 km at better than 10 m vertical resolution and captured North American biomass-burning plumes and Saharan dust, agreeing with active remote sensing on layer structure and extensive properties at r 0.74-0.81. Two campaigns and no independent in-situ profile leave the intensive retrievals unvalidated aloft. Still the clearest demonstration this quarter that a commodity OPC can do profiling work.

doi: 10.5194/amt-19-5989-2026

Contini et al. 2026 Environ Pollut

10-year trends of particle size distributions and equivalent black carbon at the Environmental-Climatic Observatory (ECO) in Lecce (Italy).

A decade (2015-2024) of co-located online PM_{2.5}, coarse PM, particle number size distribution (10-10,000 nm) and equivalent BC at a southern-Italian observatory, with offline OC/EC to derive the mass absorption cross-section. PM_{2.5} fell -2.58%/yr, eBC -3.17%/yr and total PNC -2.99%/yr (Aitken and accumulation modes), while PM_{10-2.5} was flat, held up by Saharan dust on 13.2% of days. MAC was stable at 12.4 +/- 2.4 m²/g, so falling absorption reflects falling eBC rather than changing optics; new particle formation occurred on about 25% of days. Single site, and the abstract gives neither the trend estimator nor uncertainty on the slopes. Relevance: a stable decade-long MAC is a rare field-validated constant for converting absorption to BC mass - and the fine/coarse decoupling under dust is precisely where low-cost optical sensors misreport.

doi: 10.1016/j.envpol.2026.129234

Khalili Hollo et al. 2026 ACS Environ Au

Validation and Correction of PM_{2.5} Mass Concentrations from Consumer-Grade Personal Exposure and Outdoor Fixed Monitors: A Two-Year Colocation Study.

Fifty Atmotube Pro personal monitors and five QuantAQ Modulair-PM outdoor units were co-located four times over two years with a GRIMM EDM180 before deployment in the SJEQ-Denver community study. Raw 1-h agreement was modest (Atmotube rho 0.770, RMSE 5.12 µg m⁻³; QuantAQ rho 0.720, RMSE 5.51), both under-read above 30 µg m⁻³, and unit-level response drifted between colocations; 5-11% of Atmotubes were not working at any comparison. An MLR correction on RH and dew point raised rho by 6.5-8.3% and cut RMSE by 42-50%, and reduced drift. The GRIMM is itself an optical FEM, so 'reference' carries its own humidity and composition bias, and a correction fitted to Denver's high-altitude semi-arid aerosol will not transfer. Relevance: direct evidence that periodic re-colocation, not one pre-deployment fit, is what holds a wearable fleet in calibration.

doi: 10.1021/acsenvironau.6c00020

Lawson et al. 2026 Aerosol Sci Technol

Humidity-dependent photoacoustic absorption biases and effective water mass accommodation coefficients for multicomponent aerosol particles

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed fetch and Semantic Scholar all returned nothing. The title states a laboratory characterisation of humidity-dependent bias in photoacoustic absorption measurement and effective water mass accommodation coefficients for multicomponent particles (Lawson, Langridge, Allan, Cotterell). Tiered B despite the missing abstract: photoacoustic absorption is a BC reference method, and an RH-dependent bias in the reference propagates into every sensor calibrated against it. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1080/02786826.2026.2733732

Le et al. 2026 Meas Sci Technol

Real-time monitoring of ultrafine aerosol size distributions using parallel non-bouncing impactors coupled with corona charger.

A Parallel Electro-inertial Particle Sizer: unipolar corona charger feeding six parallel non-bouncing impactor/Faraday-cup electrometer channels with kernel inversion, 10-560 nm. Impactor cut-offs landed within 5% of design; against an SMPS, unimodal and bimodal distributions over $1e4$ - $1e7$ /cm³ were reproduced with under 10% deviation in total PNC, NMAD and GSD. In a 16-day rooftop co-location, total PNC and UFP gave slope 1.023 and r 0.995, and nucleation, growth and plume events were resolved. Weak points are stated: sub-20 nm and complex multimodal structure deviate, and one site over 16 days says nothing about drift, humidity or long-term electrometer zero stability. Cost is claimed lower but not quantified. Directly relevant: a cheaper, maintenance-light sizer is what UFP network monitoring currently lacks.

doi: 10.1088/1361-6501/aeaafe

Liu et al. 2026b Atmos Environ

Morphology Change of Diesel Exhaust Black Carbon Particles during Humidification and Evaporation.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Liu, Du, Li, Zhang) and DOI on 23 Sep without an abstract. Morphology change of diesel black carbon under humidification and evaporation is the physical mechanism behind humidity-dependent bias in every optical instrument that infers mass from scattering, so the topic is directly load-bearing for low-cost sensor correction - and the deposit gives no technique, no humidity range, no restructuring metric and no numbers. Recorded on metadata with its subtopic taken from the title, flagged for retry, and explicitly not summarised: a morphology claim inferred from a title would be a fabricated result.

doi: 10.1016/j.atmosenv.2026.122360

Low-cutoff cyclone method 2026 Aerosol Sci Technol

An improved method for measuring the penetration efficiency of low cutoff cyclones.

METADATA ONLY. Metadata-only record: Taylor & Francis deposited no abstract, the landing page is bot-blocked, and neither Crossref nor a Consensus sweep of 2026 cyclone-characterisation work surfaced it. This is the most directly relevant instrumentation title of the window and the least legible. Low-cutoff cyclone penetration efficiency is the calibration that defines what a size-selective inlet actually admits, and an error there propagates into every downstream gravimetric or optical measurement behind that inlet - including the reference samplers that low-cost sensor networks are co-located against, where an inlet cut error becomes a systematic multiplicative bias in the correction factor rather than added noise. Top of the retry list; worth a manual full-text pull rather than waiting on the deposit.

doi: 10.1080/02786826.2026.2731025

Medellin MABI eBC 2026 Atmos Pollut Res

Multi-wavelength spectral attribution of equivalent black carbon in Medellin, Colombia (2010-2023): Disentangling fossil fuel and biomass burning using the MABI instrument.

METADATA ONLY. Metadata-only record; no abstract deposited or recoverable. A 14-year multi-wavelength absorption record with fossil-versus-biomass attribution would be an unusually long instrument series for a Latin American city, and the same absorption-Angstrom-exponent dependence the Sonipat paper in this issue flags as its principal uncertainty applies here, so the two would be worth reading against each other once the text is available. Nothing further can be said without it; retry list.

doi: 10.1016/j.apr.2026.103203

Papathanasiou et al. 2026 Atmos Pollut Res

Evaluating Personal Air Quality Monitors for PM_{2.5} and CO₂: Insights from a Year-Long Pilot Study.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed fetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a year-long evaluation of personal PM_{2.5} and CO₂ monitors. High priority for this watch: it pairs with the two-year Denver colocation in this issue and the retry should target instrument model, reference and drift. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.apr.2026.103210

Parks et al. 2026 Aerosol Sci Technol

Evaluating performance of real-time monitors for quantifying elemental carbon in diesel particulate matter.

METADATA ONLY. Metadata-only record: Taylor and Francis deposited title, authors (Parks, Finley, Raj, Wilson) and DOI on 23 Sep with no abstract, and doi.org returned 403 to an automated fetch, so no landing-page recovery either. This is the single most relevant title of the day for the watch - a head-to-head evaluation of real-time elemental-carbon monitors against diesel PM is exactly the reference-grade comparison that underpins occupational black-carbon sensing - and none of it can be reported: no instruments named, no reference method, no bias or precision figures, no exposure range. It is listed, not summarised, and sits at the top of the retry list.

doi: 10.1080/02786826.2026.2733733

Shen et al. 2026 Atmos Chem Phys

Apportioning light absorption of ambient aerosols to black carbon, brown carbon, and lensing effect using a PAX-ISS hybrid method: insights into absorption enhancement.

Couples photoacoustic spectroscopy (total, coated absorption) with an integrating sphere after solvent mediation (lensing-free absorption); the difference is the lensing term, and a dual-wavelength iteration splits the remainder into BC and BrC. Across seasonal Beijing samples in 2023, BC dominated absorption and BrC contributed about 10% annually. Lensing averaged 40% of total absorption but ranged 4.6-52.0% by season and contracted at short wavelengths, consistent with a BrC 'blocking' effect previously only simulated; annual absorption enhancement was 1.69 +/- 0.10. The assumption to test is that solvent mediation removes coatings without altering BC morphology or extracting chromophores unevenly - the method's validity rests on it. One city, one year. For low-cost absorption sensing, a seasonally varying 40% lensing term is the correction a filter photometer's mass-absorption cross-section silently absorbs.

doi: 10.5194/acp-26-13303-2026

Shin et al. 2026 Environ Sci Technol*A Continuous Virus Aerosol-into-Liquid Sampler (VAiLS) Integrating Twin-Vortex Condensation Growth and Wet-Cyclone Enrichment.*

A twin-vortex condensation growth unit (5 C preconditioner, wick-wetted chamber at 20–60 C) feeding a wet cyclone grows virus-sized aerosol into micrometre droplets and concentrates them continuously into liquid at 16 L/min. On KCl nanoparticles, number-weighted collection rose from 7.3% for the cyclone alone to 31.0, 64.4 and 70.4% at growth temperatures of 20, 40 and 60 C, independent of inlet humidity across 16.8–71.8%. For viruses the optimum was 40 C (MS2 93 $\pm$ 0.6%, phi6 90 $\pm$ 2.1%), giving infectious concentrations 3.9e3 and 1.2e3 times a BioSampler, with viability above 92% except phi6 at 60 C. Two surrogate viruses on a bench rig say nothing about field backgrounds or enveloped respiratory pathogens. The humidity independence is the result worth carrying: it is the failure mode of every optical bioaerosol channel.

doi: 10.1021/acs.est.6c03875

Song 2026 Atmos Environ*Relocation-associated discontinuities in Korean air-quality monitoring: a Bayesian structural time-series analysis of PM₁₀ and PM_{2.5}.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates Bayesian structural time-series detection of step changes in PM₁₀ and PM_{2.5} series when regulatory stations are relocated. Directly relevant to network metadata and to every long-term trend or epidemiology study that treats a station ID as a fixed exposure site. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122369

Thomas et al. 2026 J Test Eval*Procedural Challenges in the Analysis of Toxic Metals in Electronic Cannabinoid Delivery System Aerosols Produced for Vaping Cannabinoids.*

No subjects; a white paper reviewing ten years of trapping approaches. The central problem is that metal constituents sit in the $\mu\text{g/g}$ range, so the trap itself and the sample-preparation consumables contribute more metal than the aerosol does unless materials are chosen against a metals blank - media validated for organic constituents are simply wrong here. The second constraint is mechanical: several otherwise-clean media impose enough airflow resistance that the device will not puff-activate, so the aerosol never forms. No quantitative inter-method comparison is offered, which is the paper's limitation - it maps the failure modes without ranking the survivors. The transferable point for distributed sensing is that sampling-substrate blank, not detector sensitivity, sets the floor for any trace-metal PM measurement, which is the same constraint that governs filter-based reference sampling behind a low-cost network. Note: this record was rejected from the 19 September issue for carrying no DOI; the DOI has since been registered.

doi: 10.1520/jte20250320

Urban OOM flux 2026 Atmos Chem Phys*Urban surface-atmosphere fluxes of selected pptv-level oxygenated organic molecules (OOMs) from eddy covariance observations.*

First urban eddy covariance measurement of OOM surface-atmosphere exchange at pptv level, and the contribution is as much methodological as observational - the paper works through how mass spectral integration time and water vapour variation corrupt flux retrieval at low signal-to-noise, which is the reason these fluxes had not been measured. Sixteen OOMs resolved into deposition-dominated, emission-dominated and bidirectional classes. Campaign-mean deposition was 4.6 $\mu\text{mol/m}^2/\text{d}$, 16.3% of HNO₃ deposition and 2.2x the OOM emission flux, so the urban surface is a net sink. Formic acid dominated both directions (66.2% of deposition, 84.9% of emission), which means the 16-species budget is really a formic acid budget plus minor terms. One site, one month, hottest season - the seasonal extreme most favourable to the photochemistry being measured, so these are upper bounds. Bears on sensing indirectly but concretely: SOA formed from these precursors is secondary mass a low-cost optical sensor sees only after it has condensed, with a lag and a hygroscopicity that no dry-mass calibration captures.

doi: 10.5194/acp-26-13213-2026

Wu et al. 2026 Atmosphere*Spatiotemporal Prediction of Sandstorm Utilizing Multi-Source Data and ConvLSTM Model: A Case Study in Northwest China*

Dust forecasting for photovoltaic operations in Gansu: surface PM₁₀ and Himawari-8/9 AOD are assimilated with an ensemble Kalman filter and fed to a multi-scale ConvLSTM with a high-value-focused loss. At 4 h lead the model reached RMSE 30.1 $\mu\text{g m}^{-3}$ and MAE 16.1 $\mu\text{g m}^{-3}$, captured the evolution of an April 2024 regional event, and ran in under 30 s per simulation. Validation rests on one event, errors are not stratified by concentration (where dust peaks run to thousands of $\mu\text{g m}^{-3}$), and no persistence baseline is reported, so skill over naive forecasting is unknown. Relevance: EnKF assimilation of surface PM is the natural entry point for low-cost sensor data into nowcasts - provided the sensors' known coarse-mode undercount under dust is corrected first.

doi: 10.3390/atmos17100925

Zhang et al. 2026 Build Environ*Are expensive sensors necessary for assessing indoor environmental quality? A data-driven approach using minimal sensing.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a data-driven test of whether a reduced sensor set can stand in for a full indoor-environmental-quality instrument suite. Whether PM is among the measured parameters is not stated; that decides relevance. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.buildenv.2026.115306

Exposure assessment & modelling

34 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (34 OF 34 RECORDS IN THIS SUBTOPIC)

Ahmad et al. 2026 Preprints.org (preprint)*Characterizing Ambient Fine Particulate Matter Pollution in Jordan: National Monitoring Evidence of Spatial Inequality, PM_{2.5} Hotspots, and Temporal Trends (2018-2024)*

Preprint analysing annual PM_{2.5} across Jordan's national monitoring network, 2018–2024, by site class. Industrial-related sites averaged 26.51 $\mu\text{g m}^{-3}$ versus 29.24 $\mu\text{g m}^{-3}$ at urban-related sites (Mann-Whitney U=297, p=0.604), and the Sen's slope was -1.67 $\mu\text{g m}^{-3}/\text{yr}$ with significant governorate-level differences. The authors themselves flag that the network expanded over the period, so the declining trend is confounded by a changing site mix - a fixed-site subset analysis is the obvious missing check. Annual means only; no instrument type or completeness criteria are stated. Not peer reviewed. Relevance: a network-expansion artefact masquerading as a trend is the same failure mode that growing low-cost networks face.

doi: 10.20944/preprints202609.1911.v1

Athens NPF 2026 Atmos Res

The influence of air mass origin and ammonia emissions on new particle formation at a Mediterranean urban site (Athens, Greece).

METADATA ONLY. Metadata-only record; Elsevier deposit carried no abstract and none was recoverable. The topic is squarely in scope - ammonia-mediated new particle formation is the ultrafine source that governs urban particle *number*, and number is precisely the metric that mass-calibrated low-cost sensors do not measure and that regulatory PM_{2.5} mass does not capture. Whether this paper quantifies nucleation rates, growth rates or only classifies event days cannot be determined from the deposit. Retry list.

doi: 10.1016/j.atmosres.2026.109334

Atik et al. 2026 Environ Sci Atmos

Inter-annual trends of PM_{2.5}-bound heavy metals in Dhaka, Bangladesh: a comparative study for source apportionment and health risk assessment

Only the graphical-abstract summary was deposited: elevated-altitude PM_{2.5} sampling in Dhaka found PM_{2.5}-bound heavy metals far above global guidelines, positive matrix factorisation resolved six emission sources, and Cr, Pb, Cd and Mn drove both non-cancer and cancer risk across all age groups. No concentrations, years, sample counts or factor identities are available from the deposit, so the inter-annual trend in the title cannot be evaluated, and deterministic risk quotients from filter metals typically assume total-metal bioavailability. Elevated-altitude sampling also under-represents street-level exposure. Relevance: Dhaka is a high-priority low-cost network setting where mass sensing needs exactly this kind of periodic compositional anchoring. Admitted on the summary; re-check when the full abstract resolves.

doi: 10.1039/d6ea00084c

Bhattarai et al. 2026 Research Square (preprint)

Spatio-temporal assessment of PM_{2.5} and PM₁₀ pollution in core and peripheral municipalities of the Kathmandu valley using regulatory and low-cost sensor data

Preprint integrating regulatory monitors with a dense low-cost sensor network across core and peripheral municipalities of the Kathmandu Valley, 2023-2024. Annual mean PM_{2.5} was 40.71 $\mu\text{g m}^{-3}$ in the urban core versus 27.22 $\mu\text{g m}^{-3}$ in the periphery; the core exceeded WHO guidelines on nearly all winter and pre-monsoon days, with a core amplification factor up to 1.60 in the dry season, contrasting size dominance, and evidence of long-range transport during extreme events. The abstract gives no sensor model, no calibration or co-location statistics and no sensor count, so the core-periphery gradient cannot yet be separated from sensor bias - critical in a valley with strong humidity and dust seasonality. Not peer reviewed. Relevance: exactly the hybrid-network use case, in a data-poor basin; the calibration section decides whether it is evidence.

doi: 10.21203/rs.3.rs-10981275/v1

Broda et al. 2026 Atmosphere

Vertical and Zonal Characteristics of Transatlantic Biomass Burning Aerosol Transport During the Extreme 2023 Canadian Wildfire Season.

May-September 2023 smoke transport from Canada to Europe against a 2003-2022 baseline. Source-region smoke AOD was the highest on record in every month except August. At source the 0-2 km layer held 78-93% of smoke over western Canada; during transport most smoke rose above 2 km, the 2-5 km layer dominated over the Atlantic and the extinction-weighted mean altitude climbed from 1-2 km to 2.5-4.3 km. Monthly transport efficiency to Europe was low, and satellite AOD did not reproduce the reanalysis zonal pattern. Monthly means mask episodes, and a reanalysis is not an observation. The surface-exposure implication is the useful part: when smoke is elevated, column AOD overstates surface PM_{2.5}, so AOD-to-surface products need vertical-profile correction, and surface sensors are the check.

doi: 10.3390/atmos17100913

Chen et al. 2026 Atmos Environ

Chemical Characteristics, Source Apportionment and Toxic Potential of PM_{2.5} in Southern Taiwan: Insights from the Kao-Ping Experiment (KPEX).

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Chen, Wu, Hsu, Hsu) and DOI on 23 Sep without an abstract. The Kao-Ping Experiment is a named campaign in southern Taiwan, and the title promises speciation, receptor-model source apportionment and a toxic-potential measure in one dataset - the combination that lets a source be ranked by harm rather than by mass. None of the instruments, campaign dates, factor solution or toxicity assay can be stated from the deposit. Carried because this is the same size-composition-source axis as the stroke record in this issue and would be the natural external check on it. Retry list.

doi: 10.1016/j.atmosenv.2026.122373

Chen et al. 2026 Atmos Res

Formation mechanism of rapid daytime nitrate production in a typical semi-arid city near dust source regions

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a formation-mechanism study of rapid daytime nitrate production in a semi-arid city near dust sources. Relevant in that nitrate is semi-volatile and dust-coated nitrate shifts particle size - both known optical-sensor bias terms. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosres.2026.109361

China aerosol vertical 2026 Atmos Pollut Res

Spatiotemporal Dynamics and Vertical Distribution of Aerosols in China: Insights from Multi-Source Evidence from 2000 to 2022.

METADATA ONLY. Metadata-only record: Elsevier deposited title, journal and DOI without an abstract, and no abstract was recoverable from Crossref, Europe PMC or PubMed. A 23-year national synthesis of aerosol vertical structure is potentially a tier-A input to satellite-to-surface PM_{2.5} reconstruction - vertical profile shape is the dominant structural uncertainty when converting AOD to surface concentration - but nothing about method, platform mix or findings can be stated from the deposit, and inferring a design from a title is how this watch would start manufacturing results. Carried as a placeholder and added to the retry list; if the abstract appears it should be re-screened on its merits rather than admitted on the strength of its title.

doi: 10.1016/j.apr.2026.103198

Garcia-Vara et al. 2026 J Hazard Mater

Expanding the chemical space of urban air: Identification and semiquantification of polar emerging contaminants in PM_{2.5} through LC-HRMS non-target analysis.

PM_{2.5} filters from eight Barcelona monitoring stations spanning traffic, urban background and port sites were extracted by pressurised liquid extraction and screened by data-dependent LC-HRMS. Forty contaminants of emerging concern were annotated at confidence level 2A - industrial chemicals, personal-care products, pesticides, pharmaceuticals and drugs of abuse - 21 reported in urban air for the first time, at pg/m³ to ng/m³; dicyclohexylamine, tributyl phosphate and DEET were highest. Semiquantification without authentic standards carries large uncertainty, level-2A identity is probable not confirmed, and the abstract gives no sampling duration or sample count, so site contrasts cannot be weighed. Relevance: optical sensors report mass only; this is the composition layer that decides whether equal mass means equal hazard.

doi: 10.1016/j.jhazmat.2026.143569

Gonzalez-Mendez et al. 2026 Environ Pollut

Morpho-chemical fingerprinting and distance-dependent attenuation of airborne Particulate Matter from abandoned mine tailings in Northwestern Mexico.

Three dry-season campaigns (2022-2024) placed Sigma-2 passive samplers at nine sites from abandoned tailings through farmland to a town. Automated SEM-EDX on 3,017 particles from oxidised and non-oxidised tailings and efflorescent salts built source fingerprints, classified by a three-step ML plus compositional-screening scheme. Mine-derived particle concentrations fell steeply with distance but were still detected about 1.1 km out; power laws explained 85.8% (PM_{10-2.5}) and 82.8% (PM₈₀₋₁) of the variance. Passive Sigma-2 deposition is converted to concentration through an assumed deposition velocity, nine sites constrain a decay curve loosely, and PM_{2.5} - the respirable fraction - is not the reported metric. Relevance: single-particle fingerprints are the ground truth a low-cost optical network near a point source cannot provide.

doi: [10.1016/j.envpol.2026.129233](https://doi.org/10.1016/j.envpol.2026.129233)

Guo et al. 2026 Toxics

Seasonal Formation, Sources, and Long-Term Evolution of PM_{2.5}-Bound Water-Soluble Inorganic Ions in Nanchang, China: The Increasing Importance of Wintertime Nitrate.

Day and night PM_{2.5} samples in summer and winter in Nanchang were analysed for nine water-soluble inorganic ions with source-apportionment, trajectory and fire-hotspot analyses, and set against 2009-2026 records. WSIs averaged 8.01 +/- 3.58 µg m⁻³ in summer and 23.97 +/- 9.42 in winter (57.0% and 50.3% of PM_{2.5}); SNA made up 82.0% and 76.4% of WSIs, sulfate leading in summer and nitrate in winter, with nitrate formation favoured at night. Since 2019 the control problem has moved from sulfate to wintertime nitrate. Two seasonal campaigns at one site; campaign length is not in the abstract. Relevance: a rising ammonium-nitrate fraction is the composition shift that most inflates humidity-driven optical-sensor overestimation - calibrations fitted in the sulfate era will drift.

doi: [10.3390/toxics14090828](https://doi.org/10.3390/toxics14090828)

Hien et al. 2026 Atmos Environ

Long-term Observations of Fine Particulate Sulfate in Hanoi during 1999-2023: Impact of SO₂ Regional Emissions and Monsoon Circulations.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Hien, Vuong, Bac, Ha) and DOI on 23 Sep without an abstract; no leg recovered one. A 24-year fine-sulfate record from Hanoi set against regional SO₂ emissions and monsoon circulation would be one of the longest speciated series in Southeast Asia, and the title asserts both an emissions and a meteorological attribution. Sampling method, filter protocol, completeness and the size of any trend are all unavailable, and the title is not evidence of what the paper measured. Carried as a placeholder because long speciated records are the only way to validate the secondary fraction that low-cost optical sensors systematically misread. On the retry list.

doi: [10.1016/j.atmosenv.2026.122372](https://doi.org/10.1016/j.atmosenv.2026.122372)

Hu et al. 2026b Environ Pollut

Long-term spatiotemporal evolution, relative hotspot persistence, and predictive attribution of PM_{2.5} across China.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Hu, Shi, Zhang) and DOI on 23 Sep with no abstract recoverable from any leg. The title claims three things - long-term spatiotemporal evolution, persistence of relative hotspots, and predictive attribution of PM_{2.5} across China - of which the middle is the interesting one, since relative hotspot persistence under a falling national mean is what decides whether monitoring networks can be thinned. Dataset, period, grid resolution, attribution method and results are all unavailable. Listed on metadata; re-screen when the abstract is deposited rather than reading the title as a finding.

doi: [10.1016/j.envpol.2026.129192](https://doi.org/10.1016/j.envpol.2026.129192)

Hurynovich et al. 2026 Atmos Environ

Transferability of dust-related PM_{2.5} models under contrasting arid environmental regimes.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a test of whether dust-PM_{2.5} models fitted in one arid regime hold in another. Transferability is the same failure mode that limits sensor calibration models; worth recovering. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: [10.1016/j.atmosenv.2026.122386](https://doi.org/10.1016/j.atmosenv.2026.122386)

Ireland Census linkage 2026 Health Place

Ambient air pollution at home, work and school: Evidence from Ireland.

Whole-population census microdata linked to Irish EPA pollutant surfaces at two addresses per person. Mean annual PM_{2.5}, PM₁₀ and - most sharply - NO₂ are higher at the work, school or college address than at the residence, while O₃ runs the other way, the expected NOx-titration signature of the same urban gradient. The socioeconomic result inverts the usual assumption: among workers the residential-to-workplace contrast is *larger* for more advantaged groups, because advantaged workers commute into dense urban cores from cleaner suburbs. Residence-only assignment therefore does not simply add noise, it adds noise correlated with socioeconomic position, which is the one thing an exposure proxy must not do in a study adjusting for it. Limits: both endpoints are modelled ambient surfaces at an address, so nothing here captures infiltration, indoor sources or time-activity weighting, and the paper measures the size of a bias without estimating its effect on any health coefficient. For network design this is the strongest argument yet that siting density should follow daytime population rather than residential population.

doi: [10.1016/j.healthplace.2026.103732](https://doi.org/10.1016/j.healthplace.2026.103732)

Kagawa et al. 2026 Atmos Environ

Seasonal variations of aerosol selenium concentrations and oxidation states in central Japan.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Kagawa, Nagao, Kagami, Kida) and DOI on 23 Sep with no abstract and none recoverable elsewhere. Selenium oxidation state is a coal-combustion and long-range-transport tracer, and a seasonal speciation record from central Japan would bear on transboundary attribution in the East Asian outflow. The deposit says nothing about site, sampling duration, analytical method, detection limits or the size of the seasonal contrast, so nothing quantitative can be reported and the subtopic assignment rests on the title alone. Listed rather than summarised, on the retry list, and to be re-screened on its merits when an abstract appears.

doi: [10.1016/j.atmosenv.2026.122364](https://doi.org/10.1016/j.atmosenv.2026.122364)

Kaur et al. 2025 Atmos Pollut Res*Long-term PM_{2.5} source apportionment at 12 sites in the western United States from 2000-2020.*

Twelve monitoring sites, twenty years, five to eight resolved factors per site. PM_{2.5} fell at ten of twelve sites (-0.38 to -0.037 $\mu\text{g m}^{-3}$ per year; no decline at Boise or Commerce City), and the decline is carried by secondary ammonium nitrate, ammonium sulfate and the two EC-rich factors - exactly the sources NO_x, SO₂ and mobile-source rules targeted. EC-rich fell despite rising vehicle miles travelled, isolating per-vehicle emission control rather than reduced activity. Winter OC-rich, the biomass-burning proxy, showed the largest seasonal decline. Dust rose at most sites. The weakness is the one the authors set out to address and only partly do: CSN speciation methods changed within the record, so a factor's 20-year trajectory is partly an instrument trajectory, and reconciling PMF with CMB constrains but does not eliminate that. For distributed sensing the finding that matters is compositional: the shrinking part of western PM_{2.5} is the secondary and combustion mass an optical sensor sees reasonably well, while the growing part is dust, which nephelometric low-cost sensors under-report because coarse mineral particles scatter poorly at their sampling cut. Network bias against dust therefore worsens over time.

doi: 10.1016/j.atmosenv.2025.102779

Liu et al. 2026 Life (Basel)*Airborne Bacterial and Eukaryotic Communities in Temuco, Chile, a City Dominated by Residential Wood Combustion.*

First characterisation of airborne eukaryotic communities alongside bacteria in Temuco, a city whose PM is dominated by residential wood burning. Mean PM_{2.5} and PM₁₀ were 75.0 and 87.7 $\mu\text{g m}^{-3}$, PM_{2.5} exceeding the Chilean 50 $\mu\text{g m}^{-3}$ standard; nitrate (5.6), sulfate (2.6) and K⁺ (2.6 $\mu\text{g m}^{-3}$) were the main ions. Clostridium, Hymenobacter and Sphingomonas led the bacteria; Corioliopsis (41.0%) and Hyphodontia (26.9%) the eukaryotes. Communities varied more by season than size fraction, and RDA loaded SO₄ (0.71) among the drivers. Sample numbers and averaging periods are not in the abstract, and the ion-community correlations are descriptive. Relevance: a PM_{2.5}/PM₁₀ ratio of ~0.86 confirms a fine-mode smoke aerosol - the regime where optical sensors track reference best if humidity is corrected.

doi: 10.3390/life16091433

Liu et al. 2026a Environ Health (Wash)*Effect of Skin Types and Microenvironment of Dermal Exposure to Polycyclic Aromatic Hydrocarbon Derivatives and Heavy Metals.*

Polyethylene balloons coated with graded doses of glycerol trioleate stood in for oily, neutral and dry skin, then were deployed across four microenvironments to measure adsorption of nitrated and oxygenated PAHs and 11 metals. Adsorption followed first-order kinetics and ranked oily > neutral > dry; it was highest in dusty weather and lowest indoors on campus, with NPAHs exceeding OPAHs in campus air, and metal uptake ordered V > Zn > As > Ti, with Cu detected only during sand-blowing. A coated balloon is not skin - no sebum turnover, no perspiration, no penetration - so these are surrogate deposition rates, not doses. Useful anyway: it quantifies a route that PM mass monitoring ignores entirely, and dust episodes dominate it.

doi: 10.1021/envhealth.5c00843

Mainka & Mainka 2026 Toxics*Modelled Physical Activity Benefits and PM_{2.5} Risks in Two Urban Cyclists.*

Re-analysis of 162 previously published cycling trips by two commuters in Gliwice, each treated as a hypothetical five-day-a-week habit and compared with resting for the same time at the concurrent station concentration, using assumed ventilation and published dose-response functions. Median net RR was 0.840 (16.0% relative risk reduction; 95% UI 0.792-0.892 when the epidemiological coefficients were varied); the median break-even concentration was 196.7 $\mu\text{g m}^{-3}$, and a +/-10% personal-concentration error flipped the sign of the dose increment on 12 trips without pushing net RR above 1. Everything is modelled, n=2, and the personal PM_{2.5} came from uncalibrated optical monitors, as the authors state. Relevance: a worked example of how sensor error propagates into risk-benefit claims - here it does not change the conclusion.

doi: 10.3390/toxics14090836

Ningombam et al. 2026 Atmos Environ*Assessment of global aerosol burden over land and oceans using ground-based, satellite, and reanalysis data*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 26 Sep with no abstract; Crossref, PubMed fetch, Europe PMC and the publisher landing page returned nothing. The title indicates a global land/ocean aerosol-burden comparison across ground-based photometry, satellite retrievals and reanalysis. Retry priority: the reanalysis-versus-photometer bias by region is what decides whether reanalysis AOD is a usable prior for satellite-to-surface PM_{2.5} fusion. Nothing about data, method, sample size or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122370

Qin et al. 2026 Microorganisms*Seasonal Variation of Airborne Bacteria and Potential Pathogens Adhere to Fine and Coarse Particles in a Coastal City.*

Size-resolved airborne bacterial communities on PM_{2.5} and PM₁₀ in Qingdao were profiled by 16S sequencing across seasons. Staphylococcus, Lactobacillus, Escherichia-Shigella and Klebsiella dominated everywhere; Staphylococcus was enriched in autumn (LefSe LDA > 3.5). Summer and autumn communities diverged, but particle-size groups did not differ significantly; Chao and Shannon indices were higher on PM₁₀ than PM_{2.5}, significant only in autumn. The abstract gives no sample count or sampling duration, pathogen potential is predicted from 16S taxonomy (not viability or virulence), and no PM mass is reported. Relevance: marginal; optical sensors cannot distinguish biological particles, so this fraction sits unresolved inside every coastal PM_{2.5} reading.

doi: 10.3390/microorganisms14091924

Rana & Guzman 2026 ACS Earth Space Chem*Heterogeneous Nitrate Radical Oxidation of Catechol Produces Nitrocatechols, Oligomers, and Brown Carbon.*

Flagged backfill: Crossref created this DOI on 20 Aug; it reached this window via a 22 Sep Europe PMC deposit and had not been carried or rejected. Catechol thin films oxidised by NO₃ under dry conditions produced nitroaromatic IR bands near 1590 and 1350 cm⁻¹, a broad UV-vis absorption centred near 349 nm characteristic of brown carbon, and 4-nitrocatechol, dinitrocatechols, dimers and trimers; the proposed route runs through semiquinone radicals, then nitration and radical coupling. Thin films under dry conditions are far from real humid particles, and no yields or mass absorption efficiencies are given, so atmospheric significance is asserted rather than shown. Paired with the PAX-ISS record in this issue: nighttime BrC formation is one source of the seasonal BrC term that record apportionments.

doi: 10.1021/acsearthspacechem.6c00226

Seibert et al. 2026 Environ Pollut

National-scale NMF source apportionment reveals component-specific west-to-east asymmetries in PM_{2.5} concentrations associated with transboundary transport.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates non-negative matrix factorisation of national PM_{2.5} data resolving transboundary components with a west-east gradient. Country, input species and factor count are not in the deposit. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.envpol.2026.129243

Shen et al. 2026 Atmos Pollut Res

A Hybrid Deep Learning-Geostatistical Framework for PM_{2.5} Spatial Estimation with SHAP Interpretation

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a hybrid deep-learning and geostatistical PM_{2.5} spatial estimator with SHAP interpretation. Squarely in the exposure-modelling scope; the validation design (spatial vs random CV) is what would decide its value. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.apr.2026.103208

Sonipat ACSM campaign 2026 Atmos Chem Phys

Chemical characterization and source apportionment of carbonaceous aerosols during post-monsoon biomass burning and Diwali at an upwind site of Delhi.

25 October to 15 November 2023, spanning two severe haze episodes and Diwali. Composition-based PM_{2.5} exceeded 300 $\mu\text{g m}^{-3}$ in both episodes; organic aerosol was 65% of non-refractory PM_{2.5} with a daily mean near 140 $\mu\text{g m}^{-3}$, and oxygenated OA was 57-60% of total OA, reaching 42.8 $\mu\text{g m}^{-3}$ of more-oxidised OOA at peak haze - so the haze is substantially aged secondary mass, not fresh plume. Biomass- or solid-fuel eBC was ~78% of campaign-integrated eBC. The authors are honest that this split is sensitive to the assumed absorption Angstrom exponent pair, which is the paper's real limitation: the headline 78% is a function of a prior, and the sensitivity range is not propagated into the PMF factors. Single site, single season, three weeks. The result that matters upstream of a sensor network is that the mass is already loaded before the air reaches Delhi, so an urban-only network measures an urban increment on a large regional background and will misattribute regional transport to local sources unless it has an upwind anchor.

doi: 10.5194/acp-26-13227-2026

W. India BC-dust 2026 Environ Sci Atmos

Seasonal Variability and Interactions of Black Carbon and Mineral Dust over Semi-Arid Western India: Satellite and Ground-Based Observations.

Only the opening of the abstract was deposited, so the design is legible but the numbers are not: the study characterises seasonal co-variation of anthropogenic black carbon and natural mineral dust in a semi-arid region where both are large, using satellite retrievals alongside ground-based observations. No effect sizes, retrieval-validation statistics or station counts are recoverable from the deposit, and nothing is claimed here beyond scope. Flagged for retry rather than summarised further. The reason to carry it at all is the mixing problem it addresses: BC and dust are the two components that most degrade low-cost optical PM sensing in arid regions, BC because it absorbs rather than scatters and dust because its coarse mode falls outside a nephelometric sensor's effective cut, and a seasonal climatology of their interaction is the calibration prior a regional network would need.

doi: 10.1039/d6ea00109b

Yang et al. 2026c Atmos Chem Phys

Complexation strength between organic carbon and transition metal ions dominates the photochemical conversion of SO₂ to sulfates.

Photooxidation of SO₂ on coal-combustion organic carbon with and without Fe³⁺, Cu²⁺ and Mn²⁺. Steady-state SO₂ uptake coefficients spanned (0.39-22.24)e-6 and sulfate masses (2.36-26.05)e-3 mg. Fe³⁺ was antagonistic while Cu²⁺ and Mn²⁺ enhanced both uptake and sulfate yield; absorbance loss and fluorescence quenching confirmed metal-chromophore complexation, strongest for Fe³⁺, and sulfate mass rose linearly with hydroxyl-radical intensity. The Fe³⁺ result inverts the usual expectation that iron catalyses sulfate formation, and the proposed reason - iron binds the chromophores that generate the radicals - is testable. Films under laboratory irradiation are not ambient particles and no atmospherically scaled yields are offered. Secondary sulfate is the fraction optical sensors mis-size worst under humidity, which is why formation pathways matter here.

doi: 10.5194/acp-26-13387-2026

Youn et al. 2026 Atmos Pollut Res

Seasonal distributions and source identification of water-soluble arsenic species of PM_{2.5} in the urban atmosphere of Northeast Asia

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states seasonal distributions and source identification of water-soluble arsenic species in urban PM_{2.5}. Speciation matters because As(III)/As(V) toxicity differs; low relevance to sensing. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.apr.2026.103207

Zahid & Morris 2026 Preprints.org (preprint)

Demand or Congestion? Disentangling Event-Related Emissions Using a Directional Control on I-15.

One-minute freeway detector data from the LATTICE platform during the 2026 Electric Daisy Carnival. Northbound (arrival) lost 9.5 mph over a five-hour window; southbound lost nothing, yet CO₂ rose almost equally in both directions (+11.6% vs +11.8%), so the headline increase is weekend demand, not congestion. Using the uncongested direction as control in a difference-in-differences, only PM_{2.5} kept a substantial congestion residual (+8.9 percentage points), and residuals ranked by low-speed emission sensitivity (Spearman rho 1.00, n = 4). Emissions are modelled from speed and volume, not measured, so the PM_{2.5} result inherits the emission model's low-speed factors; n = 4 pollutants makes rho uninformative. Preprint. The control-corridor logic transfers directly to near-road sensor pairs.

doi: 10.20944/preprints202609.1747.v1

Zhang 2026 medRxiv (preprint)*Within-Year PM_{2.5} Exposure Structure Provides Mortality-Predictive Information Beyond the Annual Mean.*

Preprint. Daily modelled PM_{2.5} was linked to age-adjusted mortality for 27,289 county-years across 3,064 US counties, 2003-2011. PMRI is the largest k for which at least k days reached $k \mu\text{g m}^{-3}$ - a scale-free summary of how a year's concentrations are distributed across days. Under county-grouped 10-fold cross-validation, adding PMRI to the annual mean raised held-out R² from 0.0802 to 0.1391 (delta R² 0.0589, 95% CI 0.0499-0.0677) and reduced RMSE and MAE, holding across geographic, temporal and all six cause-specific splits. It is ecological, unadjusted for county composition, and an R² of 0.14 remains weak prediction. The relevance is direct: daily-resolution networks carry mortality-relevant structure that annual means discard.

doi: 10.64898/2026.09.20.26363510

Zhao et al. 2026 Atmosphere (Basel)*Spatial-Temporal Variability of PM_{2.5} and Ozone Pollution in the Huaihai Economic Zone of China.*

PM_{2.5} and O₃ concentrations (2000-2023) were set against MEIC inventory emissions (2000-2020) for the Huaihai Economic Zone. PM_{2.5} emissions and concentrations fell after about 2013 while O₃ rose. The raw monthly correlation between PM_{2.5} emissions and O₃ was strongly negative ($r = -0.773$) but collapsed to $r = 0.137$ after adjustment for meteorology, season and trend - the headline is that the unadjusted relationship is mostly shared seasonality. The abstract does not say whether pre-2013 PM_{2.5} is measured or reanalysis (China's national network began in 2013), and correlation on a regional inventory cannot identify chemical regime. Relevance: low; a caution that emission-concentration correlations without meteorological adjustment mislead.

doi: 10.3390/atmos17100929

Zhou et al. 2026 GeoHealth*Can Electric Vehicles Reduce Summertime Air Pollution Exposure Inequalities in Metro Atlanta? A High-Resolution Analysis*

1-km CMAQ simulations of 30% light- and heavy-duty EV adoption in metro Atlanta for summer. Domain-wide means fell 0.22 ppb NO₂, 0.94 ppb MDA8 O₃ and 0.06 $\mu\text{g m}^{-3}$ PM_{2.5}, with local NO₂ and PM_{2.5} increases near power plants; the largest reductions fell in communities with higher Asian, Hispanic and Black shares, though disparities persisted. The PM_{2.5} change is well inside CMAQ's typical error and a single season. The most transferable result is methodological: 12-km runs underestimated minority exposure by up to 25% for NO₂ and 10% for PM_{2.5} relative to 1 km. Relevance: that resolution bias is a quantitative argument for dense sensor networks in environmental-justice assessment, where coarse grids systematically understate the disparity.

doi: 10.1029/2025gh001654

da Silva et al. 2026 Toxics*Chemical Regimes and PMF-Resolved Sources Linked to Acute PM₁₀ Ecotoxicity in Luanda, Angola.*

116 daily PM₁₀ samples from Luanda (27 Jun-5 Nov 2023) were speciated, clustered by centred log-ratio composition and apportioned by PMF, with aqueous extracts tested in the Microtox Aliivibrio fischeri assay. TU15 ranged 0.84-8.54; only 3 samples were below 1. A metal-enriched regime (Zn, Pb, Cd) had the highest mean TU15 (4.22), matching a PMF non-ferrous metallurgy / scrap-burning factor that correlated most with toxicity ($\rho = 0.63$ for TU15, 0.78 for dTU), then a Ni-rich metallurgical factor ($\rho = 0.51$); marine aerosol was inversely associated. Bacterial bioluminescence of water extracts is not a human-relevant endpoint and misses insoluble fractions; correlations are univariate. Relevance: one of very few speciated PM₁₀ records from sub-Saharan Africa - baseline for any sensor network sited there.

doi: 10.3390/toxics14090773

Cardiovascular & metabolic**15 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (15 OF 15 RECORDS IN THIS SUBTOPIC)

Aysin et al. 2026 Metabolites*Effects of Traffic-Related Diesel Exhaust Exposure and Moderate Exercise on Obesity, Metabolic Dysfunction, and Inflammatory Responses in Rats.*

48 female Wistar rats randomised to six groups: diesel exhaust at a mean $300 \mu\text{g m}^{-3}$ PM_{2.5} for 2 or 4 h/day over eight weeks, with or without 30 min/day moderate treadmill exercise. Exhaust raised body weight, BMI, Lee index, visceral adiposity index, LDL, triglycerides, glucose, leptin, CRP, IL-1b, IL-6 and TNF- α and lowered HDL and insulin (all $p < 0.001$), most strongly at 4 h/day; exercise attenuated but did not abolish these changes. Only p-values are reported - no effect sizes - and whole exhaust cannot separate particles from NO₂ and CO. $300 \mu\text{g m}^{-3}$ is a high-traffic peak, not a typical mean. Relevance: the exposure-by-activity interaction argues for pairing wearable PM with accelerometry, since ventilation rate rises with exercise.

doi: 10.3390/metabo16090682

Bikdeli et al. 2026 Eur Heart J Cardiovasc Pharmacother*Coated or chewable Aspirin and a hybrid air pollution mitigation strategy in patients with atherosclerotic disease: Rationale and Design of the COATED-AIR Randomized Trial.*

Design paper for a 2x2 factorial randomised trial in patients with established ASCVD on low-dose aspirin: double-blind enteric-coated versus chewable aspirin 81 mg, and a hybrid air-pollution mitigation package versus usual care. The package combines an educational flashcard, SMS air-quality alerts, and on high-pollution days advice to limit outdoor activity or wear a KN95 mask and to raise citrus intake. No results. This is among the first outcome-powered trials of individual-level pollution mitigation, which is why it is tier B despite having no data. The weakness is built in: a bundled behavioural intervention cannot attribute any benefit to exposure reduction unless adherence and personal exposure are measured, which the abstract does not describe. The alert signal presumably comes from a monitoring network, so alert latency and spatial resolution are unreported design parameters.

doi: 10.1093/ehjcvp/pvag071

Campagna et al. 2026 Front Public Health*Tracking biochemical indicators of air pollution exposure in acute cardiorespiratory emergencies: a scoping review.*

A PCC-framework scoping review (PubMed, Scopus, Web of Science, Embase, CINAHL; 2000-2025) of studies measuring biochemical markers in emergency-department presentations after short-term outdoor air pollution. Twelve studies qualified; PM_{2.5}, PM₁₀ and NO₂ were most often linked to CRP, IL-6, D-dimer and troponins within hours to days. Twelve heterogeneous studies with no quality appraisal or pooling cannot support any causal or quantitative claim, and ED biomarkers are dominated by the presenting illness. Relevance: low; the stated need for standardised exposure windows is where hourly sensor data would add over daily station means.

doi: 10.3389/fpubh.2026.1892038

Campailla et al. 2026 Int J Cardiol Cardiovasc Risk Prev

Ambient particulate matter exposure and acute type A aortic dissection: A retrospective time-stratified case-crossover study in northern Italy.

Time-stratified case-crossover of 40 Stanford type A acute aortic dissections at a single northern-Italian centre, 2018-2022, referent days matched on weekday within month and year, temperature-adjusted. A doubling of same-day PM_{2.5} gave adjusted OR 4.78 (95% CI 1.61-14.29); 30-day PM_{2.5} and both PM₁₀ windows pointed the same way. Forty cases is the whole story - the interval spans nearly an order of magnitude, the authors label the estimates exploratory, and a single urban-background monitor stands in for everyone in the catchment. Reported here because it is a rare European estimate for a high-lethality outcome, not because the magnitude is credible. This is exactly the low-count, high-misclassification setting where a dense network changes the variance more than the point estimate.

doi: 10.1016/j.ijcrp.2026.200708

Cui et al. 2026 Environ Sci Technol

Differential Risks of Incident Heart Failure Subtypes Associated with PM_{2.5} Chemical Constituents in China between 2006 and 2021.

Kailuan prospective cohort of 101,521 adults followed a median 14.83 years (2006-2021) for incident heart failure by subtype. Every PM_{2.5} constituent was associated with higher risk across subtypes, more strongly for early- than late-onset HF. In mixture weighting, BC contributed 40.2-48.7% of the effect for overall, late-onset, HFrEF and HFmrEF; NH₄⁺ (28.2%) and BC (28.1%) led early-onset HF; OM (34.8%) led HFpEF. The abstract reports no per-constituent HRs, so magnitudes cannot be entered in the forest plot. Kailuan is a single-city cohort, so contrast comes from within-city variation, and constituents are highly collinear; mixture weights are unstable under that. Relevance: BC is measurable by low-cost aethalometers, so a BC-led HF signal favours adding absorption to network nodes.

doi: 10.1021/acs.est.6c02927

Dzhambov et al. 2026 Arh Hig Rada Toksikol

Residential exposures associated with adiposity and proinflammatory markers in obese adults with metabolic syndrome: a cross-sectional study

90 obese adults with metabolic syndrome in Bulgaria (2023-2024) had serum IL-6, TNF- α , CRP and adipokines, and adiposity measures, linked to residential NO₂, PM_{2.5}, PM₁₀, Lden noise, domestic solid-fuel burning and greenness. Per 10 $\mu\text{g m}^{-3}$ NO₂, IL-6 was 41.63% higher (95% CI 8.16-85.46), CRP 37.80% (5.89-79.32) and visceral fat 7.52% (1.30-14.12); time in nature lowered TNF- α and raised adiponectin. The PM_{2.5} and PM₁₀ associations were 'largely null, inconsistent, or occasionally protective' and are not quantified. n=90 with many exposures and outcomes and no multiplicity control; cross-sectional. Relevance: a PM null in a small, narrow-contrast sample - uninformative either way.

doi: 10.2478/aiht-2026-77-4124

Hua et al. 2026 Environ Health (Wash)

Heterogeneous Risks of Chemical Components and Sources in Size-Resolved Particulate Matter on Ischemic and Hemorrhagic Stroke.

Time-series analysis of 58,763 ischaemic and 2,588 haemorrhagic stroke hospitalisations against 41 PM components and six source factors resolved across nine size fractions below 10 μm - the finest joint size-composition-source contrast this watch has carried. Ischaemic stroke tracked components below 1.1 μm and above 3.3 μm ; haemorrhagic stroke tracked As, Cu, Fe, Mn, Ni, Pb and PAHs. Coal combustion gave RR 1.05 (95% CI 1.01-1.10) per IQR for ischaemic stroke and industrial emissions RR 1.18 (1.00-1.41) for haemorrhagic, whose lower bound sits on the null with 2,588 events. Source factors from receptor modelling are collinear by construction, so component-specific attribution is softer than the size resolution implies. Size-resolved sensing, not PM_{2.5} mass, is what this design needs at scale.

doi: 10.1021/envhealth.6c00019

Jeong et al. 2026 Environ Int

Joint effects of air pollution and temperature on glycemia: An exposome analysis in European cohorts.

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors (Jeong, Edstorp, Lovison, Pickford and colleagues) and DOI on 23 Sep with no abstract, and Crossref, Europe PMC and PubMed all returned nothing. The title states a joint air-pollution-temperature exposome analysis of glycaemia across European cohorts, which is a design this watch tracks - temperature is the covariate most likely to be doing the work in short-term PM cardiometabolic associations. Nothing about the pollutants, exposure model, cohorts, sample size or estimates can be stated from the deposit, and inferring a method from a title is not something this watch does. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.envint.2026.110538

Lin et al. 2026 Research Square (preprint)

Joint association of PM_{2.5} and AIP with advanced Cardiovascular-Kidney-Metabolic syndrome in middle-aged and older Chinese adults: a national cohort study.

CHARLS 2011-2015 participants classified by median PM_{2.5} and median atherogenic index of plasma, logistic regression for advanced (stage 3-4) cardiovascular-kidney-metabolic syndrome. High AIP was strongly associated; PM_{2.5} alone showed no positive association; joint high-PM_{2.5} and high-AIP exposure raised risk, with no additive or multiplicative interaction. Effect sizes and n are not in the abstract. The analysis is a median split of both exposures, so the 'joint' effect is almost entirely the AIP effect restated, and absence of interaction contradicts the framing. Preprint, not peer-reviewed. Carried for the null PM_{2.5} main effect, which is informative in a cohort with a wide exposure range, and as a caution against dichotomising a continuous exposure that sensor and satellite products now resolve finely.

doi: 10.21203/rs.3.rs-10585989/v1

Pellegrino et al. 2026 Sci Total Environ

A possible mediating role of SASP in the association between environmental air pollution and incident atherosclerotic cardiovascular disease

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states a mediation analysis of senescence-associated secretory phenotype markers between air pollution and incident atherosclerotic cardiovascular disease. Mechanistically interesting - ageing biology as a pathway - if the exposure is resolved below county level. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.scitotenv.2026.182315

Sun et al. 2026 Front Public Health

When air gets heavy: joint PM(2.5) and ambient humidity exposure, frailty, and stroke.

CHARLS (n=17,197) plus a single-centre hospital sample (n=782), with annual PM_{2.5} and relative humidity dichotomised at database medians. High PM_{2.5} gave HR 1.184 (95% CI 1.083-1.295) for incident stroke in CHARLS and OR 1.846 (1.247-2.733) in the hospital sample; high humidity HR 1.112; joint high-high exposure HR 1.361 (1.171-1.582). Frailty mediated 18.1% of the PM_{2.5} association. Median dichotomisation discards the exposure-response shape, the hospital arm is cross-sectional, and humidity at annual resolution is largely a climate-zone proxy confounded with region. Relevance: humidity is also the dominant bias term in optical PM sensors - an epidemiologic humidity effect estimated from sensor-derived PM would be partly an artefact, so correction must precede the health model.

doi: 10.3389/fpubh.2026.1883147

Wang et al. 2026 Front Public Health

Long-term exposure to fine particulate matter constituents and incident stroke risk among middle-aged and older adults in China: a cohort study.

CHARLS 2013-2018: 8,724 adults, 610 incident strokes over a median 5.0 years, with PM_{2.5} and five constituents from the TAP reanalysis. Per IQR, HRs were 1.55 (95% CI 1.35-1.79) for PM_{2.5}, 1.55 for sulfate, 1.37 for nitrate, 1.42 for ammonium, 1.59 for organic matter and 1.77 (1.54-2.03) for black carbon; quantile g-computation gave BC the largest mixture weight (28.6%). Susceptibility was higher above 65 and in southern China. Stroke is self-reported physician diagnosis, constituents are strongly collinear model outputs, and IQRs differ by constituent, so per-IQR HRs are not directly comparable. Still the clearest signal this week that carbonaceous particles carry the cerebrovascular risk. Relevance: argues for adding absorption (BC) channels to low-cost nodes, not just scattering mass.

doi: 10.3389/fpubh.2026.1909901

Wang et al. 2026 Bioengineering (Basel)

Long-Term Air Pollution Exposure, Cardiovascular Disease, and Cardiometabolic Risk Factors in Middle-Aged and Older Chinese Adults: A Prospective Cohort Study.

CHARLS adults aged ≥45: cross-sectional analysis of 7,420 at 2011 baseline and a prospective analysis of 8,792 followed to 2020, with CHAP-derived exposures and a GBD burden component. Incident CVD HRs were 1.09 (95% CI 1.03-1.16) for PM_{2.5}, 1.07 (1.04-1.10) for PM₁₀, 1.12 (1.00-1.24) for NO₂ and 1.08 (0.93-1.27) for O₃; exposure was also associated with hypertension, diabetes, dyslipidaemia, obesity and TyG indices. The abstract does not state the exposure increment, CVD is self-reported physician diagnosis in CHARLS, and a random forest with AUC 0.656 adds nothing. This is at least the fourth CHARLS PM-CVD paper in this watch in two months, so overlap of sample and outcome definition should be assumed. Relevance: low; single-pollutant model on 1-km reanalysis.

doi: 10.3390/bioengineering13090958

Yang et al. 2026 Environ Res

Mitochondrial DNA methylation as a bridge: Linking indoor air purification to alleviated cardiovascular impairments in young adults.

A randomised double-blind crossover of true versus sham air purification in 38 healthy young adults. Purification cut indoor PM_{2.5} by 47.45%. Each 10 µg m⁻³ rise in personal PM_{2.5} at lag 24-48 h was associated with mtATP6 hypomethylation, and mtDNA hypomethylation mediated 14.87% (p=0.04) of the PM_{2.5} association with D-dimer at lag 0 h, among 16 cardiovascular indicators tested. The mediation analyses run across many CpG sites, lags and outcomes with no stated multiplicity correction, and the headline p=0.04 would not survive one. Short crossover periods in young adults cannot show clinical benefit. Relevance: personal PM_{2.5} was the analysed exposure, so the intervention's value rests on personal-monitor accuracy - the device and its calibration are not reported in the abstract.

doi: 10.1016/j.envres.2026.125744

Zhang et al. 2026 Environ Health (Wash)

Air Pollution Exposure Induces Vascular Endothelial Cell Dysfunction via Elevated Circulating F2: Multiomics Insights from an Ex Vivo Biosensor Assay.

Travel-panel design: eight college students moved from Qingdao to the substantially more polluted Shijiazhuang, with serial serum sampling, proteomics, metabolomics and an ex vivo endothelial biosensor. Prothrombin (F2) emerged as the hub node correlated with exposure and was nominated again by weighted gene coexpression analysis; metabolomics showed bile-acid and arginine-proline perturbation, and endothelial transcriptomes showed activation. n = 8 with no counter-travel arm is the ceiling on all of it - city is confounded with diet, water, sleep and stress, and the omics search space dwarfs the sample. Hypothesis-generating only. The design itself is cheap to strengthen: a personal-sensor-instrumented reciprocal travel panel would give an exposure axis instead of a city label.

doi: 10.1021/envhealth.5c00833

Neuro / mental health**8 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Curzio et al. 2026 Toxics

The Exposome-Brain Axis: A Scoping Review of Biological Biomarkers in Environmental Neurotoxicity, Neuroinflammation, and Neurodegeneration.

PRISMA-ScR review (PubMed, Scopus, WoS, Embase, no date limit) of 52 human observational studies linking environmental exposures to objective neurobiological markers. PM_{2.5}, metals, agrochemicals and POPs were associated with changes in NfL, GFAP, sTREM2, tau and amyloid-beta, with vulnerability windows from early-life DNA methylation onward. Scoping design: no appraisal, no pooling, and PM is one exposure class among many, so the PM-specific evidence base is not separable from the abstract. Relevance: low; flags plasma NfL/GFAP as candidate short-latency outcomes for sensor-exposure panels.

doi: 10.3390/toxics14090817

Falkenstein et al. 2026 BMC Neurosci

Investigating effects of day-to-day variations in environmental exposure on the human brain: study protocol for the Day2Day Environment project.

Protocol and dataset description for a dense longitudinal study: 30 participants, 25 testing sessions each, with wearable devices, GPS-based geospatial tracking and geographic ecological momentary assessment capturing air quality, noise, light, activity, stress, cognition and affect, paired with repeated MRI (T1/T2*, quantitative multi-parameter mapping, hippocampal proton density, DTI). No results. The abstract does not name the air-quality sensor, its calibration or which pollutants are logged, which is the component this watch would need to judge. n=30 is small but the within-person repeated design is where short-term neuroplasticity signals are detectable. Relevance: one of few designs coupling personal low-cost PM tracks to imaging; worth following when the sensor methods paper appears.

doi: 10.1186/s12868-026-01052-z

Hong Kong momentary stress 2026 Environ Sci Technol

An Explainable Multi-Modal Deep Learning Framework for Investigating the Influences of Dynamic Environmental Exposure on Momentary Stress.

4,342 momentary stress reports against 125,440 minute-level environmental exposure records; the model reaches 78.26% accuracy and 0.828 AUC-ROC. Two results are worth extracting from the architecture. First, exposure in the preceding 25 minutes dominates prediction, which is a direct empirical argument for minute-resolution mobile monitoring over daily or residential-annual assignment for acute endpoints. Second, the explainability analysis recovers thresholds rather than slopes - 25 µg m⁻³ for PM_{2.5}, 55 dBA noise, 32 °C, 40-70% humidity, 25% greenery visibility - consistent with nonlinearity that a linear exposure-response term would average away. The weaknesses are structural and serious: stress is self-reported into the same phone that carries the sensors, so reporting and exposure share a device and a context; the design is correlational with no within-person randomisation of exposure; the thresholds are read off a gradient-based attribution of a black-box model, not estimated with uncertainty, so they should be treated as hypotheses; and PM_{2.5} co-varies with noise and temperature in ways a deep model can blend invisibly. Still the clearest current demonstration of what wearable-grade PM sensing buys over address-based assignment.

doi: 10.1021/acs.est.6c09466

Huang et al. 2026 Int J Mol Sci*Environmental Pollution, Biomarkers, and Dementia: An Evidence Review.*

Searched PubMed and Embase for 2020-2026 epidemiology on air pollution, environmental chemicals and dementia, with mechanistic evidence for plausibility. Long-term PM_{2.5}, NO₂ and black carbon were associated with higher dementia risk; PM₁₀ and ozone were inconclusive; phthalates, bisphenols, Pb, Cd and PFOS were linked to poorer cognition. No pooled estimate, no risk-of-bias assessment and no count of included studies in the abstract, so it adds nothing to the existing meta-analyses. Relevance: low; the BC signal is the one piece that points at a metric low-cost absorption sensors can supply.

doi: 10.3390/ijms27188224

India-Aldana et al. 2026 Environ Pollut*The chemical exposome during peri-pregnancy, maternal sleep outcomes post-pregnancy, and the potential mediating role of body mass index.*

About 393 PROGRESS mothers with 52 exposure markers (estimated air pollutants, metals, phthalates, pesticides, PFAS) in pregnancy or one month postpartum, and actigraphy plus NIH-Toolbox sleep outcomes eight years later. BWQS regression: a quartile rise in the air-pollutant mixture associated with about 8 minutes shorter sleep (beta -0.14 h, 95% CrI -0.29 to 0.00); non-essential metals lowered sleep efficiency (-0.95%, -1.81 to -0.17). No BMI mediation. The air-pollution result sits on the null boundary, the mixture's PM_{2.5} share is not separable in the abstract, and a single pregnancy exposure window is asked to explain sleep eight years later, over which residential and occupational exposure will have moved. Hypothesis-generating. The long lag is where repeated personal-sensor exposure histories would add the most.

doi: 10.1016/j.envpol.2026.129220

Oh et al. 2026 Mov Disord*Association Between Long-Term Exposure to Air Pollution and Parkinson's Disease: Administrative Cohort of Older Adults in South Korea.*

5.3 million adults aged 65 and over in the Korean National Health Insurance database, 2010-2019, with 143,557 incident Parkinson's disease cases ascertained from claims requiring hospitalisation or repeated outpatient diagnoses. Per interquartile increase, PM_{2.5} (4.3 µg m⁻³) gave HR 1.018 (95% CI 1.008-1.028) and O₃ (7.3 ppb) HR 1.019 (1.006-1.033); NO₂ was inverse after full adjustment, and no demographic or clinical effect modification reached significance. The inverse NO₂ is the tell: exposure is CMAQ output assigned by administrative district, so within-district contrast is zero and urban socioeconomic structure is absorbed into the pollutant. Claims also date diagnosis, not onset. District-resolution assignment at this cohort size is precisely what a dense sensor network would replace.

doi: 10.1002/mds.70518

Rodulfo-Cardenas et al. 2026 Neurotoxicology*APOE genotype shapes persistent behavioral and molecular alterations following early postnatal particulate matter exposure in humanized apoE3 and apoE4 mice.*

Humanized apoE3 and apoE4 mice of both sexes inhaled NIST fine PM (SRM 2787) or saline for 1 h/day across postnatal days 4-14, then underwent behavioural testing to PND96 and hippocampal/striatal expression profiling. Genotype, not PM, was the main determinant: apoE4 animals were more anxiety-like with hippocampal reorganisation of synaptic, glutamatergic, glial and redox pathways. PM produced only subtle social-behaviour effects and suggestive sex- and region-dependent marker changes. The authors state the chamber concentration was a nominal theoretical maximum of 3789 µg m⁻³ and was never measured - an unquantified dose, which makes the null-leaning PM result uninterpretable in dose-response terms. Relevance: a clean example of why exposure chambers need the same real-time mass verification as field sensors.

doi: 10.1016/j.neuro.2026.103580

Zou et al. 2026 Environ Res*Association of preconception to early childhood residential greenness with child neurodevelopmental outcomes and the mediating role of air pollution: a prospective cohort study.*

5,284 mother-child pairs in the Jiangsu Birth Cohort, with satellite NDVI from preconception to early childhood and Bayley-III screening at age three. Per 0.1 NDVI, preconception and third-trimester greenness lowered risk of the emerging/at-risk cognition category (RR 0.85, 95% CI 0.74-0.98; 0.82, 0.71-0.95) and gross motor (0.87, 0.77-0.98; 0.77, 0.68-0.88), surviving FDR at 0.10. Single-pollutant mediation assigned 68.0% (PM₁₀) and 63.5% (NO₂) of the preconception-cognition association, and 59.1% via PM_{2.5} for third-trimester gross motor. Separate single-pollutant mediators summing well past 100% show the pollutants are proxies for one another and for urbanicity; mediation of that size in a greenness model needs sequential or joint mediation to be credible. Relevance: greenness-PM trade-offs are measurable only with sub-kilometre PM surfaces.

doi: 10.1016/j.envres.2026.125746

Respiratory & allergic**10 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 10 RECORDS IN THIS SUBTOPIC)

Dinh et al. 2026 Front Med (Lausanne)*Association of air pollution with chronic obstructive pulmonary disease: anemia-related outcomes and exploratory genetic susceptibility analyses.*

Taiwan Biobank case-control of 7,566 COPD cases and 22,698 age- and sex-matched controls. Each 10 µg m⁻³ of PM_{2.5} carried OR 1.252 (95% CI 1.222-1.283) at a one-month averaging window and 1.642 (1.588-1.699) at twelve months; the within-COPD anaemia analysis showed no consistent association, and a haemoglobin polygenic score interacted with SO₂ only nominally. Those confidence intervals are implausibly narrow for 30,000 participants under correlated multi-pollutant exposure, which usually means unmodelled spatial correlation rather than precision, and COPD is defined partly by self-reported emphysema or chronic bronchitis. The monotone rise with averaging window is as consistent with confounding by place as with cumulative dose - a distinction only within-person exposure variation can settle.

doi: 10.3389/fmed.2026.1925502

Dou et al. 2026 Front Immunol*Risk factors for allergic rhinitis in children: an umbrella review of systematic review and meta-analysis.*

PROSPERO-registered umbrella review of meta-analyses to May 2025, AMSTAR and GRADE appraisal. Of 75 candidate risk factors, 22 raised paediatric allergic-rhinitis risk, 7 were protective and 46 were indeterminate; air pollution sits among the risk-raising factors with antibiotics, caesarean delivery and prenatal smoking. Evidence quality was low or very low overall, and only early-life food sensitisation reached moderate certainty - so air pollution is on the list at low certainty. Pollutant, metric and pooled effect size are not given in the abstract. Umbrella reviews inherit the heterogeneity of their inputs, and ambient-pollution meta-analyses in children pool incomparable exposure assignments. A calibration of how weak the paediatric AR evidence is, not a new estimate.

doi: 10.3389/fimmu.2026.1849271

HEPA RCT synthesis 2026 Expert Rev Respir Med*HEPA air filtration for asthma and COPD: lessons from randomized controlled trials.*

The consistent finding across trials is the one that should worry anyone who equates exposure reduction with clinical benefit: portable HEPA reliably and substantially lowers indoor PM_{2.5}, but clinical effects are heterogeneous. Several asthma trials improved symptoms, peak flow or quality of life - concentrated in highly symptomatic or higher-exposure subgroups - while most spirometric endpoints moved little or not at all. COPD evidence is thin and only suggestive. The authors attribute the gap to trial design rather than to absent efficacy, and list the specific failures: short follow-up, almost no adult studies, device specifications unreported, adherence unmeasured, no economic evaluation, and near-total absence of low-resource settings. That list is itself the contribution. Being a narrative rather than systematic synthesis with no pooled estimate and no formal risk-of-bias assessment, it cannot bound the effect, and the subgroup pattern it highlights is exactly the pattern selective reporting produces. For a sensing programme the actionable point is that unmeasured adherence and unreported device spec are measurement problems a logging sensor inside the intervention arm would largely solve.

doi: 10.1080/17476348.2026.2735502

Huang et al. 2026 Tob Induc Dis*Global, regional, and 204 countries burden of disease attributed to secondhand smoke-related COPD, 1990-2021: A secondary data analysis of the Global Burden of Disease Study 2021.*

GBD 2021 secondary analysis, 1990-2021, with an ARIMA projection to 2031. Age-standardised mortality for secondhand-smoke-attributable COPD roughly halved, 11.36 (95% UI 4.52-18.55) to 5.87 (2.29-9.72) per 100,000, and ASDR fell 234.01 (93.58-380.47) to 120.94 (47.2-197.93) - yet absolute deaths rose, the familiar ageing-and-population-growth divergence that makes rate-only reporting misleading. China carries the largest absolute mortality (117,388; 47,776-195,915), Papua New Guinea the highest rates. The SDI relationship is non-linear with an inflection at SDI 0.42. The binding limitation is inherited, not introduced: GBD secondhand-smoke exposure is modelled from survey-reported household and workplace smoking, not measured particulate or cotinine, so the uncertainty intervals understate exposure misclassification and every country contrast partly reflects differing survey quality. An ARIMA extrapolation on a modelled series compounds that. Relevant here because secondhand smoke is the one fine-particulate exposure with an intervention that works, and it anchors the indoor end of the PM dose-response.

doi: 10.18332/tid/230940

Johansson et al. 2026 Indoor Air*Associations Between Indoor Air Pollution and Asthma, Allergic Rhinitis and Eczema in a Cohort of Children in Durban, South Africa*

Cross-sectional analysis of 300 children from the MACE birth cohort in Durban with indoor PM_{2.5} measurements, ISAAC interviews and a household walkthrough. ETS exposure was associated with doctor-diagnosed asthma (aOR 3.51, 95% CI 1.20-10.30) and allergic rhinitis (aOR 21.58, 2.80-166.38); biomass fuel use with rhinitis score (aOR 4.97, 1.13-21.95) and eczema score (aOR 6.61, 1.48-29.60). The abstract reports no estimate for measured indoor or prenatal PM_{2.5}, which suggests it was null. CIs spanning two orders of magnitude show very few cases; exposure categories are self-reported. Relevance: the measured-PM result is what this watch needs, and it is absent - the categorical proxies carried the association.

doi: 10.1155/ina/8341339

Kang et al. 2026 PLoS One*Asthma Intervention with Residential Ventilation and Air Cleaner (AIRVAC) Study: A 4-arm parallel-group randomized controlled trial protocol.*

Protocol for a single-blind, sham-controlled, four-arm RCT (active or sham energy recovery ventilator; active or sham portable air cleaner) in >80 Dallas-Fort Worth households with a physician-diagnosed asthmatic aged 5-17 or adult, with up to one year pre-intervention and one year post. Primary outcome asthma control; secondary pulmonary function, asthma QoL, stress and sleep. Exposure by low-cost plus research-grade PM, NO₂, CO and VOC sensors, and mold sampling. Registered as NCT07196436, where spirometry and ACT are both listed as primary - a protocol/registry mismatch to watch. ~20 households per arm is small for ACT differences, and no PM exposure endpoint is pre-specified. Relevance: a sham-controlled trial whose exposure arm is a low-cost sensor network - the calibration protocol will determine what the IAQ results mean.

doi: 10.1371/journal.pone.0358221

Naka et al. 2026 Pediatr Allergy Immunol*Impact of particulate matter 2.5 exposure on severity of pediatric chronic rhinitis: A prospective cohort study.*

146 children with chronic rhinitis in the Bangkok Metropolitan Region followed weekly for a median 52 weeks (Jul 2024-Aug 2025), with symptom severity on a visual analogue scale and the preceding week's PM_{2.5} interpolated by inverse-distance weighting. Per 10 µg m⁻³, VAS rose by 0.260 (95% CI 0.215-0.306); the association held in both allergic and non-allergic rhinitis, with wet season, infections and medication also associated. The effect is small on a 10-point scale, the VAS is self- or parent-reported and unblinded to haze days, and IDW over sparse monitors ignores indoor exposure. Relevance: repeated weekly outcomes over a year are an ideal fit for home or school low-cost sensors, which would replace IDW with measured exposure.

doi: 10.1111/pai.70493

Samdarshi et al. 2026 Thorax*Systematic misclassification of lung function in Indian adolescents using GLI-2012 reference equations: evidence from the multicity APEAL cohort.*

Baseline of the Air Pollution Exposure on Adolescents' Lungs cohort: 4,175 healthy non-smoking 11-14-year-olds with ATS/ERS spirometry in four cities. Against GLI-2012 (Asian category), mean z-scores were -2.44 for FEV1, -1.34 for FVC and -0.78 for FEV1/FVC; 74.6% fell below the LLN for FEV1 against an expected 5%. The shift was uniform across sex, city, age, height and BMI. No PM analysis yet. It is carried because it bears directly on the outcome metrology of every South Asian PM-lung study: using GLI-2012 would label most exposed and unexposed adolescents abnormal, compressing contrasts and inflating 'impairment' prevalence. The counter-risk is that part of the deficit is itself pollution-driven, so a local reference set derived from these same cities may absorb the very effect later analyses seek.

doi: 10.1136/thorax-2026-225309

Song et al. 2026 Atmos Environ*Air Pollution Effects on Influenza-like Illness Modified by Temperature and Humidity in Central China*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title states effect modification of air-pollution associations with influenza-like illness by temperature and humidity in central China. Relevant to the humidity-confounding thread running through this issue. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.atmosenv.2026.122374

Zhang et al. 2026 Respir Med*Associations between household particulate matter and inflammatory cytokines in cystic fibrosis.*

Pilot in adults with cystic fibrosis: 4-11 days of in-home PM_{2.5} monitoring with real-time sensors, then multiplex serum cytokines. Two metrics were compared - mean PM_{2.5} and percentage of monitoring time above 35 $\mu\text{g m}^{-3}$. The mean was unrelated to any cytokine; percentage of time above 35 correlated moderately and significantly with PDGF-AA/BB, with non-significant positive trends for IL-7 and IL-8. Sample size, sensor model and any calibration are not in the abstract; Spearman correlations across a cytokine panel with no multiplicity correction; one blood draw per person. Relevance: high in kind - it is the peak-exceedance metric, only obtainable from continuous sensors, that carried the signal, not the mean a filter would give.

doi: 10.1016/j.rmed.2026.109180

Reproductive & developmental

8 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Eick et al. 2026 J Expo Sci Environ Epidemiol*Prenatal exposure to aromatic amines and associations with infant cognition in the ECHO.CA.IL cohorts.*

N=250 across two pooled prospective birth cohorts (IKIDS and Chemicals in Our Bodies). Six aromatic amines were detected in over half of second-trimester urine samples; infant visual recognition memory was measured by infrared eye tracking at 7.5 months, with log-transformed individual analytes and a molar sum entered in linear regressions and child sex examined as a modifier. The relevance to a PM watch is indirect and worth stating plainly: aromatic amines are combustion and tobacco-smoke products, cotinine is measured alongside them, so urinary AAs partly index particulate-bound combustion exposure rather than only consumer-product contact - but the study makes no air measurement and cannot separate the two routes. At N=250 with six correlated analytes and a sex-interaction analysis, this is hypothesis-generating; any single-analyte association at this sample size is underpowered for the multiplicity involved. Carried at tier C for the exposure biomarker panel, not for the cognitive estimate.

doi: 10.1038/s41370-026-00905-3

Hu et al. 2026 Reprod Toxicol*Real-world PM_{2.5} exposure triggers early testicular epigenetic responses in the absence of significant changes in measured reproductive endpoints.*

Male mice breathed real-world ambient PM_{2.5} in Hangzhou through a whole-body inhalation system for 4, 8 or 12 weeks - roughly one to three spermatogenic cycles. Sperm quality, reproductive hormones, histopathology and oxidative DNA damage were all unchanged, while global testicular 5mC fell at 4 weeks and rose at 12 weeks relative to control ($P < 0.05$), and 5hmC rose significantly at 12 weeks. The claim is that epigenetic remodelling precedes measurable injury. Two limits: the record gives no exposure concentration, so dose is unquantified, and global 5mC/5hmC has no locus resolution, so a bidirectional time course is as compatible with adaptation as with damage. Real-ambient inhalation studies live or die on continuous characterisation of what the animals actually breathe - an instrumentation problem.

doi: 10.1016/j.reprotox.2026.109366

Li et al. 2026a BMJ Open*Cohort profile: the Shenzhen Birth Cohort (SZBC) study.*

Cohort profile rather than a result: 4,208 women recruited at 19 weeks' gestation or earlier from a single Shenzhen maternity hospital since March 2018, 3,962 with complete baseline data, 3,150 deliveries, up to 20 follow-up points through age seven, with questionnaires, examinations and blood, urine and faecal samples. Findings cited to date include prenatal greenness associated with better motor development and postnatal PM_{2.5} - nitrate in particular - with delayed motor and problem-solving development. No effect sizes, intervals, exposure model or adjustment set are given, and single-hospital recruitment in a migrant-dense city limits transportability. Carried as infrastructure: a component-resolved prenatal and postnatal cohort with biosamples is the design that could eventually separate nitrate from mass.

doi: 10.1136/bmjopen-2026-121066

Tamire et al. 2026 J Expo Sci Environ Epidemiol*Prenatal fine particulate matter (PM_{2.5}) exposure differentially impairs fetal skeletal and soft-tissue growth: a longitudinal study in rural Butajira, Ethiopia.*

Pregnant women recruited at or before 24 weeks, followed over three visits, March-June 2025. An IQR rise of 27.0 $\mu\text{g m}^{-3}$ in personal PM_{2.5} was associated with a 0.43 SD fall in femur length (beta -0.43, 95% CI -0.51 to -0.35) and a 0.17 SD fall in abdominal circumference (-0.17, -0.21 to -0.13); head circumference and biparietal diameter showed no significant association under either concurrent or lagged exposure. The dissociation is the finding - skeletal and soft-tissue growth respond while cranial growth is spared, which is the brain-sparing pattern and argues against a purely nutritional confounder. Three weaknesses bite. The exposure contrast is enormous by design (an IQR of 27 $\mu\text{g m}^{-3}$ implies a biomass-cooking population far outside the range of the cohorts these coefficients will be compared against), the sample is small and the paper does not report n in the abstract, and three 24-h samples cannot represent a 16-week gestational window - classical measurement error that, being non-differential, biases toward null and so does not explain the femur result. Directly relevant to distributed sensing: the exposure instrument is a consumer-grade optical wearable, and effect sizes this large resting on three wearable-days per pregnancy is exactly the design low-cost sensing makes newly affordable in LMIC settings - and exactly where sensor humidity and biomass-smoke calibration error, unreported here, would propagate straight into beta.

doi: 10.1038/s41370-026-00966-4

Thirumalazhagan et al. 2026 Reprod Toxicol*A Comprehensive Meta-Analysis on the Impact of airborne Particulate Matter on Trophoblast Cell Function In Vitro.*

Twelve in vitro studies from 1,506 records to September 2025. PM_{2.5} reduced trophoblast viability by a pooled 20.7%, disrupted steroidogenic enzymes (CYP11A1, 3beta-HSD1, FDXR), uncoupled hCG from progesterone synthesis, induced oxidative DNA damage with ATR-mediated p53-independent apoptosis, suppressed VEGF-VEGFR2-MEK-ERK-mTOR signalling and impaired invasion via TIMP, RhoA and collagen I. The central problem is that pooling relative risks of gene expression across studies with different particle sources, extraction methods, doses and cell lines produces a number with no exposure meaning; the dose axis is absent from the abstract, and cell-line trophoblasts are not placenta. Google Scholar as a search source compromises reproducibility. Useful as a pathway catalogue for placental mechanisms behind birth-weight associations, not as quantitative evidence.

doi: 10.1016/j.reprotox.2026.109364

Wang et al. 2026 Biomolecules

Associations of ASIC2, IKZF3, and GSDMB Variants with Gestational Diabetes Mellitus and Their Interactions with Environmental Exposures Among Pregnant Women.

Case-control study of 1,564 pregnant women genotyped for ASIC2 rs12450465, IKZF3 rs2060941 and GSDMB rs2305479, with gene-environment interaction tests for PM_{2.5}, O₃ and urinary MIBP. The T-T-T haplotype was associated with GDM (OR 1.741, 95% CI 1.208-2.509; FDR p 0.0133) and, in the 289 with MIBP, OR 2.980 (1.378-6.446). Interaction signals surviving FDR mainly involved O₃, not PM_{2.5}; primary analyses were covariate-unadjusted. Candidate-gene interaction testing in one sample without replication is weak evidence, and exposure assignment is not described. Relevance: low; no PM_{2.5} main effect or robust interaction is reported.

doi: 10.3390/biom16091373

Wanta et al. 2026 Cells Tissues Organs

Ultrastructural Changes in Human Placental Tissues and Evidence of Particulate Matter Transport into the Umbilical Vein Following PM_{2.5} Exposure in Chiang Rai.

Placentas from 10 women with third-trimester exposure during Chiang Rai's February-April burning season and 10 without were examined by SEM-EDS and TEM. Exposed villi were flattened and irregular; metal-containing particles were adherent to the placental surface and inside the umbilical vein, with an elemental profile (Al, Ca, Cr, Cu, Fe, K, Mg, Mn, Na, Ni, P, Si, Ti) resembling filter-collected PM_{2.5}; syncytiotrophoblast abnormalities and barrier thickening were seen. Exposure is season of delivery, not a measured dose; n=10 per group; SEM-EDS cannot exclude contamination during processing, and these elements are also endogenous. Relevance: supports tissue translocation, but a personal-monitor exposure axis would be needed to make it dose-response.

doi: 10.1159/cto/acbag006

Zhang et al. 2026 Front Public Health

Association between ambient PM(2.5) exposure during pregnancy and risk of preterm birth: a retrospective birth cohort study in Chengdu, Southwest China.

35,136 singleton births at one Chengdu hospital, 2015-2024; 2,622 (7.46%) preterm. Mean whole-pregnancy PM_{2.5} was 49.1 µg m⁻³ (satellite) and 51.9 µg m⁻³ (monitors). Per 10 µg m⁻³, whole-pregnancy ORs were 1.089 (95% CI 1.010-1.174) satellite and 1.091 (0.996-1.195) monitor, with first- and third-trimester signals and landmark ORs 1.06-1.10 for weeks 32-35; associations were confined to late, not earlier, preterm birth. Single-hospital sampling, district-level fixed effects and residence at delivery limit the exposure contrast. The methodological result is the useful one: the two exposure products gave near-identical point estimates, with satellite giving complete coverage and narrower intervals. Relevance: a direct, if coarse, benchmark for when a sparse monitor network is adequate - the question dense sensor networks exist to answer at finer scale.

doi: 10.3389/fpubh.2026.1936547

Mechanistic toxicology**20 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (20 OF 20 RECORDS IN THIS SUBTOPIC)

Ameli et al. 2026 Environ Pollut

Acute co-exposure to urban particulate matter and aircraft noise elicits lung-centered immune-redox proteomic remodeling in mice.

DIA LC-MS/MS proteomics of lung, brain and heart from male C57BL/6J mice given three days of NIST SRM 1648a urban PM, aircraft noise, both, or control - the exposure model previously characterised by Kuntic et al. The lung dominated: PM+noise gave the broadest induction of acute-phase, phagosome, interferon and macrophage proteins, with NADPH oxidase, iron handling, barrier and endothelial/ECM changes that agree with the parent study's NOX-2, HO-1, ROS, blood-pressure and vascular endpoints. Most stress proteins scaled proportionally rather than synergistically; brain and heart responses were smaller. Dose is a standard reference material, not a measured airborne concentration in the abstract, and only males were used. Relevance: noise and PM co-vary along the same roads, so co-located acoustic and PM sensing is the matching exposure design.

doi: 10.1016/j.envpol.2026.129165

CSE HASMC isoforms 2026 PLoS One

A comprehensive full-length transcriptome landscape of cigarette smoke-exposed HASMCs reveals extensive remodeling of mRNA isoforms and regulatory networks.

Oxford Nanopore long-read sequencing of human aortic smooth muscle cells at 0.6% cigarette smoke extract, the midpoint of a biphasic viability curve (0.2% mildly proliferative, 0.4-0.8% progressively cytotoxic). 11,222 novel transcripts; 1,866 transcripts up in ferroptosis and cell division pathways, 493 down in oxidative phosphorylation and glutathione metabolism, 1,753 alternative polyadenylation events remodelling 3'UTRs, and 269 alternative splicing events hitting cell-cycle regulators and transcription factors including CREM, FOSL1 and HBP1. The claim is that high-dose cytotoxicity is executed post-transcriptionally rather than through differential gene expression alone, and the isoform resolution is what makes that visible - a short-read study would have reported the pathway shifts and missed the 3'UTR remodelling entirely. Admitted at tier C with an explicit caveat: cigarette smoke extract is not ambient PM, the cell line is not a vessel, a single 0.6% dose is not a dose-response for the mechanism claimed, and no particle mass or size metric is reported, so nothing here is transferable to an ambient exposure-response coefficient. Carried for the ferroptosis and glutathione-depletion axis, which recurs in ambient PM vascular toxicology and is worth tracking.

doi: 10.1371/journal.pone.0354604

Cherriere et al. 2026 Part Fibre Toxicol

Influence of lung overload and exposure scenarios on recovery following inhalation of an alumina nanoparticle and HCl gas mixture.

Rats inhaled alumina nanoparticles at 1-20 mg/m³ with 5 ppm HCl under two temporal patterns - four consecutive days, or once weekly for eight weeks - with assessment at 1 and 28 days post-exposure. The short scenario produced dose-dependent neutrophilic inflammation that resolved by day 28 with partial clearance recovery; the intermittent long scenario left significantly higher retained lung burdens, reduced clearance and persistent neutrophilia at day 28, with the top dose exceeding accepted overload thresholds but no remodelling or genotoxicity. The abstract gives no burden values or dispersion, so the overload claim cannot be checked against the usual volumetric criterion. The transferable point for exposure science: equal cumulative dose delivered intermittently is not equivalent, which time-averaged occupational metrics assume.

doi: 10.1186/s12989-026-00703-7

Ghio et al. 2026 Pathogens

Iron, Tuberculosis, and Particle-Related Exposures Including Smoking.

Mechanistic review from the EPA group proposing that particles - cigarette smoke, air pollution, biomass smoke, silica - complex host iron and induce a functional iron deficiency in the respiratory tract, which favours M. tuberculosis because of its many iron-acquisition pathways, explaining the epidemiological association of particle exposures with TB. A hypothesis-driven synthesis without systematic search or new data. It pairs, with opposite sign, with the Shen et al. finding in this issue that exogenous Fe relieves organic-induced iron deficiency in macrophages. Relevance: low; conceptual.

doi: 10.3390/pathogens15090982

Gradinaru et al. 2026 J Xenobiot

Atmospheric Particulate Matter as a Carrier of Pb, Cd, and Ni: From Environmental Transfer and Bioaccessibility to Molecular Toxicity and Predictive Modeling.

A narrative review tracing PM-bound lead, cadmium and nickel from emission through transport, deposition, food-chain transfer, bioaccessibility and toxicokinetics to molecular toxicity and predictive modelling. The central argument is that total PM mass and total metal concentration misrepresent biologically effective dose, which also depends on respiratory deposition, dissolution kinetics, gastrointestinal release and co-associated constituents; the three metals share oxidative, inflammatory, mitochondrial, genomic and epigenetic effects but differ in mobility and target organs. No search strategy or quality appraisal is described, so coverage cannot be judged, and no quantitative synthesis is offered. Relevance: a reminder that optical mass sensing is blind to exactly the metal fraction that drives toxicity; composition needs periodic filter sampling alongside.

doi: 10.3390/jox16050167

Gulzar et al. 2026 Atmos Pollut Res

Mechanistic Insights into PM_{2.5}-Induced Cardiopulmonary Dysfunction: Bridging Pathways to Therapeutic Interventions

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 24 Sep with no abstract; Crossref, Europe PMC, PubMed efetch and Semantic Scholar all returned nothing. The title indicates a review of PM_{2.5} cardiopulmonary mechanisms oriented to therapeutic intervention. Low priority for this watch unless it carries a structured synthesis. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.apr.2026.103204

Jiang et al. 2026 Environ Sci Technol

Nitration of Urban PM_{2.5}: A Hidden Driver of Enhanced Human Inflammation.

Longitudinal population exposomics in Beijing resolved roughly 1,000 nitrated aromatic compounds (NACs) on personal PM_{2.5}, at median concentrations above the traditional polycyclic aromatics measured alongside them. Secondary NACs formed by aqueous-phase nitration, represented by 3- and 5-nitrosalicylic acid, carried over 70% of summer NAC exposure, and their associations with lung and systemic IL-6 were 2.1 to 232 times stronger than those of PAHs or gaseous NOx. The upper end of that ratio range is doing a lot of work and no CIs are given; associations are cross-sectional within a panel and NAC exposure covaries with everything else in summer photochemistry. Still the strongest argument this quarter that NOx health effects are partly particle-phase - which means mass-only sensing misses the mechanism entirely.

doi: 10.1021/acs.est.6c06494

Lang et al. 2026 Environ Toxicol Pharmacol

Particulate matter selectively promotes macropinocytosis and malignant transformation-associated phenotypes in p53-impaired epithelial cells.

Murine mammary and hamster respiratory epithelial cells with intact or impaired p53 were exposed acutely and repeatedly to a European certified reference PM_{2.5}. PM_{2.5} activated oxidative- and cellular-stress pathways in all models, but p53-impaired cells adapted with enhanced macropinocytosis and better growth under nutrient limitation, and repeated exposure increased anchorage-independent growth in a transformed mammary line. Rodent lines, a single reference material, and doses not stated in the abstract; soft-agar growth in an already transformed line is a promotion, not initiation, readout. Relevance: low; mechanistic support for PM as a tumour promoter in p53-deficient tissue.

doi: 10.1016/j.etap.2026.105177

Lin et al. 2026 Arterioscler Thromb Vasc Biol

Air Pollution Triggers Metabolomic Signature of Mitochondrial Dysfunction in Mice and Humans: Implications for Cardiovascular Disease Development.

A cross-species design: ApoE^{-/-} mice given two weeks of diesel exhaust, and 26 healthy adults who moved from Los Angeles (PM_{2.5} 14.4 $\mu\text{g m}^{-3}$) to Beijing (67.6 $\mu\text{g m}^{-3}$) for ten weeks. Multi-platform metabolomics and lipidomics converged on long-chain dicarboxylic acids and medium-to-long-chain acylcarnitines rising in both species, consistent with impaired fatty-acid oxidation; Seahorse respirometry confirmed hepatic mitochondrial dysfunction in the mice, and in humans the shifts tracked 12-/15-LOX lipid peroxidation products and PAH exposure biomarkers. Tryptophan, pyrimidine and lysine signatures did not replicate across species. n=26 with a city-level exposure contrast that also changes diet, time zone and stress - no personal monitoring is reported. Relevance: a 4.7-fold ambient contrast is exactly where calibrated personal sensors would sharpen the dose axis.

doi: 10.1161/atvbaha.125.324277

Meganathan et al. 2026 Respir Res

Aryl hydrocarbon receptor (AhR) regulates organic dust and organic dust-associated bacterial extracellular vesicles-induced airway inflammation via ROS-NFkB signaling.

Beas2B bronchial epithelial cells were exposed to poultry-farm organic dust extract or dust-associated bacterial extracellular vesicles, with siRNA AhR knockdown, and AhR-knockout mice were challenged with the EVs. Both exposures increased AhR expression and nuclear translocation; knockdown reduced ROS and NFkB activation and attenuated IL-1b, ICAM-1, IL-6 and IL-8, though STAT3 responded in opposite directions to extract versus EVs. Dust extract is a soluble fraction, not the inhaled particle, and doses are not related to barn airborne concentrations in the abstract. Relevance: low for sensing; agricultural dust is coarse-mode, the size range where low-cost optical sensors are least reliable.

doi: 10.1186/s12931-026-03908-8

Messina et al. 2026 Front Toxicol

Organ-specific apoptotic pathways induced by micro- and nanoplastics: a systematic review of in vitro toxicological evidence.

Eighty in vitro studies, PECOS framing, PRISMA reporting, search run 6 September 2024. Apoptosis recurs across gastrointestinal, hepatic, pulmonary, renal, reproductive, placental, neural, cardiovascular, dermal and haematological models, and the mechanisms converge: oxidative stress, mitochondrial depolarisation, ER stress, calcium dysregulation, MAPK and PI3K/AKT signalling, Bcl-2 family shifts and caspase activation, with ferroptotic and autophagic crosstalk. Particle size and surface chemistry modulate potency. Two limitations bound the inference hard. First, the search predates a fast-moving literature by two years. Second - and this is the one that matters for inhalation - the included work is overwhelmingly polystyrene spheres at mass doses far above any modelled human airway deposition, so the review establishes that these pathways can be engaged, not that they are engaged at ambient airborne microplastic levels. No dose-response anchor to a measured air concentration appears anywhere in the synthesis, which is the gap the monitoring side would have to fill.

doi: 10.3389/ftox.2026.1908508

Monge et al. 2026 Cells

Heart Function Changes During Environmentally Relevant Combined Exposure to Heat, Ultrafine Carbon Black, and Ozone: Role of NOD-like Receptor X1. Female Nlr1+/+ and Nlr1-/- mice received 3 h of filtered air, heat (36 °C), ultrafine carbon black plus ozone (2.5 mg/m³ + 1 ppm), or all three, with echocardiography, BAL and cardiac mitochondrial assays at 2 and 24 h. UfCB+O₃ alone cut stroke volume, cardiac output and complex III-V activity in both genotypes; the triple exposure produced larger falls in ejection fraction and fractional shortening, more pulmonary neutrophilia and dyssynchronous strain in the knockouts, and heat alone impaired mitochondrial function only in Nlr1-/- . Doses are two to three orders above ambient and the particle and ozone arms are never separated, so particle-specific attribution is not possible. Relevance: supports co-logging temperature with PM at every sensor node, since heat modifies the cardiac response.

doi: 10.3390/cells15181699

Pei et al. 2026 Environ Sci Technol

Deciphering Allergenic Components Responsible for Differential Sensitization Potentials of Source-Specific Atmospheric Particulate Matter

h-CLAT and keratinocyte-dendritic cell co-culture tests of three reference materials: urban PM 1648a, coal fly ash 2693 and diesel particles 2975. 1648a raised CD54 in THP-1 monoculture via TNF-α and IL-1β; KC co-culture amplified DC activation for all three (CD86, CD54) through IL-1β and extracellular ATP. AhR signalling in HaCaT was required; amorphous silica drove 1648a's CD54 response, while soot and magnetite, with silica, raised CD86; PAHs acted synergistically. SRMs are aged bulk dusts, not size-fractionated ambient PM_{2.5}, and doses are not in the abstract. Relevance: low for sensing; identifies silica and soot as the sensitising components, both separable only by composition, not mass.

doi: 10.1021/acs.est.6c08887

Shen et al. 2026 Environ Sci Technol

A Brake on Toxicity: Iron Suppresses Organic-Induced Superoxide Generation in Macrophages

Macrophages were exposed to brake-wear particles (BWP), highway PM_{2.5}, and Fe alone or mixed with isoprene SOA or 9,10-phenanthrenequinone. BWPs slightly induced cellular superoxide below 10 µg/mL and suppressed it dose-dependently above; highway PM_{2.5} did neither. Fe²⁺ and Fe³⁺ suppressed baseline and organic-induced superoxide, which the authors attribute to replenishment of the labile iron pool (relieving organic-induced iron deficiency and NADPH oxidase activity) plus direct chemical quenching. In vitro doses are not related to deposited lung dose, and superoxide is one ROS endpoint - Fe drives hydroxyl radical via Fenton chemistry, which this does not address. Relevance: undermines simple 'more Fe, more toxicity' weighting in composition-based exposure metrics for non-exhaust PM.

doi: 10.1021/acs.est.6c06554

Song et al. 2026 Toxicol Lett

Lung Toxicity Induced by Co-Exposure to Particulate Matter and Indoor Microorganisms in Mice.

C57BL/6 mice received PM combined with dominant indoor airborne microorganisms by intratracheal instillation for two weeks. PM with the fungus *Bjerkandera adusta* increased lung eosinophil infiltration beyond either agent alone, with synergistic upregulation of C3 and IL-3; PM with *Bifidobacterium infantis* produced histopathology suggestive of fibrogenesis and synergistic collagen-gene induction. The PM type, dose and group sizes are not in the abstract, and instillation delivers a bolus unlike inhalation. Relevance: low for sensing; argues that indoor PM mass alone under-describes hazard where bioaerosol load is high.

doi: 10.1016/j.toxlet.2026.113207

Tzagaroulaki et al. 2026 Toxics

An A549 Cell-Based Approach Using Repeated Fluorescence Readouts for Assessing Reactive Oxygen Species Activity of Atmospheric Particulate Matter.

Method-performance characterisation of an A549/DCFH-DA ROS assay read repeatedly from 15 min to 6 h, using zymosan and NIST SRM 2584 controls, CV and signal-to-noise. At 60 min the SRM response rose across 0.02, 0.05 and 0.10 mg/mL (descriptive R² 0.90); earlier readouts were non-monotonic. Eight matched PTFE-quartz filter pairs gave zero-intercept slopes of 0.90 (mass-normalised) and 0.99 (volume-normalised), R² ~0.99. Field samples showed DCFH-DA fluorescence did not track PM mass. Three concentrations and eight pairs are thin, and DCFH-DA is prone to auto-oxidation artefacts. Relevance: the filter-substrate equivalence is useful for networks archiving PTFE filters from reference co-location sites for oxidative-potential follow-up.

doi: 10.3390/toxics14090789

Wang et al. 2026 Environ Pollut

Ozone and PM_{2.5} co-exposure exacerbates allergic rhinitis and is associated with PPAR/TRPV1-related barrier damage: Insights from nasal transcriptomic and microbial analyses.

C57BL/6J allergic-rhinitis mice in five groups (control, AR, AR+O₃, AR+PM_{2.5}, AR+PM_{2.5}+O₃). Co-exposure markedly aggravated sneezing, disrupted nasal mucosa and lowered ZO-1 and occludin; nasal transcriptomics implicated PPAR/TRPV1 signalling and 16S sequencing showed microbial dysbiosis correlated with inflammation and barrier injury. The authors state the doses were acute and high and not designed to model environmental exposure, which is the limitation - interaction at high dose need not persist at ambient levels, and group sizes and PM dose are not reported in the abstract. Associations between omics layers are correlational. Carried for the co-exposure design: O₃ and PM_{2.5} peak in different seasons in most airsheds, so their joint effect is a question sensor networks measuring both can actually address.

doi: 10.1016/j.envpol.2026.129207

Yang et al. 2026a Environ Health (Wash)

The Human Microbiome in Airborne Particulate Matter-Associated Aging-Related Chronic Diseases: An Evidence-Stratified Review.

Evidence-stratified review proposing a PM exposure-microbiome-aging framework across respiratory, gut and skin compartments, with the gut microbiota argued to be the most mechanistically developed interface. The stratification is the contribution: the authors grade epidemiological, animal and multiomics evidence separately rather than pooling them into a single causal story. What it cannot do is quantify anything - no pooled estimates, no dose-response, and the reviewed studies use incompatible PM metrics, so the framework is a research agenda rather than a finding. Carried because microbiome mediation keeps being invoked to explain PM associations that survive adjustment, and because a graded map of where the evidence actually is saves the field repeated litigation of the same weak studies.

doi: 10.1021/envhealth.5c00852

Yang et al. 2026b Environ Health (Wash)

Molecular Mechanisms of Indoor Environmental Pollution tt-DDE-Induced Blood Pressure Elevation and the Protective Role of Human-Specific Gene CHRFAM7A.

Cooking-oil fumes are stated here as roughly 90% of indoor particulate emissions, and tt-DDE is their dominant aldehyde. Inhaled tt-DDE raised blood pressure in mice through endothelial damage; CHRFAM7A-transgenic mice resisted the systolic rise and the endothelial dysfunction, with protection traced to STAT3/HO-1 activation, and in vitro tt-DDE drove KLF5 inflammatory signalling via METTL3-dependent m⁶A modification, reversible by CHRFAM7A overexpression and abolished by HO-1 inhibition. No exposure concentration, duration or blood-pressure magnitude appears in the record, and CHRFAM7A is human-specific, so the transgenic model has no wild-type counterpart. Relevance: kitchen PM sensing conflates mass with a gas-phase aldehyde that may carry the cardiovascular signal.

doi: 10.1021/envhealth.5c00703

Ye et al. 2026 Clin Exp Pharmacol Physiol*Mechanistic Study of SRSF1-Mediated Inhibition of Treg Cell Function via the mTORC1/ULK1 Signalling Axis Through Mitophagy in Asthma.*

SRSF1 was elevated in BALF cells from asthma patients and correlated negatively with Treg FOXP3 and FEV1 %pred. In OVA-asthma mice, SRSF1 upregulation in Tregs increased mTOR and ULK1-Ser757 phosphorylation and impaired mitophagy and metabolic reprogramming; SRSF1 knockdown or rapamycin restored Treg function and reduced airway inflammation, while an mTORC1 activator reversed the rescue. A PM_{2.5} arm exaggerated these changes in an SRSF1-dependent way. PM is secondary to the paper's question: dose, particle source and exposure route are absent from the abstract, and patient numbers are not given. The value is mechanistic - a Treg mitophagy axis through which PM could worsen established asthma - and it adds nothing to exposure-response. Tier C.

doi: 10.1111/1440-1681.70157

Occupational & indoor**22 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (22 OF 22 RECORDS IN THIS SUBTOPIC)

Albert et al. 2026 Innov Aging*Assessing home environmental hazards for older adults receiving in-home supportive services: the healthy homes assessment tool.*

Development and pilot validation of the Healthy Home Assessment Tool, a care-coordinator checklist for indoor air quality and moisture/mould embedded in home- and community-based services, in 41 low-income older-adult households (median age 76). Against consumer-grade IAQ monitors, 61.0% of homes exceeded $9 \mu\text{g m}^{-3}$ PM_{2.5} and 21.9% exceeded the 4 pCi/L radon action level; PM_{2.5} was higher where smoking was reported and fungal burden higher with visible mould. The comparison of indoor snapshots with an annual ambient standard overstates exceedance, and the consumer monitors are uncalibrated in the abstract. Relevance: a concrete pathway for low-cost PM sensors into social-care workflows - but the 61% figure depends entirely on a sensor whose bias is not stated.

doi: 10.1093/geroni/igag078

Build Environ 2026 (cleanroom) Build Environ*Multi-objective ventilation optimization in non-unidirectional cleanrooms: an integrated RSM-NSGA-II-TOPSIS approach.*

METADATA ONLY. The title names a response-surface plus genetic-algorithm plus TOPSIS pipeline applied to non-unidirectional cleanroom ventilation, which is a simulation-optimisation study, not a measurement one; the objectives in this literature are almost always particle count against fan energy. Relevance to this watch is narrow but real - cleanroom control is the upper bound on what mechanical ventilation can do to particle number indoors, and the trade-off surface is the same one that governs filtration interventions in ordinary buildings at far lower air-change rates. Carried at tier C on the strength of the venue and the topic, with no finding asserted.

doi: 10.1016/j.buildenv.2026.115290

Build Environ 2026 (subway) Build Environ*Spatiotemporal distribution characteristics of multiple air pollutants within subway stations.*

METADATA ONLY. The Crossref deposit carries no abstract and neither Crossref nor Europe PMC had text at the time of screening, so nothing beyond the title is asserted here. Subway microenvironments are the highest-concentration routine PM exposure most urban commuters encounter - platform PM_{2.5} commonly runs several-fold above street level and is Fe-dominated from rail-wheel-brake wear - so a spatiotemporal characterisation across a station is directly relevant to fixed-versus-personal monitor siting. Held for a deferred re-query once the publisher releases the abstract; the record is carried now rather than dropped so the DOI is on the ledger.

doi: 10.1016/j.buildenv.2026.115292

Environ Int 2026 (button mfg) Environ Int*Plastic button manufacturing as an under-recognized source of airborne multi-pollutant mixtures: process-resolved characterization.*

METADATA ONLY. No abstract in the Crossref deposit or Europe PMC at screening. The title claims a process-resolved characterisation, which in occupational aerosol work normally means emissions resolved to individual unit operations rather than a shift-average - the design that actually supports a control recommendation. Small-scale plastics fabrication is a plausible unrecognised source of mixed particulate and VOC exposure precisely because it sits below the size threshold at which most jurisdictions require monitoring. No effect size is claimed here because none is available. On the deferred re-query list.

doi: 10.1016/j.envint.2026.110529

Figueiredo et al. 2026 Environ Toxicol Pharmacol*Chemical and toxicological assessment of particulate matter in a charcoal barbecue restaurant.*

Indoor and outdoor PM₁₀ sampled over six working days plus a closure period at one charcoal-grill restaurant. Indoor PM₁₀ exceeded outdoor on every working day, carbonaceous matter reached half the mass, and K⁺, sulphate, nitrate and chloride dominated the ions; saccharides and cholesterol were the main organic tracers. Indoor extracts had higher DTT oxidative potential, lowered A549 viability and raised intracellular ROS, and ROS correlated with indoor phenolics and carboxylic acids. The design is a single site with a handful of filter days, so the correlations are descriptive, and PM₁₀ rather than PM_{2.5} blurs the combustion fraction. No worker personal sampling. The chemical profile is the useful output for sensing: biomass-like K⁺ and cholesterol from grilling is exactly the mixed smoke-and-cooking aerosol that optical sensors misread without source-specific correction.

doi: 10.1016/j.etap.2026.105173

Gaillard et al. 2026 Preprints.org (preprint)*Potential Link Between Indoor Air Quality and Coronary Artery Disease in Amsterdam, a Field Study*

Preprint: several weeks of in-home PM_{2.5}, PM₁₀ and CO₂ monitoring in 49 patients referred for coronary CT angiography in two socioeconomically contrasting Amsterdam neighbourhoods (Zuid n=24, Nieuw-West n=25), compared with CAD-RADS, calcium scores, vessel involvement and obstructive disease. No systematic neighbourhood difference in indoor air was found, but within-neighbourhood variability was large, and indoor PM showed no significant association with CAD or obstructive CAD. A weeks-long snapshot cannot represent the decades of exposure behind coronary atherosclerosis, and n=49 is badly underpowered for a null. Not peer reviewed. Relevance: the within-neighbourhood spread is the durable finding - household factors dominate indoor PM, which area-level outdoor sensing cannot capture.

doi: 10.20944/preprints202609.1981.v1

JP laboratory exposure 2026 Ann Work Expo Health*Assessment of personal exposure and spatial concentration variability in a small-scale chemical laboratory.*

Personal exposure concentrations ran consistently higher than co-located area measurements, mean values roughly an order of magnitude greater, reproduced across two independent measurement years. CFD then explains the gap mechanically: concentration varies strongly with height, so a fixed sampler placed away from the breathing zone of a person leaning over their own source cannot recover personal exposure, whatever its precision. The timing is regulatory - Japan makes personal exposure measurement mandatory from October 2026 - and the paper is effectively a demonstration that the outgoing area-based system was biased low by roughly 10x in this setting. Narrow and honest about it: one small teaching laboratory, student experiments rather than production work, passive samplers for vapour-phase analytes rather than PM. The 10x should not be transported to other geometries. It generalises to distributed PM sensing as a siting argument with a number attached - a fixed node samples the room, a person samples their own plume, and near a source the two differ by an order of magnitude, not a percentage.

doi: 10.1093/annweh/wxag082

Lee et al. 2026 J Hazard Mater*Comparative analysis of metal concentrations in respirable particles and nanoparticulate matter (NPM) from mild steel, aluminum, and stainless steel welding fumes.*

Breathing-zone respirable and nanoparticle respiratory deposition (NRD) samplers on 61 participants - 38 welders, 12 factory and 11 office controls - with ICP-OES metals and TEM/EDS morphology. Mild-steel fumes (n=31) were dominated by Mn and Fe, stainless (n=3) by Cr and Ni, aluminium (n=4) by Al; across the six metals, the nanoparticulate fraction carried 48.3% of the respirable metal mass on average, with substantial Mn, Cr and Ni in NPM. The stainless and aluminium arms are too small for inference, which the authors acknowledge. Relevance: if half the respirable metal sits below ~300 nm, optical sensors - which lose efficiency below that size - will systematically undercount the toxicologically dominant fraction in welding environments.

doi: 10.1016/j.jhazmat.2026.143540

Liu 2026 Atmosphere*Energy-Indoor Air Quality-Health Nexus in Rural Residential Retrofits: A Field-Calibrated Coupled Simulation and Multi-Criteria Evaluation for the Central Plains Transitional Climate of China*

Solid-fuel brick-concrete rural homes in Anyang, Henan: 72-h paired winter and summer measurements of PM_{2.5}, PM₁₀, CO₂, formaldehyde and TVOC calibrate a two-way EnergyPlus-CONTAM model (indoor temperature NMBE -2.1%, CV(RMSE) 6.7%). An envelope-only retrofit cut thermal loads 52.2% but raised winter PM_{2.5} 25.8% and formaldehyde 32.4% via a 62.3% fall in infiltration; adding schedule-optimised natural ventilation kept 95.6% of savings while cutting the PM_{2.5} hazard quotient 20.3%. Calibration and out-of-sample validation are reported for temperature and CO₂, not for PM, so the PM results rest on unvalidated source and deposition terms; single author, one county. Relevance: tightening envelopes raises indoor PM from solid fuel - retrofit programmes need indoor sensors to catch it.

doi: 10.3390/atmos17100921

Lv et al. 2026 J Hazard Mater*Impact of biomass stove intervention on indoor air quality in rural homes of China: Attenuation of cooking contribution.*

Advanced biomass stoves were introduced in Xuanwei households and real-time PM_{2.5} monitors logged kitchen, living room, bedroom and yard. Kitchen daily mean PM_{2.5} fell 29.3%, 55.8% and 30.7% with direct-fired, recirculating and gasifier stoves. Peak analysis showed gasifier stoves cut kitchen cooking-peak height by 59.3% and peak area by 60.2%, cooking's contribution to kitchen PM_{2.5} fell ~30%, and indoor/outdoor ratios in living rooms and bedrooms approached 1. The abstract gives no household count, no control arm and no sensor calibration - a pre/post design confounded by season and behaviour. Relevance: peak-resolved metrics from time-resolved sensors capture intervention effects that daily means understate - a methodological point for any stove or purifier evaluation.

doi: 10.1016/j.jhazmat.2026.143732

Opbroek et al. 2026 Sci Total Environ*Health effects of indoor environmental quality (IEQ): A systematic review of indoor environments in high-income settings.*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 22 Sep with no abstract, and none was recoverable from any leg. A systematic review of IEQ health effects in high-income settings would cover indoor PM alongside thermal, acoustic, lighting and ventilation parameters, but the deposit does not say which exposures, how many studies, or whether any pooling was done, and the author list is not evidence about content either. Carried as a placeholder because indoor PM is the dominant fraction of personal exposure in high-income settings; added to the retry list and to be re-screened rather than summarised from the title.

doi: 10.1016/j.scitotenv.2026.182158

Perfilyeva et al. 2026 Genes (Basel)*Genetic Susceptibility to Impaired Lung Function in Underground Coal Miners from Kazakhstan.*

GWAS of six spirometric traits (FVC, FEV1, FEV1/FVC, MEF25/50/75) in 186 underground coal miners from the Karaganda basin, adjusted for age, mining exposure, BMI, smoking and population structure, with imputation. No variant reached genome-wide significance; recurrent suggestive signals at PRKCB, FANCE and UBTD1 across correlated traits. n=186 is grossly underpowered for GWAS, dust exposure is years-in-mining rather than measured, and there is no replication cohort. Relevance: low; underscores that occupational cohorts in Central Asia still lack any measured respirable-dust exposure.

doi: 10.3390/genes17091021

Pessoa Campos et al. 2026 J Toxicol Environ Health A*Salivary 8-hydroxy-2'-deoxyguanosine as a biomarker of oxidative DNA damage in urban outdoor workers exposed to traffic-related air pollution.*

Cross-sectional comparison of outdoor street vendors against indoor control workers, with workplace PM_{2.5} and noise logged by portable instruments and salivary 8-OHdG measured by ELISA. Exposed workers had higher salivary 8-OHdG; PM_{2.5} and BMI were independent predictors in multivariate models while income and education correlated inversely. What the record does not give is the whole quantitative content: no sample size, no effect estimate, no interval, and no city. Saliva sampled once cannot separate chronic burden from that morning's traffic, and the socioeconomic gradient is entangled with the exposure by construction. Carried because the design rests entirely on portable PM_{2.5} monitors at the workplace, and its credibility therefore rests on a calibration the abstract never describes.

doi: 10.1080/15287394.2026.2734428

Qian et al. 2026 Indoor Air*Impacts of Housing Retrofits on Indoor Environmental Quality and Energy Consumption: A Review*

A review of residential retrofit studies. Retrofits cut energy use by 10-84%; indoor temperature rises for winter-targeted and falls for summer-targeted measures, but effects on RH, CO₂ and pollutants including PM_{2.5} varied with ventilation, occupant behaviour and outdoor levels. The authors' main conclusion is critical of the evidence base: most studies lacked a causal framework, long observation, adequate samples, weather adjustment, matched controls and behaviour control. No pooled estimate and no search protocol in the abstract. Read together with the Anyang simulation in this issue, the gap is clear. Relevance: long-duration, matched-control indoor PM monitoring with low-cost sensors is the cheapest way to supply the causal designs this review says are missing.

doi: 10.1155/ina/3883472

SAA wetting agent 2026 Environ Geochem Health*High-efficiency 'SAA' wetting agent for coal mine dust control: from preparation and mechanism to field application.*

A three-component surfactant blend (sodium gluconate, AEO-3, AES) lowers surface tension and contact angle on coal, raises oxygen-containing surface functional groups, and agglomerates micron-scale dust - D50 rose 22.58% over untreated. In a purpose-built simulation rig the suppression rate was 71.50%, 2.17x conventional water mist, and the formulation was then taken underground on a real working face. The mechanism-to-field arc is the strength; the evaluation is the weakness. A suppression rate measured in the authors' own custom rig has no external calibration, the field test reports no control shift or personal sampling, and no respirable-fraction or silica-specific result is given, which is what determines pneumoconiosis risk - agglomerating dust into coarser particles raises D50 partly by removing mass from exactly the respirable size range the health endpoint cares about, so the health-relevant efficacy is almost certainly different from 71.5% and is not reported. Relevant to occupational sensing as a reminder that a mass-based sensor reads an agglomeration intervention as a larger improvement than the lung sees.

doi: 10.1007/s10653-026-03491-3

Taskin and Ozkal 2026 Indoor Air*Investigating Airborne Pollutant Dynamics in a Pathology Laboratory Using an Integrated High-Resolution Sensor Network and Operational-Context Framework.*

A university hospital pathology laboratory was logged at 1-min resolution for 42 days with a myRIO-based node (sensor-indicated HCHO, MOx eTVOC, eCO₂, PM, T, RH), aggregated to 123 eight-hour blocks. HCHO was higher in working than non-working blocks (median 313 vs 186 ppb) but the work-status coefficient fell from +0.430 to +0.019 ($p = 0.897$) once block position, temperature and humidity entered; 91.1% of blocks exceeded the 100 ppb ACGIH TLV regardless. Work days differed in their 15-min maxima (1,723 vs ~530-608 ppb). PM results are not reported in the abstract and the HCHO sensor is uncorrected against a reference. Relevance: the block-aggregation and confounding analysis is the right template for any occupational sensor deployment.

doi: 10.1155/ina/9352274

Wang et al. 2026 PLoS One*Research on dust control and ventilation parameter optimization in roadhead mining based on response surface methodology.*

ANSYS Fluent simulation of a fully mechanised heading face under long-forcing, short-exhaust ventilation, with a Box-Behnken design over exhaust-duct inlet angle, exhaust-to-forcing airflow ratio, exhaust inlet diameter and forcing outlet diameter. Inlet angle and exhaust diameter dominated the dust field; multi-objective optimisation cut dust mass concentration by 58% at the lower and 34% at the upper return-air-side corner, and a 1:4 scaled physical model reproduced the simulated field to within 5%. The validation is of the flow model, not of the dust reduction in an operating mine - no in-situ gravimetric comparison, no respirable size cut, no worker exposure. Relevance is direct nonetheless: corner accumulation zones are where fixed monitors are not, and a few nodes placed by this simulation would test it cheaply.

doi: 10.1371/journal.pone.0338763

Wang et al. 2026 J Hazard Mater*Aeration change from horizontal brush to fine bubble substantially reduces the emission of inhalable antibiotic resistome from wastewater to air.*

Seasonal metagenomics of PM_{2.5} and wastewater at one WWTP operating horizontal-brush and fine-bubble aeration under matched inflow and location. PM_{2.5}-borne ARGs showed higher occurrence, mobility and risk scores under brush aeration, peaking in winter; brush aeration transferred 74.5% of wastewater ARGs and 75.0% of pathogenic resistant bacteria to air versus 57.5% and 50.0% for fine bubble. Mobile ARGs were integrase-carried and mainly MLS-resistant; six pathogens including *K. pneumoniae* expressed mobility. One plant, and 'transfer percentages' are shared-gene fractions, not emission rates; no worker exposure or PM mass is reported. Relevance: low for sensing; PM_{2.5} is the carrier, but no optical sensor distinguishes bioaerosol.

doi: 10.1016/j.jhazmat.2026.143712

Wu et al. 2026 Int J Phytoremediation*Sphagnum palustre moss bags as sentinels to monitor indoor air PAHs in university libraries.*

Sphagnum palustre moss bags were exposed for 45 days in autumn (non-heating) and winter (heating) in study rooms, e-reading rooms, stacks and outdoors at three university libraries. Total PAHs were 487-1,050 ug/kg moss in autumn and 735-3,167 ug/kg in winter, dominated by phenanthrene, naphthalene, pyrene and fluoranthene; I/O ratios pointed to indoor sources in autumn and infiltration in winter, and diagnostic ratios to petrogenic then mixed sources. Moss uptake is not converted to an air concentration, gas and particle phases are not separated, and diagnostic PAH ratios are notoriously unreliable for apportionment. Relevance: low; a cheap integrating sampler, but no route to a $\mu\text{g m}^{-3}$ exposure metric.

doi: 10.1080/15226514.2026.2734570

Xing et al. 2026 Sci Bull (Beijing)*Household air pollution associated with different heating approaches in rural Northern China.*

Year-round indoor PM_{2.5} monitoring in 948 rural North China Plain households under the Clean Heating Initiative, with exposure modelling and cost-benefit analysis. Annual indoor PM_{2.5} was $109 \pm 207 \mu\text{g m}^{-3}$ and above the WHO guideline for >90% of the year. Pelletised-biofuel households had 20-30% lower indoor PM_{2.5} and ~20% lower population exposure than traditional solid-fuel users, and an estimated 8% lower attributable premature mortality; benefit-cost ratios were 4.39-8.64 for pellets versus 0.76-2.20 for full electrification. Fuel type was not randomised, the SD twice the mean signals heavy episodic tails, and the monitor type and its correction are not given in the abstract. Relevance: at $n=948$ year-round this is the scale at which household LCS networks become the exposure instrument of record.

doi: 10.1016/j.scib.2026.09.018

ZCND aerogel filter 2026 J Colloid Interface Sci

Biomimetic dual-network nanofibrous aerogels with robust interfacial assembly for multifunctional air filtration.

A zein/curcumin nanofibre plus down-fibre aerogel reached 99.51%, 99.74% and 99.93% efficiency for PM₁, PM_{2.5} and PM₁₀ at 27.39 Pa pressure drop, with 335.85 g/m² dust-holding capacity, and - the result that distinguishes it - retained 98.67% PM₁ efficiency after 12 h at 90% relative humidity. It also adsorbs ammonia (104.95 cm³/g), shows visual NH₃ sensing, is antibacterial, and biodegrades in soil burial faster than commercial media. The target is livestock housing, where PM, NH₃ and bioaerosol co-occur and humidity is chronically high, so the multifunctionality is matched to a real setting rather than assembled for novelty. Standard caveats for this literature apply and are not minor: bench aerosol challenge under controlled conditions, no barn deployment, 12 h is not a service life, and nothing on mechanical durability under pulse cleaning or cost at scale. Noted here mainly because humidity-resistant PM₁ capture is the same physical problem, inverted, that defeats nephelometric low-cost sensors in barns - both fail when water condenses on the particle, one by fouling and one by over-reading.

doi: [10.1016/j.jcis.2026.141640](https://doi.org/10.1016/j.jcis.2026.141640)

Zigabe et al. 2026 Environ Pollut

Determinants of early-life household air pollution exposure (HAP): carbon monoxide (CO) and PM_{2.5} concentrations in Bukavu, eastern Democratic Republic of Congo (DRC).

MUTOTO pregnancy-cohort sub-study, 405 households, March 2025 to January 2026. Mean 24-h household PM_{2.5} was 53.8 µg m⁻³ (SD 34.2), maximum household mean 244.3; personal CO averaged 5.3 ppm (SD 8.7), maximum 65.1. Biomass fuel and cooking frequency drove both; 96% of kitchens had no chimney, and biomass use rose from 56% to 87% across CO exposure groups. The limitation is metrological: PurpleAir optical counts in dense biomass smoke are not reported as corrected, and uncorrected PMS sensors typically overestimate in smoke while underestimating coarse mass - the reported means could be biased in either direction without a gravimetric subsample. Single 24-h samples also cannot capture seasonal variability. For this watch it is the model deployment: low-cost PM plus CO loggers at cohort scale in a setting with no reference network.

doi: [10.1016/j.envpol.2026.129170](https://doi.org/10.1016/j.envpol.2026.129170)

Burden, policy & mitigation**17 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (17 OF 17 RECORDS IN THIS SUBTOPIC)

Aerosol Air Qual Res 2026 Aerosol Air Qual Res

Cross-national Evaluation of STEM-Based Indoor Air Quality (IAQ) Education for Sustainable Development.

METADATA ONLY. A cross-national evaluation of STEM-delivered indoor air quality education; no sample, instrument or outcome is recoverable from the deposit. Logged at tier C because low-cost sensor deployments in schools are the standard vehicle for exactly this kind of programme, and whether the measurement component changes behaviour or only knowledge is an open question this watch has no evidence on either way. Nothing is claimed here. Deferred re-query.

doi: [10.1007/s44408-026-00168-5](https://doi.org/10.1007/s44408-026-00168-5)

Ann Glob Health 2026 Ann Glob Health

Disclosure without accountability: Air pollution reporting by major South African industrial companies.

METADATA ONLY. Europe PMC held no abstract at screening. The title states the finding as a thesis - corporate air-pollution reporting that discloses without enabling accountability - which in the self-reported-emissions literature usually means figures are published at an aggregation or cadence that cannot be checked against any independent measurement. South Africa's Highveld is one of the few heavily industrialised regions where that gap is measurable, because ambient monitoring exists alongside the disclosures. This is precisely the gap independent low-cost networks are deployed to close, so the record is carried despite being unreadable. Deferred re-query.

doi: [10.5334/aogh.5437](https://doi.org/10.5334/aogh.5437)

Bouchriti et al. 2026 Environ Monit Assess

Spatial analysis of the burden of lung cancer attributable to long-term PM(2.5) exposure in Morocco: a national assessment using the WHO AirQ+ model.

WHO AirQ+ applied to 2019 city-specific PM_{2.5} for 46 Moroccan urban and rural areas. PM_{2.5} ranged from 7.1 µg m⁻³ in coastal El Jadida (RR 1.02; attributable proportion 2.35%) to 28.4 µg m⁻³ in Errachidia (RR 1.30, 95% CI 1.17-1.42; AP 23.29%); Casablanca's moderate RR of 1.06 still yielded 24 attributable cases. The RRs are the model's concentration-response function evaluated at each city's mean, not observed associations, and a single year of concentration data - sources not specified, and Morocco's monitoring is sparse - drives everything. Southern desert areas likely carry a dust-dominated PM_{2.5} whose lung-cancer potency differs from the combustion-derived PM behind the CRF. Relevance: the uncertainty sits almost entirely in the exposure input, where a sensor network would change the answer most.

doi: [10.1007/s10661-026-15942-6](https://doi.org/10.1007/s10661-026-15942-6)

Ding et al. 2026 Environ Sci Technol

Which Roads Should Cities Control First? A Health-Risk Framework for Mobile-Source Air Pollution.

METADATA ONLY. Metadata-only record: ACS deposited title, authors (Ding, Zhang, Li, Chen, Hu) and DOI on 22 Sep without an abstract, and none was recoverable from Crossref, Europe PMC or PubMed. The title promises a prioritisation framework that ranks road segments by population health risk rather than by emissions alone - squarely the question a near-road sensor network is built to answer. Nothing about pollutants, exposure model, city or validation can be stated from the deposit, and inferring a method from a title is not something this watch does. Placed on the retry list; re-screen on its merits when the abstract appears, and check whether the risk weights use measured or modelled concentrations.

doi: [10.1021/acs.est.6c12287](https://doi.org/10.1021/acs.est.6c12287)

Dong et al. 2026 Nat Hum Behav

Impacts of PM_{2.5} air pollution on high-skilled worker productivity in China.

Individual scholars' publication counts linked to annual PM_{2.5} at their city, with an instrumental-variable design. Each 1 µg m⁻³ rise in annual PM_{2.5} reduced publications by 0.23% (95% CI -0.28% to -0.17%); scaled nationally, a 1% PM_{2.5} rise costs about 1,053 papers. Effects were larger in heavy-pollution years and among younger, more productive scholars; mechanism analyses point to fewer working hours, more hospital visits and weaker collaboration. The inference rests on the exclusion restriction, which the abstract does not name - common instruments (thermal inversions, upwind emissions) can also shift temperature and mobility. Annual city means are coarse against the indoor office environment where this work actually happens. Relevance: a cognitive-labour damage function in a high-exposure setting, and a case for workplace indoor monitoring rather than ambient assignment.

doi: [10.1038/s41562-026-02576-4](https://doi.org/10.1038/s41562-026-02576-4)

Fregni et al. 2026 J Am Heart Assoc

Stroke Burden Attributable to Risk Factors in the Americas, 1990 to 2021: A Temporal Trends Analysis From the GBD Study 2021.

GBD 2021 stroke deaths and DALYs attributable to 23 modifiable risk factors across 39 countries and territories of the Americas, with jointpoint trends 1990-2021. The 23 factors together accounted for 78.8% of stroke deaths; high systolic pressure led (54.8% of deaths). Household air pollution from solid fuels showed the steepest decline of any factor (deaths AAPC -5.09%, 95% CI -5.17 to -5.02; DALYs -4.92%). Ambient PM_{2.5} is not reported in the abstract. GBD attributable fractions inherit modelled exposure and meta-analytic relative risks; this is a re-cut of existing estimates, not new evidence, and the narrow CIs are regression CIs on modelled series. Relevance: low; background showing HAP falling fastest while ambient PM is not separated.

doi: 10.1161/jaha.125.047114

Kan et al. 2026 Environ Pollut

From Emission-Dominated Mitigation to Meteorology-Modulated Secondary Pollution: Decadal Insights from Beijing's Short-Term Interventions.

A natural experiment repeated a decade apart: aerosol composition and VOCs during the 2025 Beijing parade controls compared with the 2015 parade. Traffic controls in 2025 cut NO₂ by 41.6% and aromatic VOCs by 14.8%, yet PM_{2.5} rose 20.9%; in 2015 comparable controls took PM_{2.5} from 55.7 to 27.0 µg m⁻³. ML attribution assigned ~90% of the 2015 drop to emission controls and the 2025 rise, led by sulfate and SOA, to warmer, more humid conditions, with biogenic, oxygenated and VCP emissions up. Two short control episodes are anecdotes with good instruments; the ML deweathering is only as good as its training-period representativeness. Relevance: short-term PM interventions evaluated by any network now need co-measured RH and composition to be interpretable.

doi: 10.1016/j.envpol.2026.129240

Kwon & Kim 2026 BMC Public Health

Mortality and morbidity consequences of particulate matter in South Korea: evidence from transboundary air pollution.

Mortality 2015-2022, morbidity 2016-2019, national. The instrument is the useful part: local PM_{2.5} is endogenous to local economic activity, so the authors instrument it with transboundary transport - wind direction crossed with upwind Chinese monitors - which shifts Korean PM_{2.5} without plausibly acting on Korean health through any other channel. A 1 µg m⁻³ increase raises 65+ mortality by 0.17% and respiratory hospitalisations by 5.1%; acute upper-respiratory visits rise 1.2%. The observed 5.58 µg m⁻³ national decline over 2015-2022 is valued at USD 14.14 billion on avoided life-years alone. Two weaknesses: the exclusion restriction fails if the same synoptic conditions that import Chinese PM independently affect health (cold-air outbreaks, humidity), which the abstract does not address; and the monetisation counts only life-years, so it is a floor, not an estimate. No confidence intervals are reported for the elasticities, so none are plotted. The design is worth borrowing: for any dense sensor network sited in a downwind receptor region, upwind transport is a ready-made instrument for exposure.

doi: 10.1186/s12889-026-29307-8

Li et al. 2026 Sci Bull (Beijing)

Clean air paradox: exacerbating urban-rural inequality in PM_{2.5} exposure.

Income-stratified urban and rural PM_{2.5} exposure across mainland China, 2005-2019, combining modelled exposure with field-measured source-specific toxicity profiles. Despite the Clean Air Actions, the urban-rural exposure gap widened by 55.6%; rural residents had 3.2-fold higher exposure in 2019, rising to 15.0-fold once source-specific toxicity was applied, >85.2% of it from residential burning. A 20% residential-emission cut achieved 88.9% inequality mitigation in developed regions, with smaller gains from a 60% cut in low-income areas. The 15-fold figure rests entirely on the toxicity weighting, whose metric (oxidative potential or other) the abstract does not name. Relevance: rural indoor burning is where ambient networks are blind and where household sensors have the largest marginal value.

doi: 10.1016/j.scib.2026.09.015

Li et al. 2026b Environ Res

Contributions of local household fuel burning and regional transport to ambient PM_{2.5} in the Qinghai-Tibet Plateau.

A residential emission inventory for the Qinghai-Tibet Plateau rebuilt from survey household-energy data for 1960-2019 and run through GEOS-Chem with a normalised marginal attribution. Residential emissions rose to about 1990 then declined with partial clean-energy transition, dominated by inefficient animal-dung combustion except for SO₂, which tracked coal. Residential sources are over 80% of primary PM_{2.5} emissions on the plateau, yet all local sources explain only 17% of ambient PM_{2.5} in Tibet against 46% in Qinghai, with South Asian transport supplying 65% of Tibet's annual ambient PM_{2.5}. Survey-derived inventories over six decades carry large unstated uncertainty and the attribution inherits the model's transport errors. The policy implication is sharp and testable with a sparse plateau sensor network: cleaning local stoves cannot fix Tibetan ambient PM_{2.5}.

doi: 10.1016/j.envres.2026.125734

Mundu et al. 2026 J Air Waste Manag Assoc

Fossil fuel use, urban air pollution and health in rapidly growing African cities: Implications for sustainable and low-emission urban development.

PRISMA-reported critical review and policy synthesis on fossil-fuel sources, urban air quality, health impacts and governance in fast-growing African cities. Diesel road transport, backup generators and household energy emerge as the dominant PM, NO_x, black carbon and VOC sources; observed levels often exceed WHO guidelines; monitoring networks are sparse and unevenly distributed. No pooled estimates, no quality appraisal of the included studies is described in the abstract, and the recommendations (electrification, clean cooking, mini-grids) are generic. Its value is as a map, not as evidence. The monitoring-gap finding is the direct link to this watch: generator-dominated diurnal profiles and diesel BC are two conditions where uncorrected optical sensors are least reliable, so an African LCS rollout needs co-located BC or reference anchors from the start.

doi: 10.1080/10962247.2026.2737847

N fertilizer NH₃-PM_{2.5} 2026 Atmos Res

Integrating nitrogen fertilizer management for mitigating agricultural ammonia emissions and PM_{2.5} pollution.

METADATA ONLY. Metadata-only record; no abstract deposited or recoverable from any leg. Agricultural ammonia is the control lever with the best cost-per-microgram in most secondary-inorganic-dominated airsheds and the one least often regulated, so a fertilizer-management framing is squarely in scope for the burden and mitigation track. Whether the paper is a modelling assessment, a field trial or a policy review is not determinable from the deposit. Retry list. Note it sits alongside the Athens new-particle-formation record in this same issue - two independent ammonia-centred papers deposited the same day, both illegible, which is a coverage gap rather than a coincidence worth reading into.

doi: 10.1016/j.atmosres.2026.109343

PPS antifouling filter 2026 Environ Sci Technol*Robust and Self-Cleaning Fluorine-Free Superhydrophobic PPS Filters for High-Humidity Flue Gas Filtration.*

Light-induced covalent grafting of a triple-cross-linked low-surface-energy network onto polyphenylene sulfide fibre membranes produces a fluorine-free antifouling bag-filter medium. Against the unmodified control it extends mean pulse-jet cleaning cycle to 550.84 s, lowers mean residual pressure drop to 280.69 Pa and raises quality factor to 0.044 /Pa under simulated humid dynamic filtration. The fluorine-free part is the policy-relevant part - PTFE-based antifouling treatments are the incumbent and are under PFAS restriction, so a non-fluorinated substitute that survives wet dust addresses a regulatory constraint on stationary-source PM control rather than just an engineering one. Bench-scale, simulated flue gas, no capture-efficiency-versus-size curve reported and no coating-lifetime data beyond the test campaign, so durability - the property the whole design targets - is demonstrated over hours rather than the months a baghouse runs. Tier C and upstream of this watch's usual scope; carried under mitigation because primary PM emission control determines the ambient field every downstream sensor measures.

doi: 10.1021/acs.est.6c08122

Simoes et al. 2026 J Xenobiot*Air Pollution, Temperature, and Demographic Vulnerability as Determinants of Cause-Specific Mortality in Portugal.*

Nationwide monthly data for Portugal, 2010-2022, with GEE models of malignant, circulatory and respiratory mortality against NO₂, PM_{2.5} and temperature, exposures coded as concentration above WHO 2021 guideline values. The signal is NO₂, not PM: respiratory mortality RR 1.014 (95% CI 1.009-1.019) per 1 µg m⁻³ NO₂ above guideline, circulatory 1.005; PM_{2.5} appears mainly as a modifier attenuating the NO₂ association with tumour mortality, and pollutant findings were not robust in sensitivity analyses. A reported RR of 142.6 for males 65+ is a predicted risk ratio against a reference group, not an exposure effect. Monthly national aggregation is ecological and thresholding at guidelines discards most variance. Relevance: minimal; logged for the PM_{2.5}-modifies-NO₂ finding, which would need personal-scale data to test.

doi: 10.3390/jox16050172

Turkate et al. 2026 Am J Public Health*Implementing Community-Centered Air Quality Law: Success Stories From California's Assembly Bill 617.*

Mixed-methods comparison of community emissions reduction programmes (CERPs) in nine California AB 617 communities: over 800 strategies coded into 25 groups and rated on capacity for health protection, adoption pathway and ease of implementation, with funding expenditure as the progress proxy. Success tracked creative programme development, interagency coordination and community capacity. Expenditure is a process measure, not an air-quality or health outcome, and 'success stories' framing invites selection. Relevance: AB 617 is the largest statutory deployment of community air monitoring in the US, yet the abstract evaluates strategies without any measured concentration change - the monitoring data that could close that loop exist and are not used here.

doi: 10.2105/ajph.2026.308696

Wan et al. 2026 Environ Sci Technol*Three-Phase Transition in China's Future Air Quality under Ambitious Mitigation Pathways Revealed by Large-Ensemble AI Projections.*

A hybrid deep-learning surrogate (hyDL) of a chemical transport model cut compute by >99% while reproducing meteorology and emission sensitivities ($R > 0.9$), enabling daily ~50 km projections for China 2015-2060 under six emission pathways, each driven by 7-11 climate models. Megacities showed an exponential O₃-PM_{2.5} relationship implying three phases under ambitious mitigation: PM_{2.5} falls while O₃ persists (current), co-improvement, then O₃ gains with marginal PM_{2.5} returns. With population ageing, attributable mortality stays flat to ~2035 even on the cleanest pathway. A surrogate trained on the historical emission space is extrapolating when pathways leave it, and $R > 0.9$ on sensitivities is not validation of 2060 chemistry. Relevance: the surrogate approach is how sensor-network data could be assimilated at useful cost.

doi: 10.1021/acs.est.6c04494

Yang et al. 2026 Thorax*PM_{2.5}, O₃ and survival gains from China's clean air actions among patients with lung cancer: a 20-year registry-based cohort study in Beijing.*

172,845 lung cancer patients diagnosed 2003-2022 in the Beijing Cancer Registry, with three-year pre-diagnosis exposure from ChinaHighAir-Pollutants. Per 10 µg m⁻³, PM_{2.5} gave HR 1.111 (95% CI 1.083-1.140) and O₃ HR 1.117 (1.070-1.166) for all-cause mortality, with a significant multiplicative interaction concentrated above roughly 105 µg m⁻³ of O₃. Counterfactual simulation against 2013 levels attributed 13,229 avoided deaths over 2016-2022 and a 7.8-month gain in five-year extended expected lifespan - of which rising O₃ and the interaction offset about 40%. Exposure is a reanalysis product at residence at diagnosis with no mobility and no treatment covariates, and the survival gains are simulated, not observed. The interaction is the operational point: a PM_{2.5}-only network cannot rank policy benefit in an ozone-rising regime.

doi: 10.1136/thorax-2026-225199

Other clinical endpoints**8 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Ali et al. 2026 Toxics*The Impact of Polycyclic Aromatic Hydrocarbons Exposure from Seasonal Biomass Burning on Oxidative Stress and Genotoxicity Biomarkers in Patients with Diabetes and Hypertension.*

152 adults in Chiang Mai (88 healthy, 35 diabetes, 15 hypertension, 14 both) sampled before (December) and after (May) the burning season. Urinary 1-OHP rose from 0.21 +/- 0.50 to 0.87 +/- 1.25 µmol/mol creatinine and serum 8-OHdG from 51.04 +/- 24.90 to 66.08 +/- 27.82 pg/mL (both p<0.001), with the rise significant in diabetic and hypertensive groups; MDA did not change overall (6.09 to 6.00 nmol/mL). No ambient PM was measured, so season is the exposure and every other seasonal change is confounded; subgroups of 14-35 are small. Relevance: a reusable design - adding low-cost PM_{2.5} at participants' homes would turn the pre/post contrast into a dose-response.

doi: 10.3390/toxics14090771

Hong et al. 2026 Front Public Health*Associations of temperature, particulate matter, and greenness with human papillomavirus positivity at screening: an analysis of Guangdong Province (2016-2022) using the treed distributed lag nonlinear model.*

Screening records for 1,287,412 women in Guangdong, 2016-2022, analysed with treed distributed-lag nonlinear models against biweekly temperature, PM_{2.5} and NDVI. High PM_{2.5} (97.5th percentile, 38.0 µg m⁻³ vs median 20.3) was associated with higher odds of HPV positivity over lag weeks 1-16 (cumulative OR 1.172, 95% CrI 1.051-1.283); low temperature and low NDVI also carried associations. The authors report that temporal sensitivity analyses attenuated the PM_{2.5} association. HPV positivity reflects acquisition months to years earlier and screening attendance is seasonal, so a weeks-scale lag structure is biologically implausible and likely confounded by attendance timing. Relevance: low; included as a caution on DLNM results with outcomes whose latency does not match the lag window.

doi: 10.3389/fpubh.2026.1842611

Huang et al. 2026 Int J Epidemiol*Short-term health impacts of air pollution in Canada: a nonlinear multi-pollutant Bayesian case-crossover framework.*

Cause-specific daily mortality 2001-2015 and hospitalisation 2001-2018 across 20 Census Divisions covering over half of Canada's population. A 10-unit contrast in the combined PM_{2.5}+NO₂ metric gave +1.5% non-accidental mortality (95% CrI 0.3 to 2.2), +0.6% hospitalisation (0.0 to 1.3) and +2.7% senior respiratory hospitalisation (0.9 to 4.5). Several PM_{2.5} and NO₂ curves were nonlinear; O₃ was near-linear. The methodological gain is real: exposure uncertainty from imputation is propagated rather than ignored, and CD-level curves are partially pooled. The threat is interpretive: a summed PM_{2.5}+NO₂ index has no single-pollutant counterpart, so the coefficient cannot be dropped into a PM_{2.5}-only burden model, and the CrIs for cause- and age-specific strata are wide. For sensing: imputed monitor gaps are treated as an exposure-error source, which is exactly where dense low-cost networks would shrink the posterior.

doi: 10.1093/ije/dyag194

Huang et al. 2026b Front Pediatr*The role and management of dietary and environmental factors in pediatric atopic dermatitis.*

Stage 1: 535 paediatric atopic-dermatitis cases diagnosed 2022-2024 and 535 age- and sex-matched controls, conditional logistic regression. PM_{2.5} above 35 µg m⁻³ carried an adjusted OR of 2.48 (95% CI 1.80 to 3.50), alongside early milk/egg sensitisation (3.18), winter indoor humidity below 40% (1.92) and S. aureus colonisation above 50% (4.08). Stage 2 was a 66-child non-randomised intervention comparison on diet and home care, with no air-quality component. The PM estimate is the weakest element: the exposure source, spatial resolution and averaging window are not given, a single dichotomy at 35 µg m⁻³ discards the gradient, and matched hospital controls invite selection on healthcare use. Treat the OR as hypothesis-grade. Relevance to sensing is indirect - indoor humidity and PM_{2.5} are measured by the same low-cost units, so a home-sensor design could separate them properly.

doi: 10.3389/fped.2026.1862285

Kelishadi et al. 2026 Int J Prev Med*The Effect of Synbiotic Supplementation on Malondialdehyde Levels in Adolescents Exposed to Air Pollution: A Randomized, Single-Blind, Controlled Clinical Trial.*

Seventy healthy 10-18-year-olds randomised, single-blind, to synbiotics (n = 38) or omega-3 (n = 32) for 8 weeks; outcome serum malondialdehyde. MDA fell within the synbiotic arm (148.8 to 51.5), but the between-arm difference was null (P = 0.75); only after covariate adjustment did synbiotic use associate with MDA reduction. That is the result that matters, and it is null on the pre-specified contrast. The comparator is an active antioxidant, so the trial cannot test synbiotics against nothing; there is no placebo, no measured exposure and the MDA standard deviations exceed the post-treatment mean. It cannot inform a PM dose-response. Carried because nutritional mitigation of PM oxidative stress keeps recurring in the trial registry; an exposure-stratified design using personal sensors is what this line of work lacks.

doi: 10.4103/ijpvm.ijpvm_246_25

King 2026 Br J Anaesth*Outdoor air pollution as a potentially modifiable perioperative risk factor.*

METADATA ONLY. Metadata-only record: PubMed/Elsevier deposited title, authors and DOI on 25 Sep with no abstract; Crossref, PubMed efetch and the publisher landing page returned nothing, and Europe PMC was down (HTTP 503) for the whole run. The title indicates a single-author piece framing ambient air pollution as a modifiable perioperative risk factor. Likely editorial; low priority unless it reports primary data. Nothing about exposure metric, sample size, instrument or estimates can be stated from the deposit, and this watch does not infer a method from a title. On the retry list; re-screen on its merits when the abstract appears.

doi: 10.1016/j.bja.2026.08.047

Leonforte et al. 2026 Toxics*Environmental Contaminants and Osteoporosis-Related Outcomes: A Systematic Review and Meta-Analysis.*

PRISMA review (PubMed, Scopus, WoS; 2016-Apr 2026; PROSPERO CRD420261460078) of 48 studies and 4,510,868 participants on environmental contaminants and BMD/osteoporosis, REML random-effects pooling where ≥3 studies existed. Air pollutants were the most consistent class. PM_{2.5} was associated with osteoporosis (OR 1.24, 95% CI 1.02-1.51) and PM₁₀ more weakly (OR 1.17, 0.96-1.42). The authors report extreme heterogeneity and sensitivity to influential studies; the abstract gives no exposure increment, so the ORs pool incommensurate contrasts. Relevance: low for sensing; a musculoskeletal endpoint that is starting to accumulate PM evidence.

doi: 10.3390/toxics14090790

Zhu et al. 2026 Environ Res*Long-term exposure to particulate matter and survival among patients with gastric cancer: A hospital-based cohort study.*

4,441 stage I-III gastric cancer patients resected 2012-2022, median follow-up 3.83 years, 1,318 deaths (919 gastric-cancer-specific). Per 1 SD increase, all-cause mortality HRs were 1.15 (95% CI 1.06-1.25) for PM_{2.5}, 1.14 (1.06-1.24) for PM₁₀ and 1.12 (1.03-1.21) for the coarse fraction; gastric-cancer-specific subdistribution HRs were larger - 1.31 (1.20-1.44), 1.28 (1.17-1.40) and 1.17 (1.07-1.28) - with Fine-Gray models treating non-gastric deaths as competing events, which is the right handling and is often skipped in this literature. Counterfactual PAFs under lower historical exposure are reported. The inference is threatened most by exposure assignment: CHAP is a 1 km satellite-and-model reanalysis assigned at the residential address, so measurement error is spatially structured, and pre-index exposure is a proxy for decades of residence and occupation that the 36-month window does not capture. Reverse causation is limited by the pre-surgery window but not eliminated - sicker patients may relocate. That the cause-specific effect exceeds the all-cause effect is internally coherent but unusual enough to want replication outside Zhejiang.

doi: 10.1016/j.envres.2026.125689

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Curzio et al.	The Exposome-Brain Axis: A Scoping Review of Biological Biomarkers in Environmental Neurotoxicity, Neuroinflammation, and Neurodegeneration.	Toxics	Scoping review	Ambient PM _{2.5} among other exposures (as reported by included studies)	Global / multi-region (review)

First author	Focus	Journal	Design	Metric	Region
Falkenstein et al.	Investigating effects of day-to-day variations in environmental exposure on the human brain: study protocol for the Day2Day Environment project.	<i>BMC Neurosci</i>	Study protocol	Wearable air-quality, noise and light sensors with GPS tracking over 25 sessions	Not stated (protocol)
Hong Kong momentary stress	An Explainable Multi-Modal Deep Learning Framework for Investigating the Influences of Dynamic Environmental Exposure on Momentary Stress.	<i>Environ Sci Technol</i>	Ecological momentary assessment with mobile sensing, multi-modal deep learning	Min-level mobile sensor PM _{2.5} alongside noise, temperature, humidity, greenery, street-view and remote-sensing imagery	Hong Kong
Huang et al.	Environmental Pollution, Biomarkers, and Dementia: An Evidence Review.	<i>Int J Mol Sci</i>	Narrative review	Long-term PM _{2.5} , PM ₁₀ , NO _x /NO ₂ , BC, O ₃ (as reported by included studies)	Global / multi-region (review)
India-Aldana et al.	The chemical exposome during per-pregnancy, maternal sleep outcomes post-pregnancy, and the potential mediating role of body mass index.	<i>Environ Pollut</i>	Prospective cohort (mixtures)	Estimated air-pollutant mixture during pregnancy within a 52-marker chemical exposome	Mexico City, Mexico (PROGRESS cohort)
Oh et al.	Association Between Long-Term Exposure to Air Pollution and Parkinson's Disease: Administrative Cohort of Older Adults in South Korea.	<i>Mov Disord</i>	Nationwide cohort	District-level annual PM _{2.5} , NO ₂ and O ₃ from CMAQ, 1-year moving average, time-varying	South Korea
Rodulfo-Cardenas et al.	APOE genotype shapes persistent behavioral and molecular alterations following early postnatal particulate matter exposure in humanized apoE3 and apoE4 mice.	<i>Neurotoxicology</i>	Animal model	NIST SRM 2787 fine PM, whole-body inhalation 1 h/day PND4-14, nominal max 3789 µg/m ³ (not measured)	Laboratory (humanized apoE mice)
Zou et al.	Association of preconception to early childhood residential greenness with child neurodevelopmental outcomes and the mediating role of air pollution: a prospective cohort study.	<i>Environ Res</i>	Prospective birth cohort	PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ as single-pollutant mediators of NDVI	China (Jiangsu)
Cardiovascular & metabolic					
Aysin et al.	Effects of Traffic-Related Diesel Exhaust Exposure and Moderate Exercise on Obesity, Metabolic Dysfunction, and Inflammatory Responses in Rats.	<i>Metabolites</i>	Animal model	Diesel exhaust, mean 300 µg/m ³ PM _{2.5} , 2 or 4 h/day for 8 weeks	Laboratory (Wistar rats)
Bikdeli et al.	Coated or chewable Aspirin and a hybrid air pollution mitigation strategy in patients with atherosclerotic disease: Rationale and Design of the COATED-AIR Randomized Trial.	<i>Eur Heart J Cardiovasc Pharmacother</i>	Trial / intervention	Real-time ambient air-quality alerts driving behavioural and KN95 responses	Not stated (design paper; sites not in abstract)
Campagna et al.	Tracking biochemical indicators of air pollution exposure in acute cardiorespiratory emergencies: a scoping review.	<i>Front Public Health</i>	Scoping review	Short-term ambient PM _{2.5} , PM ₁₀ , NO ₂ (as reported by included studies)	Global / multi-region (review)
Campailla et al.	Ambient particulate matter exposure and acute type A aortic dissection: A retrospective time-stratified case-crossover study in northern Italy.	<i>Int J Cardiol Cardiovasc Risk Prev</i>	Case-crossover	Daily mean PM _{2.5} and PM ₁₀ from one urban-background station, same-day and 30-day windows	Italy (Piacenza)
Cui et al.	Differential Risks of Incident Heart Failure Subtypes Associated with PM _{2.5} Chemical Constituents in China between 2006 and 2021.	<i>Environ Sci Technol</i>	Prospective cohort	Long-term residential PM _{2.5} constituents (BC, OM, NH ₄ ⁺ , others)	China (Kailuan, Tangshan)
Dzhambov et al.	Residential exposures associated with adiposity and proinflammatory markers in obese adults with metabolic syndrome: a cross-sectional study	<i>Arh Hig Rada Toksikol</i>	Observational - cross-sectional	Modelled residential NO ₂ , PM _{2.5} , PM ₁₀ ; traffic noise; solid-fuel use; greenness	Bulgaria

First author	Focus	Journal	Design	Metric	Region
Hua et al.	Heterogeneous Risks of Chemical Components and Sources in Size-Resolved Particulate Matter on Ischemic and Hemorrhagic Stroke.	<i>Environ Health (Wash)</i>	Time-series	41 chemical components and 6 emission sources across 9 aerodynamic size fractions below 10 μm	China
Jeong et al.	Joint effects of air pollution and temperature on glycemia: An exposome analysis in European cohorts.	<i>Environ Int</i>	Metadata only (no abstract)	Metadata only - air pollution and temperature jointly, metrics not recoverable	Europe (cohorts not recoverable from metadata)
Lin et al.	Joint association of PM _{2.5} and AIP with advanced Cardiovascular-Kidney-Metabolic syndrome in middle-aged and older Chinese adults: a national cohort study.	<i>Research Square (preprint)</i>	Cross-sectional survey	Satellite-derived residential PM _{2.5} split at the median	China (national, CHARLS)
Pellegrino et al.	A possible mediating role of SASP in the association between environmental air pollution and incident atherosclerotic cardiovascular disease	<i>Sci Total Environ</i>	Metadata only (no abstract)	Metadata only - air pollution, SASP mediation, incident ASCVD	Not recoverable from metadata
Sun et al.	When air gets heavy: joint PM(2.5) and ambient humidity exposure, frailty, and stroke.	<i>Front Public Health</i>	Prospective cohort (CHARLS)	Annual PM _{2.5} and relative humidity at residence, dichotomised at median	China (nation-wide)
Wang et al.	Long-term exposure to fine particulate matter constituents and incident stroke risk among middle-aged and older adults in China: a cohort study.	<i>Front Public Health</i>	Prospective cohort (CHARLS)	TAP-dataset PM _{2.5} and five constituents (SO ₄ , NO ₃ , NH ₄ , OM, BC) at residence	China (nation-wide)
Wang et al.	Long-Term Air Pollution Exposure, Cardiovascular Disease, and Cardiometabolic Risk Factors in Middle-Aged and Older Chinese Adults: A Prospective Cohort Study.	<i>Bioengineering (Basel)</i>	Prospective cohort (CHARLS)	CHAP long-term PM _{2.5} , PM ₁₀ , NO ₂ , O ₃ at residence	China (CHARLS, national)
Yang et al.	Mitochondrial DNA methylation as a bridge: Linking indoor air purification to alleviated cardiovascular impairments in young adults.	<i>Environ Res</i>	Randomised crossover chamber	Personal PM _{2.5} under true vs sham indoor air purification (47.45% indoor reduction)	Not stated (young-adult panel)
Zhang et al.	Air Pollution Exposure Induces Vascular Endothelial Cell Dysfunction via Elevated Circulating F2: Multiomics Insights from an Ex Vivo Biosensor Assay.	<i>Environ Health (Wash)</i>	Panel study	Ambient PM _{2.5} , PM ₁₀ , SO ₂ , NO ₂ and CO contrast between two cities, no personal sampling	China (Qingdao to Shijiazhuang)
Reproductive & developmental					
Eick et al.	Prenatal exposure to aromatic amines and associations with infant cognition in the ECHO.CA.IL cohorts	<i>J Expo Sci Environ Epidemiol</i>	Prospective birth cohort (mixtures)	Second-trimester urinary aniline, o/m-toluidine, 4-chloroaniline, o-anisidine, 4,4'-methylenedianiline, cotinine	Illinois and California, USA
Hu et al.	Real-world PM _{2.5} exposure triggers early testicular epigenetic responses in the absence of significant changes in measured reproductive endpoints.	<i>Reprod Toxicol</i>	Murine inhalation exposure	Real-world ambient PM _{2.5} , whole-body inhalation, 4, 8 and 12 weeks	China (Hangzhou)
Li et al.	Cohort profile: the Shenzhen Birth Cohort (SZBC) study.	<i>BMJ Open</i>	Prospective birth cohort	Prenatal and postnatal PM _{2.5} with component resolution (nitrate singled out); model not stated	China (Shenzhen)
Tamire et al.	Prenatal fine particulate matter (PM _{2.5}) exposure differentially impairs fetal skeletal and soft-tissue growth: a longitudinal study in rural Butajira, Ethiopia.	<i>J Expo Sci Environ Epidemiol</i>	Community-based longitudinal cohort, linear mixed-effects	Repeated 24-h personal PM _{2.5} , Atmotube Pro wearable monitor, at 20, 30 and 36 weeks gestation	Rural Butajira, Ethiopia
Thirumalazhagan et al.	A Comprehensive Meta-Analysis on the Impact of airborne Particulate Matter on Trophoblast Cell Function In Vitro.	<i>Reprod Toxicol</i>	Systematic review + random-effects meta-analysis	PM _{2.5} extracts applied to trophoblast cell lines across 12 in vitro studies	Laboratory (pooled in vitro studies)

First author	Focus	Journal	Design	Metric	Region
Wang et al.	Associations of ASIC2, IKZF3, and GSDMB Variants with Gestational Diabetes Mellitus and Their Interactions with Environmental Exposures Among Pregnant Women.	<i>Biomolecules</i>	Case-control	Pregnancy PM _{2.5} and O ₃ (method not stated); urinary MIBP in a subset	China
Wanta et al.	Ultrastructural Changes in Human Placental Tissues and Evidence of Particulate Matter Transport into the Umbilical Vein Following PM _{2.5} Exposure in Chiang Rai.	<i>Cells Tissues Organs</i>	Biomonitoring + in vitro	Third-trimester high-PM _{2.5} season (Feb-Apr) vs no high exposure; SEM-EDS, TEM	Thailand (Mae Chan, Chiang Rai)
Zhang et al.	Association between ambient PM(2.5) exposure during pregnancy and risk of preterm birth: a retrospective birth cohort study in Chengdu, Southwest China.	<i>Front Public Health</i>	Retrospective cohort	Satellite-derived vs monitor-based PM _{2.5} across gestational windows	China (Chengdu)
Respiratory & allergic					
Dinh et al.	Association of air pollution with chronic obstructive pulmonary disease: anemia-related outcomes and exploratory genetic susceptibility analyses.	<i>Front Med (Lausanne)</i>	Case-control (registry)	Residential 1-, 3-, 6- and 12-month PM ₁₀ , PM _{2.5} , NO ₂ , NO, NOx, SO ₂ , CO and O ₃ at recruitment	Taiwan
Dou et al.	Risk factors for allergic rhinitis in children: an umbrella review of systematic review and meta-analysis.	<i>Front Immunol</i>	Umbrella review / meta-analysis	Air pollution as one of 75 pooled risk factors (metric not specified in abstract)	Global / multi-region
HEPA RCT synthesis	HEPA air filtration for asthma and COPD: lessons from randomized controlled trials.	<i>Expert Rev Respir Med</i>	Narrative synthesis of randomized controlled trials	Portable HEPA filtration as intervention; indoor PM _{2.5} as the mediating exposure	Multi-country trial evidence (PubMed and Scopus to 5 May 2026)
Huang et al.	Global, regional, and 204 countries burden of disease attributed to second-hand smoke-related COPD, 1990-2021: A secondary data analysis of the Global Burden of Disease Study 2021.	<i>Tob Induc Dis</i>	Descriptive burden analysis, GBD 2021	Secondhand tobacco smoke as a fine-particulate exposure, GBD comparative risk assessment	Global, 204 countries
Johansson et al.	Associations Between Indoor Air Pollution and Asthma, Allergic Rhinitis and Eczema in a Cohort of Children in Durban, South Africa	<i>Indoor Air</i>	Birth-cohort sub-study with indoor measurement	Indoor PM _{2.5} measurements; prenatal PM _{2.5} ; ETS, biomass fuel, dampness by walkthrough	South Africa (Durban)
Kang et al.	Asthma Intervention with Residential Ventilation and Air Cleaner (AIR-VAC) Study: A 4-arm parallel-group randomized controlled trial protocol.	<i>PLoS One</i>	Trial / intervention	Indoor/outdoor PM, NO ₂ , CO, VOCs and mold by low-cost and research-grade sensors	USA (Dallas-Fort Worth, Texas)
Naka et al.	Impact of particulate matter 2.5 exposure on severity of pediatric chronic rhinitis: A prospective cohort study.	<i>Pediatr Allergy Immunol</i>	Prospective cohort	1-week mean ambient PM _{2.5} by inverse-distance weighting of monitors	Thailand (Bangkok)
Samdarshi et al.	Systematic misclassification of lung function in Indian adolescents using GLI-2012 reference equations: evidence from the multicity APEAL cohort.	<i>Thorax</i>	Cross-sectional (cohort baseline)	None at this analysis stage (APEAL air-pollution cohort baseline)	Delhi, Mumbai, Bengaluru and Mysuru, India
Song et al.	Air Pollution Effects on Influenza-like Illness Modified by Temperature and Humidity in Central China	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - air pollution x temperature/humidity, influenza-like illness	China (Central China)
Zhang et al.	Associations between household particulate matter and inflammatory cytokines in cystic fibrosis.	<i>Respir Med</i>	Real-time sensor campaign	In-home real-time PM _{2.5} sensors, 4-11 days; mean and % time > 35 µg/m ³	Not stated (single-centre CF pilot)
Mechanistic toxicology					
Ameli et al.	Acute co-exposure to urban particulate matter and aircraft noise elicits lung-centered immune-redox proteomic remodeling in mice.	<i>Environ Pollut</i>	Animal model	NIST SRM 1648a urban PM, 3 days, alone and with aircraft noise	Laboratory (C57BL/6J mice)

First author	Focus	Journal	Design	Metric	Region
CSE HASMC isoforms	A comprehensive full-length transcriptome landscape of cigarette smoke-exposed HASMCs reveals extensive remodeling of mRNA isoforms and regulatory networks.	<i>PLoS One</i>	In vitro long-read transcriptomics, dose-response	Cigarette smoke extract 0.2-0.8% on human aortic smooth muscle cells (combustion-aerosol surrogate)	In vitro
Cherriere et al.	Influence of lung overload and exposure scenarios on recovery following inhalation of an alumina nanoparticle and HCl gas mixture.	<i>Part Fibre Toxicol</i>	Animal inhalation exposure + dosimetry	Alumina nanoparticles 1-20 mg/m ³ co-exposed with 5 ppm HCl, nose-only inhalation	Laboratory (rat inhalation, France)
Ghio et al.	Iron, Tuberculosis, and Particle-Related Exposures Including Smoking.	<i>Pathogens</i>	Mechanistic review	Cigarette smoke, ambient PM, biomass smoke, silica (conceptual)	Global / multi-region (review)
Gradinaru et al.	Atmospheric Particulate Matter as a Carrier of Pb, Cd, and Ni: From Environmental Transfer and Bioaccessibility to Molecular Toxicity and Predictive Modeling.	<i>J Xenobiot</i>	Narrative review	PM-bound Pb, Cd and Ni across the source-to-effect continuum	Global / multi-region
Gulzar et al.	Mechanistic Insights into PM _{2.5} -Induced Cardiopulmonary Dysfunction: Bridging Pathways to Therapeutic Interventions	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - PM _{2.5} cardiopulmonary mechanisms (review by title)	Not recoverable from metadata
Jiang et al.	Nitration of Urban PM _{2.5} : A Hidden Driver of Enhanced Human Inflammation.	<i>Environ Sci Technol</i>	Exposome-wide association	Personal PM _{2.5} filters, 1000 nitrated aromatic compounds quantified by HRMS	China (Beijing)
Lang et al.	Particulate matter selectively promotes macropinocytosis and malignant transformation-associated phenotypes in p53-impaired epithelial cells.	<i>Environ Toxicol Pharmacol</i>	In vitro (cell lines)	European certified reference PM _{2.5} , acute and repeated exposure	Laboratory (murine mammary, hamster respiratory cells)
Lin et al.	Air Pollution Triggers Metabolomic Signature of Mitochondrial Dysfunction in Mice and Humans: Implications for Cardiovascular Disease Development.	<i>Arterioscler Thromb Vasc Biol</i>	Animal + in vitro	Diesel exhaust (ApoE ^{-/-} mice, 2 wk) and a natural LA-to-Beijing move (PM _{2.5} 14.4 → 67.6 µg/m ³ , 10 wk)	USA and China (Los Angeles-Beijing travel panel)
Meganathan et al.	Aryl hydrocarbon receptor (AhR) regulates organic dust and organic dust-associated bacterial extracellular vesicles-induced airway inflammation via ROS-NFκB signaling.	<i>Respir Res</i>	Animal + in vitro	Poultry-farm organic dust extract and dust-associated bacterial EVs (in vitro, murine)	Laboratory (Beas2B cells, AhR knockout mice)
Messina et al.	Organ-specific apoptotic pathways induced by micro- and nanoplastics: a systematic review of in vitro toxicological evidence.	<i>Front Toxicol</i>	Systematic review (PRISMA)	Micro- and nanoplastic particles across cell and organoid models (airborne fraction not separated)	Not applicable (literature synthesis)
Monge et al.	Heart Function Changes During Environmentally Relevant Combined Exposure to Heat, Ultrafine Carbon Black, and Ozone: Role of NOD-like Receptor X1.	<i>Cells</i>	Animal model	Ultrafine carbon black 2.5 mg/m ³ + O ₃ 1 ppm, with or without 36 °C heat, 3 h	Laboratory (Nlrp1 knockout mice)
Pei et al.	Deciphering Allergenic Components Responsible for Differential Sensitization Potentials of Source-Specific Atmospheric Particulate Matter	<i>Environ Sci Technol</i>	In vitro exposure	NIST SRMs: urban PM 1648a, coal fly ash 2693, diesel exhaust particles 2975	Laboratory (THP-1, HaCaT co-culture)
Shen et al.	A Brake on Toxicity: Iron Suppresses Organic-Induced Superoxide Generation in Macrophages	<i>Environ Sci Technol</i>	In vitro exposure	Brake-wear particles, highway PM _{2.5} , Fe ²⁺ /Fe ³⁺ with isoprene SOA or 9,10-PQN	Laboratory (macrophage cell culture)
Song et al.	Lung Toxicity Induced by Co-Exposure to Particulate Matter and Indoor Microorganisms in Mice.	<i>Toxicol Lett</i>	Animal experiment	PM plus dominant indoor microorganisms by intratracheal instillation, 2 weeks	Laboratory (C57BL/6 mice)

First author	Focus	Journal	Design	Metric	Region
Tzagkaroulaki et al.	An A549 Cell-Based Approach Using Repeated Fluorescence Readouts for Assessing Reactive Oxygen Species Activity of Atmospheric Particulate Matter.	<i>Toxics</i>	Analytical method validation	NIST SRM 2584 and field PM _{2.5} (urban-background, high-altitude) on PTFE and quartz	Greece
Wang et al.	Ozone and PM _{2.5} co-exposure exacerbates allergic rhinitis and is associated with PPAR/TRPV1-related barrier damage: Insights from nasal transcriptomic and microbial analyses.	<i>Environ Pollut</i>	Animal model	Acute high-dose PM _{2.5} and O ₃ , single and combined, in OVA-sensitised mice	Laboratory (murine)
Yang et al.	The Human Microbiome in Airborne Particulate Matter-Associated Aging-Related Chronic Diseases: An Evidence-Stratified Review.	<i>Environ Health (Wash)</i>	Mechanistic review	Airborne PM (size fractions not separated across the reviewed literature)	Global / multi-region (literature synthesis)
Yang et al.	Molecular Mechanisms of Indoor Environmental Pollution tt-DDE-Induced Blood Pressure Elevation and the Protective Role of Human-Specific Gene CHRFAM7A.	<i>Environ Health (Wash)</i>	Animal + in vitro	trans,trans-2,4-decadienal, the dominant aldehyde in cooking-oil fumes, by inhalation	Laboratory (transgenic mice and endothelial cells, China)
Ye et al.	Mechanistic Study of SRSF1-Mediated Inhibition of Treg Cell Function via the mTORC1/ULK1 Signalling Axis Through Mitophagy in Asthma.	<i>Clin Exp Pharmacol Physiol</i>	Animal model	PM _{2.5} instillation arm added to OVA asthma model (dose not reported in abstract)	Laboratory (murine plus patient BALF)
Occupational & indoor					
Albert et al.	Assessing home environmental hazards for older adults receiving in-home supportive services: the healthy homes assessment tool.	<i>Innov Aging</i>	Cross-sectional survey	Consumer-grade indoor PM _{2.5} and radon monitors vs care-coordinator hazard ratings	USA (older-adult households)
Build Environ 2026 (cleanroom)	Multi-objective ventilation optimization in non-unidirectional cleanrooms: an integrated RSM-NSGA-II-TOPSIS approach	<i>Build Environ</i>	Metadata only (no abstract)	Not recoverable from metadata	Not recoverable from metadata
Build Environ 2026 (subway)	Spatiotemporal distribution characteristics of multiple air pollutants within subway stations	<i>Build Environ</i>	Metadata only (no abstract)	Not recoverable from metadata	Not recoverable from metadata
Environ Int 2026 (button mfg)	Plastic button manufacturing as an under-recognized source of airborne multi-pollutant mixtures: process-resolved characterization	<i>Environ Int</i>	Metadata only (no abstract)	Not recoverable from metadata	Not recoverable from metadata
Figueiredo et al.	Chemical and toxicological assessment of particulate matter in a charcoal barbecue restaurant.	<i>Environ Toxicol Pharmacol</i>	Measurement campaign	Simultaneous indoor/outdoor PM ₁₀ filters: ions, OC/EC, organic speciation, OP (AA, DTT)	Portugal
Gaillard et al.	Potential Link Between Indoor Air Quality and Coronary Artery Disease in Amsterdam, a Field Study	<i>Preprints.org (preprint)</i>	Long-term indoor monitoring campaign	Multi-week in-home PM _{2.5} , PM ₁₀ and CO ₂ monitoring	Netherlands (Amsterdam)
JP laboratory exposure	Assessment of personal exposure and spatial concentration variability in a small-scale chemical laboratory.	<i>Ann Work Expo Health</i>	Repeated field measurement with CFD simulation	Paired fixed-point area sampling and personal passive sampling, plus computational fluid dynamics	Japan (university teaching laboratory, 2024 and 2025 campaigns)
Lee et al.	Comparative analysis of metal concentrations in respirable particles and nanoparticulate matter (NPM) from mild steel, aluminum, and stainless steel welding fumes.	<i>J Hazard Mater</i>	Field measurement	Breathing-zone respirable and NRD nanoparticle samplers; Mn, Fe, Cr, Ni, Cu, Al by ICP-OES	Not stated (manufacturing facilities)
Liu	Energy-Indoor Air Quality-Health Nexus in Rural Residential Retrofits: A Field-Calibrated Coupled Simulation and Multi-Criteria Evaluation for the Central Plains Transitional Climate of China	<i>Atmosphere</i>	Repeated field measurement with CFD simulation	72-h paired before/after indoor PM _{2.5} , PM ₁₀ , CO ₂ , HCHO, TVOC; EnergyPlus-CONTAM co-simulation	China (Anyang, Henan)

First author	Focus	Journal	Design	Metric	Region
Lv et al.	Impact of biomass stove intervention on indoor air quality in rural homes of China: Attenuation of cooking contribution.	<i>J Hazard Mater</i>	Trial / intervention	Real-time PM _{2.5} monitors in kitchen, living room, bedroom and yard; pre/post stove change	China (Xuanwei, Yunnan)
Opbroek et al.	Health effects of indoor environmental quality (IEQ): A systematic review of indoor environments in high-income settings.	<i>Sci Total Environ</i>	Metadata only (no abstract)	Metadata only - indoor environmental quality parameters (scope not recoverable)	Not recoverable from metadata
Perfilyeva et al.	Genetic Susceptibility to Impaired Lung Function in Underground Coal Miners from Kazakhstan.	<i>Genes (Basel)</i>	Cross-sectional (cohort baseline)	Occupational coal mine dust (mining exposure as covariate)	Kazakhstan (Karaganda coal basin)
Pessoa Campos et al.	Salivary 8-hydroxy-2'-deoxyguanosine as a biomarker of oxidative DNA damage in urban outdoor workers exposed to traffic-related air pollution.	<i>J Toxicol Environ Health A</i>	Cross-sectional (comparative community)	Workplace PM _{2.5} and noise measured with portable instruments; no personal sampling	Not stated (urban street-vending and indoor control workplaces)
Qian et al.	Impacts of Housing Retrofits on Indoor Environmental Quality and Energy Consumption: A Review	<i>Indoor Air</i>	Narrative review	Indoor PM _{2.5} , VOCs, NO ₂ , HCHO, radon, CO ₂ , T and RH before and after retrofit	Global / multi-region
SAA wetting agent	High-efficiency 'SAA' wetting agent for coal mine dust control: from preparation and mechanism to field application.	<i>Environ Geochem Health</i>	Laboratory formulation and mechanism study with underground field trial	Respirable coal dust; median particle size D50 and dust suppression rate	China (3-1802 working face, deep coal seam)
Taskin and Ozkal	Investigating Airborne Pollutant Dynamics in a Pathology Laboratory Using an Integrated High-Resolution Sensor Network and Operational-Context Framework.	<i>Indoor Air</i>	Real-time sensor campaign	NI myRIO multi-sensor node, 1-min HCHO, eTVOC, eCO ₂ , PM, T, RH; 42 days	Not stated (university hospital pathology laboratory)
Wang et al.	Research on dust control and ventilation parameter optimization in roadhead mining based on response surface methodology.	<i>PLoS One</i>	CFD simulation + scaled-model validation	Simulated dust mass concentration at return-air-side corners of a heading face	China (underground coal heading face)
Wang et al.	Aeration change from horizontal brush to fine bubble substantially reduces the emission of inhalable antibiotic resistome from wastewater to air.	<i>J Hazard Mater</i>	Field measurement	PM _{2.5} metagenomics (ARGs, pathogens) under two aeration modes, seasonal	Not stated (single WWTP)
Wu et al.	Sphagnum palustre moss bags as sentinels to monitor indoor air PAHs in university libraries.	<i>Int J Phytoremediation</i>	Field measurement	Sphagnum moss bags, 45-day deployments, total PAHs (ug/kg moss), indoor and outdoor	Not stated (three university libraries)
Xing et al.	Household air pollution associated with different heating approaches in rural Northern China.	<i>Sci Bull (Beijing)</i>	Long-term field measurement campaign	Year-round indoor PM _{2.5} monitoring in 948 households, plus exposure modelling	China (North China Plain, rural)
ZCND aerogel filter	Biomimetic dual-network nanofibrous aerogels with robust interfacial assembly for multifunctional air filtration.	<i>J Colloid Interface Sci</i>	Materials synthesis and bench filtration characterization	PM ₁ , PM _{2.5} and PM ₁₀ filtration efficiency, pressure drop, dust-holding capacity under controlled humidity	Bench (live-stock housing application)
Zigabe et al.	Determinants of early-life household air pollution exposure (HAP): carbon monoxide (CO) and PM _{2.5} concentrations in Bukavu, eastern Democratic Republic of Congo (DRC).	<i>Environ Pollut</i>	Cross-sectional (cohort baseline)	24-h household PM _{2.5} by PurpleAir Zen; 24-h personal CO by Lascar EL-USB-CO	Bukavu, Democratic Republic of Congo

First author	Focus	Journal	Design	Metric	Region
Sensing, forecasting & instrumentation					
AIRWISE Delhi	Improving aerosol-radiation interactions in the operational forecasting system - AIRWISE.	<i>Atmos Chem Phys</i>	Operational NWP sensitivity experiments with observational evaluation	Species-resolved complex refractive indices driving aerosol-radiation interaction in an operational forecast	Delhi, India (October 2023 - January 2024)
Anwer et al.	Artificial Intelligence in Sand and Dust Storm Research: A Systematic and Bibliometric Review	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - AI methods for sand and dust storm detection/forecasting (by title)	Not recoverable from metadata
Barth et al.	Online characterisation of oxidative potential and reactive oxygen species of fresh and aged aircraft turbine and ship engine particulate emissions.	<i>Environ Sci Atmos</i>	Chamber emission characterisation	Online OP and particle-bound ROS of fresh and photochemically aged aircraft-turbine and ship-engine exhaust	Chamber / laboratory (engine test facility)
Broda et al.	A new balloon-borne system for measuring the vertical variability of aerosol optical properties using low-cost sensors.	<i>Atmos Meas Tech</i>	Instrument development	Sensirion SPS30 low-cost OPC, calibrated to scattering coefficient, Angstrom exponent and effective radius	Poland (southeastern)
Contini et al.	10-year trends of particle size distributions and equivalent black carbon at the Environmental-Climatic Observatory (ECO) in Lecce (Italy).	<i>Environ Pollut</i>	Long-term field measurement campaign	Online PM _{2.5} , PM _{10-2.5} , PNSD 10-10,000 nm, eBC; offline OC/EC for MAC, 2015-2024	Italy (Lecce)
Khalili Hollo et al.	Validation and Correction of PM _{2.5} Mass Concentrations from Consumer-Grade Personal Exposure and Outdoor Fixed Monitors: A Two-Year Colocation Study.	<i>ACS Environ Au</i>	Sensor co-location / calibration	Atmotube Pro (n=50) and QuantAQ Modulair-PM (n=5) vs GRIMM EDM180, 1-h PM _{2.5}	USA (Denver, Colorado)
Lawson et al.	Humidity-dependent photoacoustic absorption biases and effective water mass accommodation coefficients for multicomponent aerosol particles	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - photoacoustic absorption bias vs RH, multicomponent aerosol	Laboratory (not recoverable from metadata)
Le et al.	Real-time monitoring of ultrafine aerosol size distributions using parallel non-bouncing impactors coupled with corona charger.	<i>Meas Sci Technol</i>	Instrument development	Particle number size distribution 10-560 nm; total PNC and UFP (PM _{0.1})	Single urban rooftop (location not stated)
Liu et al.	Morphology Change of Diesel Exhaust Black Carbon Particles during Humidification and Evaporation.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - black carbon particle morphology through humidification and evaporation cycles	China (laboratory, site not recoverable)
Low-cutoff cyclone method	An improved method for measuring the penetration efficiency of low cutoff cyclones.	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - penetration efficiency measurement for low cut-point cyclone inlets	Not recoverable
Medellin MABI eBC	Multi-wavelength spectral attribution of equivalent black carbon in Medellin, Colombia (2010-2023): Disentangling fossil fuel and biomass burning using the MABI instrument.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - Multi-wavelength Absorption Black carbon Instrument, spectral source attribution, 14-year record	Medellin, Colombia
Papathanasiou et al.	Evaluating Personal Air Quality Monitors for PM _{2.5} and CO ₂ : Insights from a Year-Long Pilot Study.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - personal PM _{2.5} /CO ₂ monitors, year-long pilot (by title)	Not recoverable from metadata

First author	Focus	Journal	Design	Metric	Region
Parks et al.	Evaluating performance of real-time monitors for quantifying elemental carbon in diesel particulate matter.	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - real-time elemental carbon monitors against diesel particulate matter	Not stated (abstract not deposited)
Shen et al.	Apportioning light absorption of ambient aerosols to black carbon, brown carbon, and lensing effect using a PAX-ISS hybrid method: insights into absorption enhancement.	<i>Atmos Chem Phys</i>	Multi-instrument field evaluation	Photoacoustic absorption (PAX) plus solvent-mediated integrating-sphere absorption of ambient filter samples	Beijing, China
Shin et al.	A Continuous Virus Aerosol-into-Liquid Sampler (VAiLS) Integrating Twin-Vortex Condensation Growth and Wet-Cyclone Enrichment.	<i>Environ Sci Technol</i>	Laboratory sampler validation	Number-weighted collection efficiency for KCl nanoparticles and MS2/phi6 bioaerosols at 16 L/min	Laboratory (bioaerosol test rig, South Korea)
Song	Relocation-associated discontinuities in Korean air-quality monitoring: a Bayesian structural time-series analysis of PM ₁₀ and PM _{2.5} .	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - station relocation step changes in PM ₁₀ /PM _{2.5} records (by title)	South Korea (national monitoring network, by title)
Thomas et al.	Procedural Challenges in the Analysis of Toxic Metals in Electronic Cannabinoid Delivery System Aerosols Produced for Vaping Cannabinoids.	<i>J Test Eval</i>	Method review	Aerosol trapping media for ICP-MS determination of metals in e-cigarette and cannabis vaping aerosol	United States (ASTM D37.08 work group)
Urban OOM flux	Urban surface-atmosphere fluxes of selected pptv-level oxygenated organic molecules (OOMs) from eddy covariance observations.	<i>Atmos Chem Phys</i>	Eddy covariance flux measurement, method development	Iodide-adduct CIMS eddy covariance fluxes of 16 oxygenated organic molecules (SOA precursors)	Central China megacity, hottest month
Wu et al.	Spatiotemporal Prediction of Sandstorm Utilizing Multi-Source Data and ConvLSTM Model: A Case Study in Northwest China	<i>Atmosphere</i>	Deep-learning forecasting	Surface PM ₁₀ with Himawari-8/9 AOD assimilated by EnKF into ConvLSTM	China (Gansu)
Zhang et al.	Are expensive sensors necessary for assessing indoor environmental quality? A data-driven approach using minimal sensing.	<i>Build Environ</i>	Metadata only (no abstract)	Metadata only - minimal-sensing IEQ assessment (by title)	Not recoverable from metadata
Exposure assessment & modelling					
Ahmad et al.	Characterizing Ambient Fine Particulate Matter Pollution in Jordan: National Monitoring Evidence of Spatial Inequality, PM _{2.5} Hotspots, and Temporal Trends	<i>Preprints.org (preprint)</i>	Monitoring-record analysis	Annual PM _{2.5} from Jordan's national monitoring network, 2018-2024	Jordan
Athens NPF	The influence of air mass origin and ammonia emissions on new particle formation at a Mediterranean urban site (Athens, Greece).	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only - new particle formation events, ammonia, back-trajectory air-mass origin	Athens, Greece
Atik et al.	Inter-annual trends of PM _{2.5} -bound heavy metals in Dhaka, Bangladesh: a comparative study for source apportionment and health risk assessment	<i>Environ Sci Atmos</i>	Source apportionment	Elevated-altitude PM _{2.5} filter sampling, heavy metals, PMF	Bangladesh (Dhaka)
Bhattarai et al.	Spatio-temporal assessment of PM _{2.5} and PM ₁₀ pollution in core and peripheral municipalities of the Kathmandu valley using regulatory and low-cost sensor data	<i>Research Square (preprint)</i>	Network monitoring + burden	Regulatory monitors integrated with a high-resolution low-cost sensor network, daily PM _{2.5} /PM ₁₀ , 2023-2024	Nepal (Kathmandu Valley)
Broda et al.	Vertical and Zonal Characteristics of Transatlantic Biomass Burning Aerosol Transport During the Extreme 2023 Canadian Wildfire Season.	<i>Atmosphere</i>	Multi-platform observation synthesis	NAAPS reanalysis smoke AOD (550 nm), MODIS fire and AOD, modelled plume injection heights	North America to Europe (transatlantic)

First author	Focus	Journal	Design	Metric	Region
Chen et al.	Chemical Characteristics, Source Apportionment and Toxic Potential of PM _{2.5} in Southern Taiwan: Insights from the Kao-Ping Experiment (KPEX).	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - speciated PM _{2.5} with source apportionment and a toxicity metric	Taiwan (Kao-Ping region)
Chen et al.	Formation mechanism of rapid day-time nitrate production in a typical semi-arid city near dust source regions	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only - daytime particulate nitrate formation near dust sources	Not recoverable from metadata
China aerosol vertical	Spatiotemporal Dynamics and Vertical Distribution of Aerosols in China: Insights from Multi-Source Evidence from 2000 to 2022.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - column and vertically resolved aerosol from multiple platforms, 2000-2022	China (national)
Garcia-Vara et al.	Expanding the chemical space of urban air: Identification and semiquantification of polar emerging contaminants in PM _{2.5} through LC-HRMS non-target analysis.	<i>J Hazard Mater</i>	Field measurement	Filter PM _{2.5} at 8 stations (traffic, urban background, port); LC-HRMS NTA	Spain (Barcelona)
Gonzalez-Mendez et al.	Morpho-chemical fingerprinting and distance-dependent attenuation of airborne Particulate Matter from abandoned mine tailings in Northwestern Mexico.	<i>Environ Pollut</i>	Source apportionment	Sigma-2 passive samplers, 9 sites; automated SEM-EDX single-particle (3,017 particles)	Mexico (northwestern)
Guo et al.	Seasonal Formation, Sources, and Long-Term Evolution of PM _{2.5} -Bound Water-Soluble Inorganic Ions in Nanchang, China: The Increasing Importance of Wintertime Nitrate.	<i>Toxics</i>	Field observations + box model + thermodynamic equilibrium	Day/night PM _{2.5} filters, nine WSIs; 2009-2026 historical records	China (Nanchang, Jiangxi)
Hien et al.	Long-term Observations of Fine Particulate Sulfate in Hanoi during 1999-2023: Impact of SO ₂ Regional Emissions and Monsoon Circulations.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - fine particulate sulfate time series, 1999-2023	Vietnam (Hanoi)
Hu et al.	Long-term spatiotemporal evolution, relative hotspot persistence, and predictive attribution of PM _{2.5} across China.	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - gridded long-term PM _{2.5} with hotspot persistence and attribution	China
Hurynovich et al.	Transferability of dust-related PM _{2.5} models under contrasting arid environmental regimes.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - dust PM _{2.5} model transferability (by title)	Not recoverable from metadata
Ireland Census linkage	Ambient air pollution at home, work and school: Evidence from Ireland.	<i>Health Place</i>	Population-wide administrative linkage, cross-sectional	EPA modelled annual ambient PM _{2.5} , PM ₁₀ , NO ₂ and O ₃ assigned to residential and to work/school address	Republic of Ireland (2022 Census of Population)
Kagawa et al.	Seasonal variations of aerosol selenium concentrations and oxidation states in central Japan.	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - aerosol selenium concentration and speciation by oxidation state	Japan (central)
Kaur et al.	Long-term PM _{2.5} source apportionment at 12 sites in the western United States from 2000-2020.	<i>Atmos Pollut Res</i>	Source apportionment	Speciated PM _{2.5} (CSN/IMPROVE), Positive Matrix Factorization and Chemical Mass Balance jointly	Utah, California, Nevada, Idaho, Colorado (USA)
Liu et al.	Airborne Bacterial and Eukaryotic Communities in Temuco, Chile, a City Dominated by Residential Wood Combustion.	<i>Life (Basel)</i>	Bioaerosol monitoring campaign	PM _{2.5} and PM ₁₀ filters, WSIs; 16S and 18S rRNA sequencing	Chile (Temuco)
Liu et al.	Effect of Skin Types and Microenvironment of Dermal Exposure to Polycyclic Aromatic Hydrocarbon Derivatives and Heavy Metals.	<i>Environ Health (Wash)</i>	Bench / device experiment	Particle-bound NPAHs, OPAHs and 11 metals adsorbed onto glycerol-trioleate-coated surrogate skin	China (campus indoor and outdoor sites, including dust events)

First author	Focus	Journal	Design	Metric	Region
Mainka & Mainka	Modelled Physical Activity Benefits and PM _{2.5} Risks in Two Urban Cyclists.	<i>Toxics</i>	Health impact model	Trip-resolved PM _{2.5} from portable optical monitors (162 trips) vs fixed reference station	Poland (Gliwice)
Ningombam et al.	Assessment of global aerosol burden over land and oceans using ground-based, satellite, and reanalysis data	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - global AOD from ground network, satellite and reanalysis (by title)	Not recoverable from metadata
Qin et al.	Seasonal Variation of Airborne Bacteria and Potential Pathogens Adhere to Fine and Coarse Particles in a Coastal City.	<i>Microorganisms</i>	Bioaerosol monitoring campaign	Size-segregated PM _{2.5} and PM ₁₀ filter samples, 16S rRNA sequencing	China (Qingdao)
Rana & Guzman	Heterogeneous Nitrate Radical Oxidation of Catechol Produces Nitrocatechols, Oligomers, and Brown Carbon.	<i>ACS Earth Space Chem</i>	Chamber / laboratory	Condensed-phase products of NO ₃ oxidation of catechol thin films (FTIR, UV-vis, ESI-MS, UHPLC-MS)	Laboratory (thin-film reactor)
Seibert et al.	National-scale NMF source apportionment reveals component-specific west-to-east asymmetries in PM _{2.5} concentrations associated with trans-boundary transport.	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - national NMF source apportionment of PM _{2.5} (by title)	Not recoverable from metadata
Shen et al.	A Hybrid Deep Learning-Geostatistical Framework for PM _{2.5} Spatial Estimation with SHAP Interpretation	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - hybrid DL + geostatistical PM _{2.5} surface with SHAP	Not recoverable from metadata
Sonipat ACSM campaign	Chemical characterization and source apportionment of carbonaceous aerosols during post-monsoon biomass burning and Diwali at an upwind site of Delhi.	<i>Atmos Chem Phys</i>	Intensive field campaign, PMF source apportionment	Composition-based PM _{2.5} (ToF-ACSM non-refractory PM _{2.5} plus multi-wavelength Aethalometer eBC)	Sonipat, Haryana, India (upwind receptor for Delhi-NCR)
W. India BC-dust	Seasonal Variability and Interactions of Black Carbon and Mineral Dust over Semi-Arid Western India: Satellite and Ground-Based Observations.	<i>Environ Sci Atmos</i>	Multi-year satellite and ground observational synthesis	Columnar and surface black carbon with mineral dust loading, satellite retrievals against ground stations	Semi-arid western India
Yang et al.	Complexation strength between organic carbon and transition metal ions dominates the photochemical conversion of SO ₂ to sulfates.	<i>Atmos Chem Phys</i>	Heterogeneous kinetics experiment	SO ₂ uptake coefficients and sulfate mass on coal-combustion organic carbon under irradiation	Laboratory (irradiated OC films)
Youn et al.	Seasonal distributions and source identification of water-soluble arsenic species of PM _{2.5} in the urban atmosphere of Northeast Asia	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Metadata only - water-soluble As speciation in PM _{2.5}	Northeast Asia (not further recoverable)
Zahid & Morris	Demand or Congestion? Disentangling Event-Related Emissions Using a Directional Control on I-15.	<i>Preprints.org (preprint)</i>	Method / algorithm	Link-level CO ₂ , NO _x and PM _{2.5} emissions estimated from one-minute free-way detector data	Las Vegas, Nevada, USA (I-15)
Zhang	Within-Year PM _{2.5} Exposure Structure Provides Mortality-Predictive Information Beyond the Annual Mean.	<i>medRxiv (preprint)</i>	Metric development + retrospective application	Daily modelled PM _{2.5} summarised as the magnitude-rank index (PMRI) alongside the annual mean	United States (3,064 counties)
Zhao et al.	Spatial-Temporal Variability of PM _{2.5} and Ozone Pollution in the Huaihai Economic Zone of China.	<i>Atmosphere (Basel)</i>	Monitoring-record analysis	PM _{2.5} and O ₃ concentration data 2000-2023 against MEIC emissions 2000-2020	China (Huaihai Economic Zone)

First author	Focus	Journal	Design	Metric	Region
Zhou et al.	Can Electric Vehicles Reduce Summertime Air Pollution Exposure Inequalities in Metro Atlanta? A High-Resolution Analysis	<i>GeoHealth</i>	Chemical transport modelling	CMAQ at 1 km vs 12 km, NO ₂ , MDA8 O ₃ and PM _{2.5} under 30% EV adoption	USA (Atlanta)
da Silva et al.	Chemical Regimes and PMF-Resolved Sources Linked to Acute PM ₁₀ Ecotoxicity in Luanda, Angola.	<i>Toxics</i>	Source apportionment	116 daily PM ₁₀ filter samples, speciated; PMF; Microtox aqueous-extract toxicity	Angola (Luanda)
Burden, policy & mitigation					
Aerosol Air Qual Res 2026 (IAQ education)	Cross-national Evaluation of STEM-Based Indoor Air Quality (IAQ) Education for Sustainable Development	<i>Aerosol Air Qual Res</i>	Metadata only (no abstract)	Not recoverable from metadata	Cross-national (countries not recoverable from metadata)
Ann Glob Health 2026 (disclosure)	Disclosure without accountability: Air pollution reporting by major South African industrial companies.	<i>Ann Glob Health</i>	Metadata only (no abstract)	Not recoverable from metadata	South Africa
Bouchriti et al.	Spatial analysis of the burden of lung cancer attributable to long-term PM(2.5) exposure in Morocco: a national assessment using the WHO AirQ+ model.	<i>Environ Monit Assess</i>	Health impact model	City-specific annual PM _{2.5} (2019) for 46 areas into WHO AirQ+	Morocco
Ding et al.	Which Roads Should Cities Control First? A Health-Risk Framework for Mobile-Source Air Pollution.	<i>Environ Sci Technol</i>	Metadata only (no abstract)	Metadata only - road-segment mobile-source pollution with a health-risk weighting	Not recoverable from metadata
Dong et al.	Impacts of PM _{2.5} air pollution on high-skilled worker productivity in China.	<i>Nat Hum Behav</i>	Ecological panel	Annual city-level ambient PM _{2.5} (mean 45.9 µg/m ³), instrumented	China (national)
Fregni et al.	Stroke Burden Attributable to Risk Factors in the Americas, 1990 to 2021: A Temporal Trends Analysis From the GBD Study 2021.	<i>J Am Heart Assoc</i>	GBD secondary analysis	GBD 2021 household air pollution from solid fuels (one of 23 risk factors)	Western Hemisphere, 39 countries and territories (GBD 2021)
Kan et al.	From Emission-Dominated Mitigation to Meteorology-Modulated Secondary Pollution: Decadal Insights from Beijing's Short-Term Interventions.	<i>Environ Pollut</i>	Measurement campaign	Aerosol composition and VOCs, 2025 vs 2015 Victory Day Parade controls; ML attribution	China (Beijing)
Kwon & Kim	Mortality and morbidity consequences of particulate matter in South Korea: evidence from transboundary air pollution.	<i>BMC Public Health</i>	Instrumental-variable panel analysis	Local monitored PM _{2.5} , instrumented by wind direction interacted with upwind Chinese monitor levels	South Korea
Li et al.	Clean air paradox: exacerbating urban-rural inequality in PM _{2.5} exposure.	<i>Sci Bull (Beijing)</i>	Exposure model (secondary data)	Modelled PM _{2.5} exposure, income-stratified, weighted by field-measured source-specific toxicity	China (mainland)
Li et al.	Contributions of local household fuel burning and regional transport to ambient PM _{2.5} in the Qinghai-Tibet Plateau.	<i>Environ Res</i>	CTM simulation + source attribution	Survey-based residential emission inventory 1960-2019 with GEOS-Chem normalised marginal attribution	China (Qinghai-Tibet Plateau)
Mundu et al.	Fossil fuel use, urban air pollution and health in rapidly growing African cities: Implications for sustainable and low-emission urban development.	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	Review of urban PM, NO _x , BC and VOC evidence; source-sector attribution	Africa (urban, multi-country)
N fertilizer NH ₃ -PM _{2.5}	Integrating nitrogen fertilizer management for mitigating agricultural ammonia emissions and PM _{2.5} pollution.	<i>Atmos Res</i>	Metadata only (no abstract)	Metadata only - agricultural NH ₃ emissions and secondary inorganic PM _{2.5} response	Not recoverable

First author	Focus	Journal	Design	Metric	Region
PPS antifouling filter	Robust and Self-Cleaning Fluorine-Free Superhydrophobic PPS Filters for High-Humidity Flue Gas Filtration.	<i>Environ Sci Technol</i>	Materials synthesis with simulated dynamic filtration testing	Industrial flue-gas dust; cleaning cycle, residual pressure drop, quality factor under humid conditions	Bench (industrial flue gas application)
Simoes et al.	Air Pollution, Temperature, and Demographic Vulnerability as Determinants of Cause-Specific Mortality in Portugal.	<i>J Xenobiot</i>	Ecological panel	Monthly NO ₂ and PM _{2.5} exceedance above WHO 2021 guideline values	Portugal
Turkate et al.	Implementing Community-Centered Air Quality Law: Success Stories From California's Assembly Bill 617.	<i>Am J Public Health</i>	Qualitative / ethnographic study	Community emissions reduction programme strategies across nine AB 617 communities	USA (California)
Wan et al.	Three-Phase Transition in China's Future Air Quality under Ambitious Mitigation Pathways Revealed by Large-Ensemble AI Projections.	<i>Environ Sci Technol</i>	Data-fusion / ML surrogate model	Hybrid deep-learning CTM surrogate, daily PM _{2.5} and O ₃ , 50 km, 2015-2060	China (national)
Yang et al.	PM _{2.5} , O ₃ and survival gains from China's clean air actions among patients with lung cancer: a 20-year registry-based cohort study in Beijing.	<i>Thorax</i>	Prospective cohort (registry linkage)	Three-year pre-diagnosis mean PM _{2.5} and O ₃ from the ChinaHigh-AirPollutants reanalysis	China (Beijing)
Other clinical endpoints					
Ali et al.	The Impact of Polycyclic Aromatic Hydrocarbons Exposure from Seasonal Biomass Burning on Oxidative Stress and Genotoxicity Biomarkers in Patients with Diabetes and Hypertension.	<i>Toxics</i>	Repeated-measures biomarker	Biomass-burning season (pre Dec vs post May); urinary 1-OHP as PAH dose	Thailand (Chiang Mai)
Hong et al.	Associations of temperature, particulate matter, and greenness with human papillomavirus positivity at screening: an analysis of Guangdong Province (2016-2022) using the treed distributed lag nonlinear model.	<i>Front Public Health</i>	Retrospective cross-sectional (health screening)	Biweekly mean PM _{2.5} at examination date, treed DLNM over 24 weeks	China (Guangdong)
Huang et al.	Short-term health impacts of air pollution in Canada: a nonlinear multi-pollutant Bayesian case-crossover framework.	<i>Int J Epidemiol</i>	Case-crossover	Daily PM _{2.5} , NO ₂ and O ₃ from imputed fixed-site monitoring, jointly modelled with nonlinear exposure-response	Canada (20 Census Divisions)
Huang et al.	The role and management of dietary and environmental factors in pediatric atopic dermatitis.	<i>Front Pediatr</i>	Case-control	Ambient PM _{2.5} dichotomised at 35 µg/m ³ (source and averaging period not stated)	Not stated (single-centre hospital series)
Kelishadi et al.	The Effect of Synbiotic Supplementation on Malondialdehyde Levels in Adolescents Exposed to Air Pollution: A Randomized, Single-Blind, Controlled Clinical Trial.	<i>Int J Prev Med</i>	Trial / intervention	Residence in a polluted setting; no individual PM measurement reported	Not stated (single urban setting)
King	Outdoor air pollution as a potentially modifiable perioperative risk factor.	<i>Br J Anaesth</i>	Metadata only (no abstract)	Metadata only - ambient air pollution and perioperative risk (commentary / review by title)	Not recoverable from metadata
Leonforte et al.	Environmental Contaminants and Osteoporosis-Related Outcomes: A Systematic Review and Meta-Analysis.	<i>Toxics</i>	Systematic review + random-effects meta-analysis	Ambient PM _{2.5} , PM ₁₀ , PM ₁ and gases among ten contaminant classes	Global / multi-region (review)
Zhu et al.	Long-term exposure to particulate matter and survival among patients with gastric cancer: A hospital-based cohort study.	<i>Environ Res</i>	Retrospective cohort	36-month pre-surgery residential PM _{2.5} , PM ₁₀ and PM _{2.5-10} from the ChinaHigh-AirPollutants (CHAP) reanalysis	Zhejiang, China

Provenance and method

Period **20–26 September 2026**, Sunday to Saturday, **seven daily issues, 162 records, 162 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. The only live queries made for this issue were the ClinicalTrials.gov reconciliation (COATED-AIR resolution and `analyze_endpoints`), which are state maintenance. Tier mix: **25 A, 64 B, 73 C**. The W39 weekly was due on the 26 September run; that run and those for 27 and 28 September did not fire, and the rollup was built on the **29 September backfill run** after the 26 September daily issue, which it includes. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.